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ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
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Foreword

This Technical Specification (TS) has been produced by ETSI Technical Committee Railway Telecommunications (RT).

The present document is part 2 of a multi-part deliverable. Full details of the entire series can be found in part 1 [26].

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document specifies the protocol conformance testing for testing a Future Railway Mobile Communication System (FRMCS) Server for compliance to railway specific functionality over 5G. The present document addresses only FRMCS Server-Client communication scenarios for:

- Specific Mission Critical Push-To-Talk (MCPTT) functionality.
- Mission Critical Ad Hoc Group Communication (AHGC) and Mission Critical Ad Hoc Group Emergency Alert (AHGEA).
- MCData including SDS, FD, QoS, MP-QUIC and MP-TCP as well IP-connectivity supporting datagram and streaming mode.
- Enhanced MCPTT functionality including geographical based features.
- Interception.
- Inter-FRMCS-Server functionality.

It does not cover e.g. MCPTT Server-5G Core, MCPTT Server-SIP Core, etc. scenarios which involve interfaces which implementation may widely vary.

The present document contains:

- the overall test structure;
- the test configurations;
- the conformance requirement and reference to the core specifications;
- the Test purposes; and
- a brief description of the test procedure, the specific test requirements and short message exchange table.

The present document is valid for FRMCS Clients implemented according to 3GPP releases starting from Release 18.

The present document does not specify the protocol conformance testing for the 5G Core bearers which carry the MCPTT data sent or received by the MCPTT Server. The specification of such testing is out of the scope of RAN5.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 122 179 (V19.3.0): "LTE; 5G; Mission Critical Push to Talk (MCPTT); Stage 1 (3GPP TS 22.179 version 19.3.0 Release 19)".

[2] ETSI TS 122 280 (V19.7.0): "LTE; 5G; Mission Critical Services Common Requirements (MCCoRe); Stage 1 (3GPP TS 22.280 version 19.7.0 Release 19)".

[3] ETSI TS 122 282 (V19.0.0): "LTE; 5G; Mission Critical (MC) data (3GPP TS 22.282 version 19.0.0 Release 19)".

[4] ETSI TS 123 280 (V19.8.1): "LTE; Common functional architecture to support mission critical services; Stage 2 (3GPP TS 23.280 version 19.8.1 Release 19)".

[5] ETSI TS 123 379 (V18.12.0): "LTE; Functional architecture and information flows to support Mission Critical Push To Talk (MCPTT); Stage 2 (3GPP TS 23.379 version 18.12.0 Release 18)".

[6] ETSI TS 138 508 (V18.6.0): "LTE; 5G; 5GS; User Equipment (UE) conformance specification; Part 2: Common Implementation Conformance Statement (ICS) proforma (3GPP TS 38.508-2 version 18.6.0 Release 18)".

[7] ETSI TS 136 579-3 (V15.0.0): "LTE; Mission Critical (MC) services over LTE; Part 3: Mission Critical Push To Talk (MCPTT) Server Application conformance specification (3GPP TS 36.579-3 version 15.0.0 Release 15)".

[8] ETSI TS 124 229 (V19.5.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3 (3GPP TS 24.229 version 19.5.0 Release 19)".

[9] IETF RFC 3856: "A Presence Event Package for the Session Initiation Protocol (SIP)".

[10] IETF RFC 6665: "SIP-Specific Event Notification".

[11] IETF RFC 1889: "RTP: A Transport Protocol for Real-Time Applications".

[12] ETSI TS 124 380 (V18.6.0): "LTE; Mission Critical Push To Talk (MCPTT) media plane control; Protocol specification (3GPP TS 24.380 version 18.6.0 Release 18)".

[13] ETSI TS 123 289 (V18.8.0): "5G; Mission Critical services over 5G System; Stage 2 (3GPP TS 23.289 version 18.8.0 Release 18)".

[14] ETSI TS 137 579-5 (V18.0.0): "LTE; 5G; Mission Critical (MC) services; Part 5: Abstract test suite (ATS) (3GPP TS 37.579-5 version 18.0.0 Release 18)".

[15] ETSI TS 133 180 (V19.4.0): "LTE; 5G; Security of the Mission Critical (MC) service (3GPP TS 33.180 version 19.4.0 Release 19)".

[16] IETF RFC 3261: "SIP - Session Initiation Protocol".

[17] IETF RFC 3903: "Session Initiation Protocol (SIP) Extension for Event State Publication (PUBLISH)".

[18] IETF RFC 3556: "Session Description Protocol (SDP) Bandwidth Modifiers for RTP Control Protocol (RTCP) Bandwidth".

[19] IETF RFC 3550: "RTP: A Transport Protocol for Real-Time Applications".

[20] ETSI TS 124 379 (V19.4.0): "LTE; Mission Critical Push To Talk (MCPTT) call control; Protocol specification (3GPP TS 24.379 version 19.4.0 Release 19)".

[21] ETSI TS 124 582 (V19.0.0): "LTE; Mission Critical Data (MCData) media plane control; Protocol specification (3GPP TS 24.582 version 19.0.0 Release 19)".

[22] ETSI TS 129 582 (V19.0.0): "LTE; 5G; Mission Critical Data (MCData) interworking with Land Mobile Radio (LMR) systems; Stage 3 (3GPP TS 29.582 version 19.0.0 Release 19)".

[23] ETSI TS 133 310 (V19.5.0): "Universal Mobile Telecommunications System (UMTS); LTE; 5G; Network Domain Security (NDS); Authentication Framework (AF) (3GPP TS 33.310 version 19.5.0 Release 19)".

[24] IETF RFC 2408: "Internet Security Association and Key Management Protocol (ISAKMP)".

[25] IETF RFC 8446: "The Transport Layer Security (TLS) Protocol Version 1.3".

[26] ETSI TS 104 069-1 (V1.1.1): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Services over 5G; Part 1: Application conformance specification for client".

[27] ETSI TS 103 765-3 (V1.1.1): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Building Blocks and Functions; Part 3: Train On-Board functions and interfaces".

[28] ETSI TS 103 765-1 (V1.1.1): "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Building Blocks and Functions; Part 1: Transport Stratum".

[29] IETF RFC 4443: "Internet Control Message Protocol (ICMPv6)".

[30] IETF RFC 8200: "Internet Protocol, Version 6 (IPv6) Specification".

[31] IETF RFC 9000: "QUIC_ A UDP-Based Multiplexed and Secure Transport".

[32] IETF RFC 9897: "Datagram Congestion Control Protocol (DCCP) Extensions for Multipath Operation with Multiple Address".

[33] ETSI TS 124 282 (V19.5.0): "LTE; Mission Critical Data (MCData) signalling control; Protocol specification (3GPP TS 24.282 version 19.5.0 Release 19)".

[34] ETSI TS 137 579-1 (V18.0.0): "LTE; 5G; Mission Critical (MC) services; Part 1: Common test environment (3GPP TS 37.579-1 version 18.0.0 Release 18)".

[35] IETF RFC 3706: "A Traffic-Based Method of Detecting Dead Internet Key Exchange (IKE) Peers".

[36] ETSI TS 123 282 (V19.10.0): "LTE; Functional architecture and information flows to support Mission Critical Data (MCData); Stage 2 (3GPP TS 23.282 version 19.10.0 Release 19)".

[37] IETF RFC 2784: "Generic Routing Encapsulation (GRE)".

[38] IETF RFC 792: STD 5: "Internet Control Message Protocol".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

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[i.1] ETSI TS 104 070: "Rail Telecommunications (RT); FRMCS conformance specification; Test Applicability and Implementation Conformance Statement (ICS) proforma specification (Release 19)".

[i.2] ETSI TS 136 579-4 (V16.8.0): "LTE; Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification (3GPP TS 36.579-4 version 16.8.0 Release 16)".

[i.3] ETSI TR 121 905 (V18.0.0): "Digital cellular telecommunications system (Phase 2+) (GSM); Universal Mobile Telecommunications System (UMTS); LTE; 5G; Vocabulary for 3GPP Specifications (3GPP TR 21.905 version 18.0.0 Release 18)".

[i.4] ETSI TR 103 791: "Rail Telecommunications (RT); Future Railway Mobile Communication System (FRMCS); Terminology for FRMCS specifications".

[i.5] ETSI TS 129 513 (V19.5.0): "5G; 5G System; Policy and Charging Control signalling flows and QoS parameter mapping; Stage 3 (3GPP TS 29.513 version 19.5.0 Release 19)".

[i.6] Recommendation ITU-T G.711 (06/1990): "Pulse code modulation (PCM) of voice frequencies".

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the present document, the terms given in ETSI TS 122 280 [2] (3GPP TS 22.280), ETSI TS 123 280 [4] (3GPP TS 23.280) and the following apply:

ad hoc group communication: combining of a multiplicity of MC service users into a group for the duration of a communication

NOTE: When the communication is released, the group no longer exists. If the communication is associated with an alert, then the group continues to exist until the alert is also cancelled.

ad hoc group emergency alert: emergency alert addressed to multiple MC service users which are part of a group

NOTE: When the alert is cancelled, the group no longer exists. If the alert is associated with a communication, then the group continues to exist until the communication is also cancelled.

functional Alias: user selectable alias that is tied to the assignment or task of the user

NOTE: A MCX User can activate one or multiple functional aliases at the same time. The activation of the functional alias(es) will take place after the user has signed in to the MCX Service using the MCX User ID.

group communication: one-to-many or many-to-many communication using an MCX Service

NOTE: The combination of different MC service users into a static group. MC service users have to affiliate for using the group service and need to de-affiliate in case they want to leave. In case of a communication, all affiliated MC service users will be involved.

MCX service emergency alert: notification from the MCX UE to the MCX Service that the MCX User has an emergency condition

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in ETSI TR 121 905 [i.3] (3GPP TR 21.905) and the following apply:

NOTE: An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in ETSI TR 121 905 [i.3] (3GPP TR 21.905).

5QI	5G Quality Indicators
ACK	Acknowledge
AES	Advanced Encryption Standard
AH	Authentication Header
AHG	Ad Hoc Group communication
AHGC	Ad Hoc Group Communication
AHGEA	Ad Hoc Group Emergency Alert
AKA	Authentication and Key Agreement
API	Application Programming Interface
APN	Access Point Node
ARP	Address Resolution Protocol
CBC	Cipher Block Chaining Mode

CNT	Counter
CO	Call Origination
CT	Call Termination
CTRL	Control
DL	Downlink
DRB	Data Radio Bearer
DSCP	Differentiated Services Code Point
ECGI	E-UTRAN Cell Global Identification
ESP	Encrypted Encapsulation Protocol
E-UTRA	Evolved Universal Terrestrial Radio Access
FA	Functional Alias
FC	Function Code
FD	File Distribution
FFS	For Further Study
FRMCS	Future Railway Mobile Communication System
FSMPM	Finite State Multipath Manager
FSMPU	Finite State Multipath User Plane
GCF	Global Certification Forum
gNodeB	Next Generation Node Base Station
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GRE	Generic Routing Encapsulation
GUI	Graphical User Interface
HTML	Hyper Text Markup Language
HTTP	Hyper Text Transfer Protocol
HTTPS	HTTP Secure
ICMP	Internet Control Message Protocol
ICS	Implementation Conformance Statement
ID	Identity
IdMS	Identity Management System
IDSN	Identity Domain and Services Network
IKE	Internet Key Exchange
IMS	IP Multimedia Subsystem
IMSI	International Mobile Subscriber Number
InK	Integrity Key
IP	Internet Protocol
IPConn	IP Connectivity
IPEG	In-Progress Emergency Group
IPEPC	In-Progress Emergency Private Call
IPIG	In-Progress Imminent peril Group
IPSEC	Internet Protocol Security
ISAKMP	Internet Security Association and Key Management Protocol
IUT	Implementation Under Test
IXIT	Implementation eXtra Information for Testing
KE	Key Exchange
KMS	Key Management System
MAC	Media Access Control
MBMS	Multimedia Broadcast and Multicast Service
MBSFN	Multimedia Broadcast multicast service Single Frequency Network
MC	Mission Critical
MCData	Mission Critical Data
MCPT	Mission Critical Push-to-Talk
MCPTT group ID	MCPTT group Identity
MCPTT	Mission Critical Push To Talk
MCS	Mission Critical Service
MCX	Mission Critical X, with X = PTT or X= Video or X= Data
MEA	MCPTT Emergency Alert
MEG	MCPTT Emergency Group
MEGC	MCPTT Emergency Group Call
MEPC	MCPTT Emergency Private Call
MEPP	MCPTT Emergency Private Priority
MES	MCPTT Emergency State

MIG	MCPTT Imminent peril Group
MIGC	MCPTT Imminent peril Group Call
MIME	Multipurpose Internet Mail Extensions
MODP	MODular exPonetial group
MONP	MCPTT Off-Network Protocol
MP	Multipath
MPEA	MCPTT Private Emergency Alert
MPF	Multipath Control Function
MPTCP	Multipath TCP
MRSP	Message Session Relay Protocol
MSRP	Message Session Relay Protocol
NAS	Non-Access Stratum
NAT	Network Address Translation
NOK	Not OK
NR	New Radio
OB	On-board
PAI	P-Asserted-Identity
PCC	Policy and Charging Control
PCF	Policy Control Function
PCMU	Pulse Code Modulation µ-Law
P-CSCF	Proxy-Call Session Control Function
PDR	Packet Detection Rule
PDU	Protocol Data Unit
PIDF	Presence Information Data Format
PIXIT	Protocol Implementation eXtra Information for Testing
PLMN	Public Land Mobile Network
POC	Push-to-Talk over Cellular
PSK	Phase Shift Keying
PTT	Push-to-Talk
QCI	QoS Class Identifier
QFI	QoS Flow Identifier
QoS	Quality of Service
QUIC	Quick UDP Internet Connection
RRC	Radio Resource Configuration
RTCP	Real-time Transport Control Protocol
RTP	Real-time Transport Protocol
SA	Security Association
SAI	Service Area Identifier
SCP	Secure Copy Protocol
SDI	Session Description Information
SDP	Session Description Protocol
SDS	Short Data Service
SFTP	Secure File Transfer Protocol
SHA2	Secure Hash Algorithm - 2 (Standard)
SIM	Subscriber Identity Node
SIP	Session Initiation Protocol
SLAAC	Stateless Address Autoconfiguration
SMF	Session Management Function
SPD	Security Policy Database
SS	System Configuration
SSRC	Synchronization SouRCe
TAP	Transferred Account Procedure
TC	Test Case
TCP	Transport Control Protocol
TGI	Temporary MCPTT Group Identity
TLS	Transport Layer Security
TMGI	Temporary Mobile Group Identity
TP	Test Purpose
TP	Transmission Point
TrK	Transport Key
UDM	Unified Data Management
UDP	User Datagram Protocol

UE	User Entity
UE-DL	User Entity - Downlink
UL	Uplink
UPF	User Plane Function
URI	Uniform Resource Identifier
URL	Uniform Resource Link
VAS	Voice Application System
XAUTH	Extended Authentication
XML	Extensible Markup Language

4 General

4.1 Test methodology

4.1.1 Testing of optional functions and procedures

Any function or procedure, which is optional, may be subject to a conformance test if it is implemented in the MCX Server for MCPTT and MCData.

A declaration by the Client supplier (to use the Implementation Conformance Statement (ICS) proforma specified in ETSI TS 104 070 [i.1] is used to determine whether an optional function/procedure has been implemented.

4.1.2 Test interfaces and facilities

Descriptions of the MCPTT Server to MCPTT Server interface is specified in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1).

The general components required for this conformance test are given in Figure 4.1-1. The MC service client carried by a MC service UE which is capable to connect via 5G NR network to a MCx Service system.

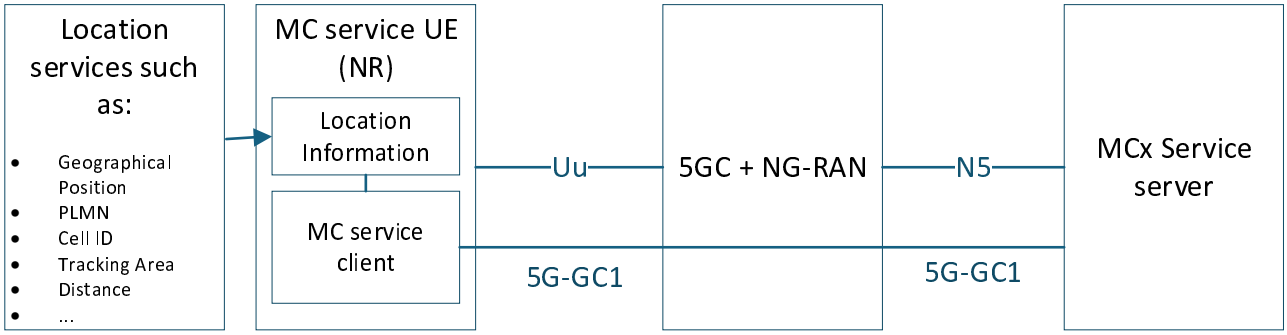
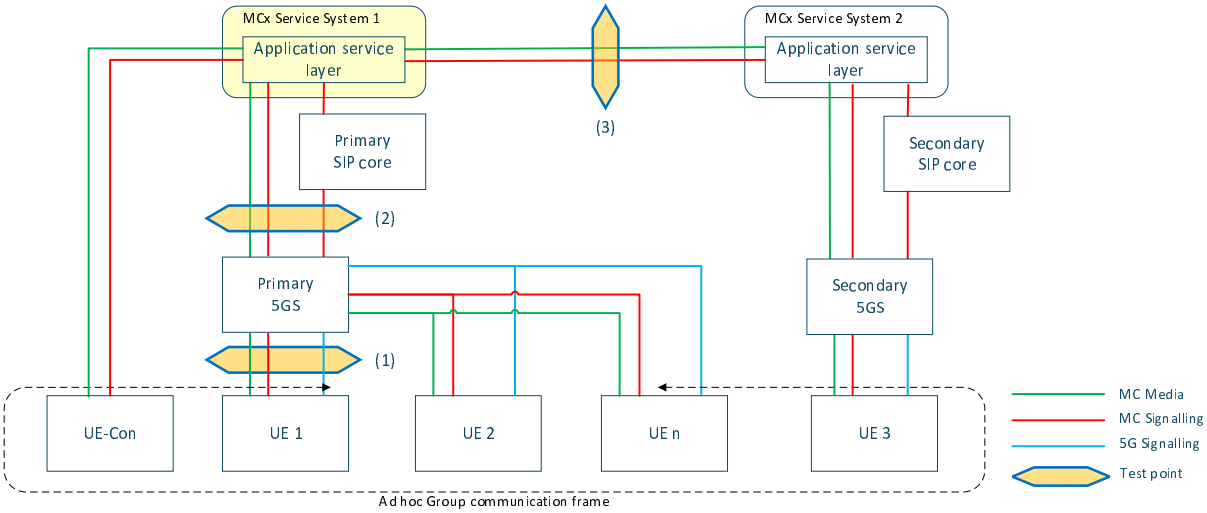


Figure 4.1-1: General configuration and interfaces

4.1.3 MC Conformance testing test points overview

Figure 4.1-2 provides a general overview of all MCx players which may have a role in different conformance testing scenarios together with virtual test points representing the information flow which is intended for conformance testing. The figure is mainly for descriptive purposes and may not necessarily represent a real MCx deployment or implementation.



NOTE 1: For the Ad hoc Group communication multiple UE will be required. They should be connected to the 5G system and provide similar interfaces for MC media plane, MC signalling and 5G signalling. The entity in light yellow will be the Implementation Under Test (IUT). Entities with white fill will be either simulated or IUT or real implementations depending on the test scenario. UE3 is connected to the partner MCx Service system and can be simulated or present as physical device.

NOTE 2: <UE n> identifies a various number of supporting devices. These can be simulated or present as physical system. To prevent any misinterpretations in the test steps because different additional numbers and different configurations could occur, "n" is replaced by a1, b2, c3 and so on, while the character indicates an individual configuration and the digit gives the ongoing number of the device. Same character indicates equivalent parameter configurations (but e.g. different SIP/IMS user accounts for sure).

NOTE 3: The Ad hoc Group communication frame can vary during the tests depending on the number of participants given by the test scope (e.g. pre-configured or list based ad hoc group communication).

NOTE 4: UE-Con is the controller/dispatcher. It is connected directly to the server application function but integrated in the configuration.

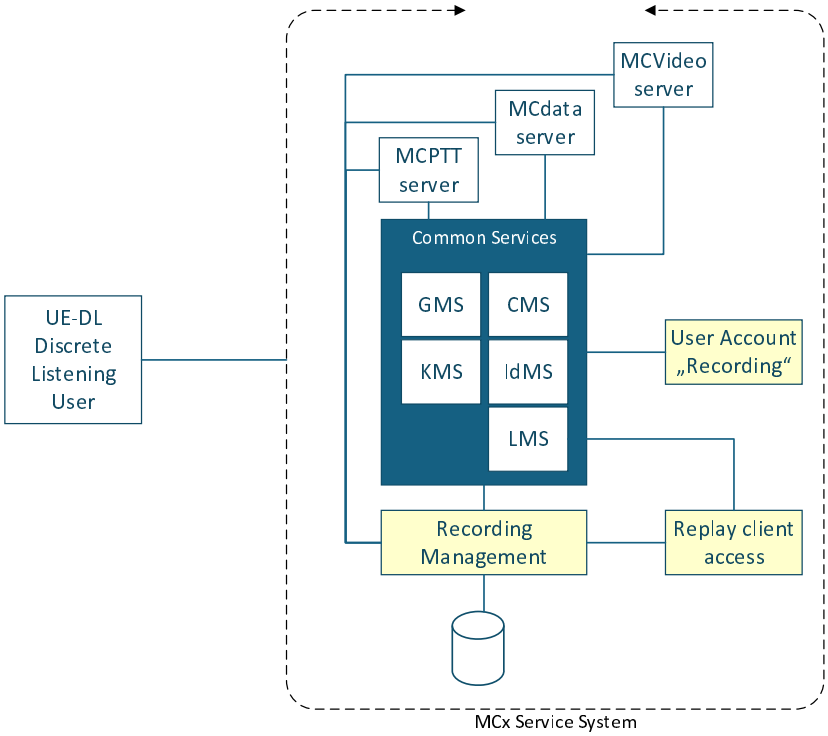
NOTE 5: Number of MC service user UE can vary depending on the test case. For this, entity UE n symbolizes several similar UE with similar interfaces.

NOTE 6: MCx Service system 2 can be homed in a different domain or on a different home server.

Figure 4.1-2: MC Conformance testing test points model

4.1.4 MC Conformance testing - configuration for recording management and discrete listening

To ensure the functionality for recording and discrete listening the conformance test environment shall be configured according to Figure 4.1-3.

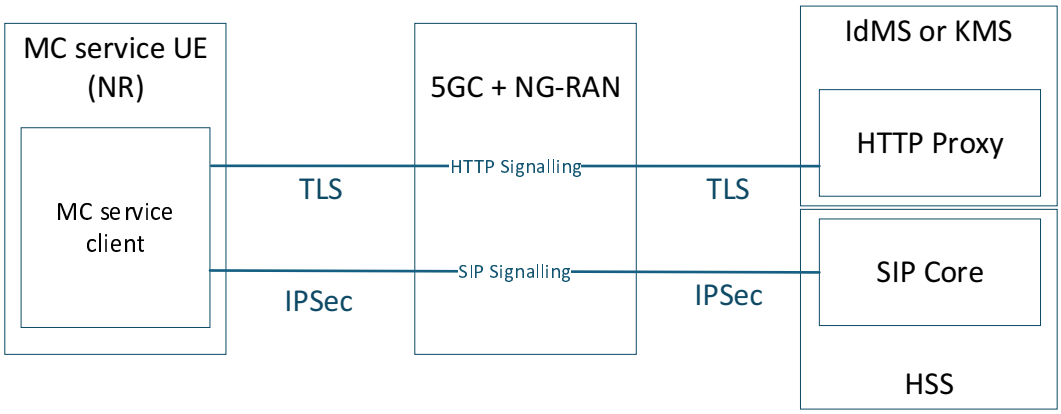


NOTE 1: User account "recording" should be an independent user access which allows only the configuration of the recording function and the allocation of a discrete listening UE. No other functionality should be possible.
NOTE 2: The storage period for the recorded interception data should not violate national data retention rules.
NOTE 3: User account "Replay Client" is only for analysing recorded session data including useful data.
NOTE 4: UE-DL is the discrete listening UE. It can be a physical UE connected to the 5G system according to Figure 4.1-2 or it can be simulated or connected directly to the server application function but integrated in the configuration.

Figure 4.1-3: Configuration for recording and discrete listening functionality

4.1.5 MC Conformance testing - configuration for ad hoc group communication security

To ensure the functionality for recording and discrete listening the conformance test environment shall be configured according to Figure 4.1-4.



NOTE 1: For HTTP usage the HTTP-1 interface is required. The authentication should be performed between HTTP client in the MC UE and the HTTP server (IdMS or KMS).
NOTE 2: The PSK TLS based authorization methods are given with ETSI TS 133 310 [23] (3GPP TS 33.310).
NOTE 3: The mutual authentication procedure between UE and SIP core should support IMS AKA and mechanisms, described in ETSI TS 133 310 [23] (3GPP TS 33.310), clause 6.2.

Figure 4.1-4: Communication Security

4.1.6 MC Conformance testing - configuration for IP Connectivity (IPConn)

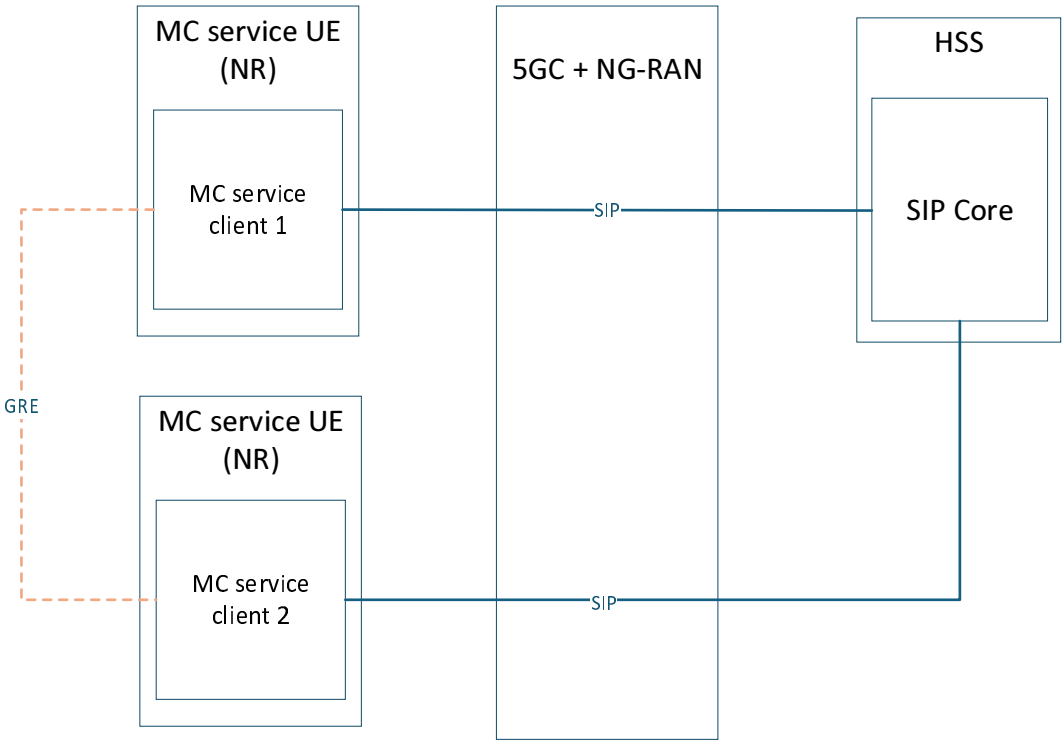


Figure 4.1-5: IPConn communication connections

4.2 Implicit testing

For some 3GPP MCPTT protocol features conformance is not verified explicitly in the present document. This does not imply that correct functioning of these features is not essential, but that these are implicitly tested to a sufficient degree in tests, which are not explicitly dedicated to test the feature.

4.3 Repetition of tests

As a rule, the test cases specified in the present document are highly reproducible and do not need to be repeated unless otherwise stated.

4.4 Handling of differences between conformance requirements in different releases of cores specifications

The conformance requirements, which determine the scope of each test case, are explicitly copy-pasted from relevant core specifications in the dedicated to each test with the title 'Conformance requirements'.

NOTE: When in the copy/pasted text there are references to other specifications the reference numbers will not match the reference numbers used in the present specification. This approach has been taken to allow easy copy and then search for conformance requirements in those specifications.

When differences between conformance requirements in different releases of the core specifications have impact on the Pre-test conditions, Test procedure sequence or/and the Specific message contents, the Conformance requirements related to different releases are specified separately with clear indication of the Release of the specification from which they were copied.

When there is no Release indicated for a conformance requirement text, this should be understood either as the Conformance requirements in the latest version of the specification with release = the TC Applicability release. This can be found in ETSI TS 136 579-4 [i.2] (3GPP TS 36.579-4), Table 4-2: Applicability of tests and additional information for testing, column 'Release'), or, as the Conformance requirements in the latest version of the specification of the release when the feature was introduced to the core specifications.

4.5 Reference conditions

The reference environments used by all signalling and protocol tests are specified in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1). Where a test requires an environment that is different, this will be specified in the test itself.

4.6 Generic setup procedures

A set of basic generic procedures for MCPTT Client-Server and MCPTT Client-Client communication are given in clause 5.

5 Generic functionality

5.1 Functional aliasing server side

5.1.1 Test Purpose (TP)

(1)

```
with {Configuration of functional aliases}
ensure that {
    when {MCX Service System provides a configuration interface and login role allows the check of
the FA configuration settings and access to the content management}
    then {The configuration for FA contains the options for creation and deletion and management
activated and passive FA and an indication for simultaneous usage by multiple MCX service users and
number of simultaneous users per FA can be configured and an indication is provided, how many of the
configured FA are in use and an indication is implemented, if a FA can be taken over or not}
            }
```

(2)

```
with {FA activation and collision handling}
ensure that {
    when {MCX Service System provides a configuration interface and login role allows the check of
the FA configuration settings and access for the activity and collision management}
    then {MCX Service administrator can activate and deactivate FA and permit or prevent
interrogation and permit or prevent overtaking of a FA by another user}
            }
```

5.1.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-005]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to create, delete, manage functional aliases, and for each functional alias indicate either if it can be simultaneously active for multiple MCX Users up to a per-alias configurable number, or if it is allowed to be taken over by an authorised MCX User, or none of these two options."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-012]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to authorise a MCX User to activate, deactivate, interrogate and take over a functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-020]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure for a particular functional alias or a specific element or combination of specific elements in the functional alias, a set of functional aliases under the same authority to which a Private Communication (without Floor control) can be made by an MCX User registered to this particular functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-021]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure for a particular functional alias or a specific element or combination of specific elements in the functional alias, a set of functional aliases under the same authority from which Private Communication (without Floor control) is allowed to an MCX User registered to this particular functional alias."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-026]]

"The MCX Service shall provide the functional alias(es) of the transmitting MCX Service Group Member that enter the communication late in accordance with clause 5.3."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-028]]

"The MCX Service shall provide the functional alias(es) associated with the MCX Service User ID, of the transmitting Participant to the receiving MCX Users (UEs) unless the transmitting Participant's identity is restricted in accordance with clause 6.4.3."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.9a-030]]

"The MCX Service shall provide a mechanism for the enforcement of uniqueness of a functional alias within a Mission Critical organization in accordance with clause 6.9."

5.1.3 Test description

5.1.3.1 Pre-test conditions

System Configuration:

MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (MCx Service system) also consists of all sub-systems of the MCx user service:

- MCx service server is configured to support the following features:
  - HTTPS access to the configuration management;
  - user and role management;
  - file distribution system (e.g. SCP or SFTP).

IUT:

- MCx_S1 (MCx Service system 1) is the acting user.
- UE1 is connected to PLMN1.
- The IUT (MCx Service system 1) is connected to UE1 (MCx client) and UE2 (MCx client).
- MCx service registration and affiliation are executed.

Preamble:

- The MCX Client has been provided either with the address information of the server from which the client can retrieve the MCX UE initial configuration document ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) (steps 1a1-1a2 of procedure 'MCX Initial Configuration and User Authentication', Table 5.3.2.3.1-1) or directly with the Initial UE Configuration Data as specified in Table 5.5.8.1-1.

- The UE User is provided with username/password for user authentication (px_MCX_User_A_username, px_MCX_User_A_password as provided in ETSI TS 137 579-5 [14] (3GPP TS 37.579-5) Table 9.2-1: MCX Client Common PIXIT).
- The UE is provided with the names and values of the Transport Key (TrK) and the Integrity Key (InK), since the KMS shall encrypt the key material sent to the client with the TrK and sign the response with the TrK or the InK according to ETSI TS 133 180 [15] (3GPP TS 33.180).
- The IUT (UE1) is connected to PLMN of the primary 5GS. UE3 is connected to the PLMN of the secondary 5GS.

5.1.3.2 Test procedure sequence

Table 5.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3				
1	MCx Service system provides a configuration interface for feature activation in respect to the FA	--		1	-
2a	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. With correct credentials the access to the FA configuration and content management will be granted	-->	HTTPS link to configuration interface	1	P
2b	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. With correct credentials the access to the FA configuration and content management will be granted	-->	XML access (e.g. via SCP or SFTP connection)	1	P
3	Configuration settings for FA. Following options are mandatory	--		1	-
3a	Check: An option for creation of a FA is available	-->	STRING for creation of FA	1	P
3b	Check: An option for deleting of a FA is available	-->	STRING deleted	1	P
3c	Check: An option for activation of a FA is available	-->	BOOL (activation true)	1	P
3d	Check: An option for deactivation of a FA is available	-->	BOOL (activation false)	1	P
3e	Check: An FA can be indicated for simultaneous use by multiple MCx service users. Bind to this option the maximum number of simultaneous users can be configured (note 1)	-->	INTEGER for maximum number of MCx Service users	1	P
3f	Check: For each configured FA an indicator is given for the current number of users (note 1)	-->	INTEGER for count of FA allocations	1	P
3g	Check: For each configured FA a variable is available for taking over the FA by another MCx Service user	-->	BOOL (true/false) for FA take over	1	P
3h	Check: For each configured FA a variable is available for activation of interrogation is available (note 2)	-->	BOOL (true/false) for FA interrogation	2	P
4	Collision management of the MCx Service system	--		2	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
5	UE1 publish the basic POC settings	<--	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]		--
6	MCx Service system confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)		P
6a	Check: UE1 register FA. **FA not in use.** FA feature list:{**not configured**} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH according ETSI TS 124 379 [20] (3GPP TS 24.379) clause 9A.2.1.2, IETF RFC 3856 [9] and IETF RFC 6665 [10]	2	--
7a	UE1 receive STATUS message	-->	STATUS 403 FORBIDDEN		P
6b	Check: UE1 register FA. FA not in use. FA feature list:{**configured** | **activation=false**} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH	-	-
7b	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 401 UNAUTHORIZED	2	P
8b	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9b	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 403 FORBIDDEN	2	P
6c	Check: UE1 register FA (note 3). FA not in use. FA feature list:{configured | **activation=true** | MAX_{user}=1 | CNT_{user}=0 | takeover=false | interrogation=false} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH	-	-
7c	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 401 UNAUTHORIZED	2	P
8c	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9c	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 200 OK	2	P
10c	MCx Service system responds NOTIFY message after successful activation of the FA	-->	NOTIFY	2	P
6d	Check: UE1 register FA (note 3). **FA in use.** FA feature list:{configured | **activation=true** | MAX_{user}=1 | CNT_{user}=0 | takeover=false | interrogation=false} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH	-	-
7d	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 401 UNAUTHORIZED	2	P
8d	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9d	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 403 FORBIDDEN	2	P
6e	Check: UE1 register FA (note 3). **FA not in use.** FA feature list:{configured | activation=true | MAX_{user}=1 | CNT_{user}=0 | takeover=false | **interrogation=true**} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH change to CTN_{user}=1	-	-
7e	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 200 OK	2	P
8e	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
9e	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 200 OK	2	P
10e	MCx Service system responds NOTIFY message after successful activation of the FA	-->	NOTIFY	2	P
6f	Check: UE1 register FA (note 3). **FA in use.** FA feature list:{configured | activation=true | MAX_{user}=1 | **CNT_{user}=1** | takeover=false | **interrogation=true**} IUT responds with error code immediately to the SUBSCRIBE request	-->	SIP PUBLISH	-	-
7f	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 200 OK	2	P
8f	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9f	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 403 FORBIDDEN	2	P
10f	MCx Service system responds NOTIFY message after successful activation of the FA	-->	NOTIFY	2	P
6g	Check: UE1 register FA (note 3). **FA in use.** FA feature list:{configured | activation=true | MAX_{user}=1 | **CNT_{user}=1** | **takeover=true** | **interrogation=true**} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH change to CTN_{user}=1	-	-
7g	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 200 OK	2	P
8g	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9g	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 200 OK	2	P
10g	MCx Service system responds NOTIFY message after successful activation of the FA	-->	NOTIFY	2	P
6h	Check: UE1 register FA (note 3). **FA in use.** FA feature list:{configured | activation=true | **MAX_{user}=5** | **CNT_{user}=1** | takeover=false | **interrogation=true**} IUT responds with error code immediately to the SUBSCRIBE request	<--	SIP PUBLISH change to CTN_{user}=2	-	-
7h	MCx Service system responds STATUS message responding the current usage of the FA	-->	STATUS 200 OK	2	P
8h	UE1 send PUBLISH for registering a FA	<--	SIP PUBLISH	-	-
9h	MCx Service system responds STATUS message responding the request for registering a FA	-->	STATUS 200 OK	2	P
10h	MCx Service system responds NOTIFY message after successful activation of the FA	-->	NOTIFY	2	P

NOTE 1: In case, only one user is allowed, the variable shall set/indicate to 1.
NOTE 2: In case interrogation is enabled, the UE will be added to a list of possible interested parties for a FA. MCx Service system add the requester to a list which is continuously updated by the MCx Service system. In case the interrogation is deactivated, the UE will receipt a STATUS 403 FORBIDDEN according to test case 5.1.
NOTE 3: The flag {activation=true} activates also variables {maximum number of users per FA (MAX_{user}) | current FA user count (CNT_{user})| FA takeover | interrogation}.

5.1.3.3 Specific message contents

Table 5.1.3.3-1: SIP PUBLISH from the UE
(Table 5.1.3.2-1; step 5, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.11-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), clause 5.5.2.11; condition POC-SETTINGS-EVENT, MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"PUBLISH"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16] IETF RFC 3909 [17]	
MIME Multipart Media Encapsulation				
application/poc-settings+xml	<answer-mode>	Call answer mode		
	<user-profile-index>	User profile		
application/mcptt-Params+xml	<mcptt-access-token>	Authentication token		
	<mcptt-client-id>	MCPTT client ID		

Table 5.1.3.3-2: SIP PUBLISH from the UE
(Table 5.1.3.2-1; step 6a, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.11-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), clause 5.5.2.11; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"SUBSCRIBE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/mcpttinfo+xml	<mcpttURI>	MCPTT user identity		MCPTT
<entry>				
<anyExt>	<manual-deactivation-not-allowed-if-location-criteria-met>	FA is location depending		True/False
	<FunctionalAliasList>	Functional alias subscription list (PIDF (FA-ID))		FUNCTIONAL ALIASING

Table 5.1.3.3-3: SIP STATUS (step 6, Table 5.1.3.2-1;
step 9c, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.17-1)

Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	200		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(OK)			

Table 5.1.3.3-4: SIP STATUS from the MCx Service system
(step 7b, Table 5.1.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.18-1)

Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	401		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Unauthorized)			

Table 5.1.3.3-5: SIP STATUS from the MCx Service system
(step 9b, Table 5.1.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.18-1)

Derivation Path: IETF RFC 3261 [16]; condition STATUS-CODE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Status-Line				
Status-Code	403		ETSI TS 124 229 [8] (3GPP TS 24.229)	
Reason-Phrase	(Forbidden)			

Table 5.1.3.3-6: SIP NOTIFY
(step 10c, Table 5.1.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.8-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.8; condition PRESENCE-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"NOTIFY"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language	<mcpttURI>	MCPTT user identity		
application/event-type+xml	<presence>	PIDF	PIDF according to [7] Table 5.5.3.5.1-1	PRESENCE-EVENT
	<xcap-diff>	xcap-div	xcap-diff as described in [7] Table 5.5.3.12-1	CONFIG. GROUP-CONFIG
	<poc-settings>		POC-Settings as described in [7] Table 5.5.3.11.2-1	POC-SETTINGS-EVENT

5.2 Floor Control in general

5.2.1 Test Purpose (TP)

(1)

```
with {Floor control request/granted/release}
ensure that {
    when {MCX service user can handle the floor control and MCX service user is in an active
communication with floor control and ongoing incoming or outgoing communication with floor control}
    then {MCX service user requests the floor control and the MCX service system response result
"granted" and "release" after successful usage}
            }
```

(2)

with {MCX service user can handle the floor control}
ensure that {
    when {MCX service user A and MCX service user B are in an active communication with floor control and MCX service user requests the floor control}
    then {MCX service system reject the request with status "busy, overriding not allowed" (floor used by another MCX service user) **or** override the floor control (without revoke of floor control)}
            }

(3)

with {Floor control revoke floor control}
ensure that {
    when {MCX Service system can revoke the floor control}
    then {Floor control is released after floor control timer expiry **or** revoke by MCX Service system **and** information for idle floor to all the participants and in multi-talker case idle floor will be noticed by all participants}
            }

(4)

with {MCX system can queue the floor control requests by multiple users}
ensure that {
    when {MCX service user A and MCX service user B are in an active communication with floor control, and other MCX service user have requested the floor and multi-taker floor control is not activated by the MCX service system}
    then {MCX service system detects the several floor requests and queue them and MCX service user request the own position in the floor control queue}
    else when {MCX service user A and MCX service user B are in an active communication with floor control, and other MCX service user have requested the floor and multi-taker floor control is not activated by the MCX service system and floor control queue contains entries and talker releases the floor control}
    then {MCS service system provides notification to all MCX service users and next MCX service user **or** MCX service user with the highest priority in the floor control queue becomes active talker}
            }

(5)

with {Floor priorities with migrating MCX user (multiple MC systems)}
ensure that {
    when {Multiple MCX service users are configured with different floor control priorities and MCX service user with active communication and floor control queuing during the migration}
    then {Active floor control migrates from MCX service system A to another MCX service system B and the priority of the floor control and the position in the floor control queue shall not be changed during migration}
            }

(6)

with {Active ad hoc group communication involving multiple participants with the authorization to use multi-talker control}
ensure that {
    when {Number of MCPTT Users requesting the permission to talk exceeds}
    then {The MCPTT Service system provides a mechanism to override and MCPTT User will receive a notification e.g. override}
            }

(7)

with {Active ad hoc group communication involving multiple participants with the authorization to use multi-talker control and maximum number of simultaneous talkers can be changed by an authorized MCPTT User}
ensure that {
    when {Number of MCPTT Users requesting the permission to talk exceeds and an authorized MCPTT User change the maximum number of simultaneous talkers during active ad hoc group communication}
    then {Additional MCPTT Users authorized for multi-talker service can talk and further MCPTT Users can request the permission to talk until the new maximum number of simultaneous talkers exceed}
            }

(8)

```
with {An MCPTT system configured for a pre-arranged group where a designated controller has the
privilege for implicit floor establishment upon call initiation}
ensure that
    when {The authorized controller initiates a pre-arranged group call with implicit floor control}
    then {The group call is successfully established, and the floor is automatically and immediately
granted to the initiating controller without the need for an explicit and implicit floor request
(PTT) procedure and controller's audio is successfully streamed to all called participants}
```

5.2.2 Conformance requirements

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-001]]

"If the number of MCPTT Users requesting the permission to talk exceeds the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control, the MCPTT Service shall apply the override mechanism."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-006]]

"The MCPTT Service shall allow an authorised MCPTT User to change the maximum number of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

5.2.3 Test description

5.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508), clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx service server is configured to support the following features:
  - implicit and explicit floor control;
  - multiple user floor control;
  - multi-talker floor control;
  - prioritization of floor control requests.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control.
- The IUT (MCx_S1) is the acting user.

Preamble:

- The UE1 is connected to PLMN1.
- The MCx Service server IUT (MCx_S1) is connected to the UE1 (MCx client), UE2 (MCx client) and UE3 (MCx client).
- MCx service registration and affiliation are executed on MCx_S1.

- MCx Service system (MCx_S2) is connected to MCx_S1.
- MCs Service user – controller function (UE-Con).

5.2.3.2 Test procedure sequence

Table 5.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), clause 5.1.3	--		-	-
1a	UE originating case. UE1 send INVITE	<--	SIP INVITE according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	-
2a	MCx_S1 responds to the INVITE request (note 1)	-->	100 (TRYING)	-	P
3a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
4a	Call established confirmation from MCx_S1	-->	200 (OK)	-	P
1b	UE terminating case. UE receive INVITE from MCx_S1	-->	SIP INVITE according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	P
2b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	P
3b	Call established confirmation towards MCx_S1	<--	200 (OK)	-	-
5	Check: Encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request - Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	--	Media format specific parameters: fmtp: mc_implicit_request	-	-
6	UE1 receives RTCP (MCPT) Floor message by a remote participant. MCx client on UE1 indicate the occupation status (note 2)	<--	RTCP (MCPT) according to IETF RFC 1889 [11]	1	-
7	UE1 REQUEST the floor; following information included: Subtype **0 Floor Request** {Floor priority | user ID | track info | floor indicator | FA | location}	<--	RTCP (MCPT) Floor REQUEST according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.4	-	-
8a	Check: MCx_S1 indicate GRANTED the floor for UE1; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.5	1	P
8b	Check: MCx_S1 indicate TAKEN the floor for UE1; the following information included: Subtype **2 Floor Taken** {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	-->	RTCP (MCPT) Floor TAKEN According to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.9	1	P
8c	Check: MCx_S1 indicate DENY the floor for UE1; the following information included: Subtype **3 Floor Deny** {Reject Cause (note 3) | User granted the floor | floor indicator}	-->	RTCP (MCPT) Floor DENY Reject causes according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.2	2	P
8e	Check: MCx_S1 indicate RELEASE the floor for UE1; the following information included: Subtype **4 Floor Release** {user ID | track info | floor indicator}	-->	RTCP (MCPT) Floor RELEASE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.7	2	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
8f	Check: MCx_S1 indicate IDLE the floor for UE1; the following information included: Subtype **5 Floor Idle** {SSRC of floor control server | message sequence | track info | floor indicator}	-->	RTCP (MCPT) Floor IDLE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.8	1	P
8g	Check: MCx_S1 indicate REVOKE the floor for UE1; the following information included: Subtype **6 Floor Revoke** {SSRC of floor control server | reject cause (note 3) | track info | floor indicator}	-->	RTCP (MCPT) Floor REVOKE according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.10	3	P
8h	Check: MCx_S1 indicate ACK the floor for UE1; the following information included: Subtype **10 Floor Ack** {SSRC of sender | source field | message type | track info | location indicator}	-->	RTCP (MCPT) Floor ACK according to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.13	1	P
9	Check: UE1 terminates the call by sending BYE	<--	SIP BYE	-	-
10	MCx Service system responds with STATUS message to the call termination request	-->	SIP STATUS (note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
11a	UE originating case. UE1 send INVITE	<--	SIP INVITE	-	-
12a	MCx_S1 responds to the INVITE request (note 1)	-->	100 (TRYING)	-	P
13a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
14a	Call established confirmation from MCx_S1	-->	200 (OK)	-	P
11b	UE terminating case. UE receive INVITE from MCx_S1	-->	SIP INVITE	-	P
12b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
13b	Call established confirmation towards MCx_S1	<--	200 (OK)	-	-
15	Minimum two user, UE2 & UE3 are involved into the active communication and activate FLOOR control; Multiple access granted Check: UE1 opens encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request – Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	--	Media format specific parameters: fmtp: mc_implicit_request	-	-
16a	UE1 request the floor	<--	RTCP (MCPT) Floor REQUEST	-	-
16b	All UE2 and UE3 are granted for multi-talker FLOOR control. Check: MCx_S1 indicate GRANTED the floor for UE2 and UE3; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED	4	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
17a	Check: UE1 receives RTP data (voice) from both, UE2 & UE3; RTP stream contains payload type information	-->	RTP stream	5	P
17b	Check: RTP data (voice) can be received from both, UE2 & UE3; RTP stream contains payload type information	<--	RTP stream	5	P
18a	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. Queued by the MCx_S1 Check marked elements: MCx_S1 sends TAKEN message to UE1; the following information included: Subtype **2 Floor Taken** {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | **list of granted users | list of audio SSRC | list of FA** | location info | list of locations of granted users}	-->	RTCP (MCPT) Floor TAKEN	4	P
18b	Check: UE1 transmit no RTP data	--	No outgoing RTP stream	-	-
19	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. All UE have released FLOOR control Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
20	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. All UE have released FLOOR control. Check: MCx_S1 sends GRANTED message to UE1: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED	4	P
21	Check: RTP data (voice) can be received from UE1; RTP stream contains payload type information	--	RTP stream	-	-
22	Check: UE1 terminates the call by sending BYE	<--	SIP BYE	-	-
23	MCx Service system responds with STATUS message to the call termination request	-->	SIP STATUS (note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
24a	UE originating case. UE1 send INVITE	<--	SIP INVITE	-	-
25a	MCx_S1 responds to the INVITE request (note 1)	-->	100 (TRYING)	-	P
26a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
27a	Call established confirmation from MCx_S1	-->	200 (OK)	-	P
24b	UE terminating case. UE receive INVITE	-->	SIP INVITE	-	P
25b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
26b	Call established confirmation towards MCx_S1	<--	200 (OK)	-	-
	UE2 & UE3 are granted for multi-talker FLOOR control Check: UE1 additionally request the floor	<--	RTCP (MCPT) Floor REQUEST		
28	Check: MCx_S1 indicate GRANTED the floor for UE2 and UE3; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED	6	P
29	Check: UE1 send RTP data (voice); RTP stream contains payload type information	--	RTP stream	-	-
30	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. Queued by the MCx_S1	<--	RTCP (MCPT) Floor REQUEST	-	-
31a	UE1 with no permission for multi-talker FLOOR control. FLOOR control activated. Queued by the MCx_S1 Check marked elements: MCx_S1 sends TAKEN message to UE1; the following information included: Subtype **2 Floor Taken** {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | **list of granted users |** list of audio SSRC | list of FA | location info | list of locations of granted users}	-->	RTCP (MCPT) Floor TAKEN	6	P
31b	UE1 migrate from MCx_S1 to MCx_S2 while FLOOR control is activated Check: MCx_S2 sends TAKEN message to UE1; the following information included: Subtype **2 Floor Taken** {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | **list of granted users |** list of audio SSRC | list of FA | location info | list of locations of granted users}	-->	RTCP (MCPT) Floor TAKEN	6	P
31c	Compare elements in **list of granted users**. Position of UE1 shall not be changed	--	No differences	-	-
32a	UE2 & UE3 are granted for multi-talker FLOOR control Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
32b	UE2 & UE3 are granted for multi-talker FLOOR control Check: MCx_S1 indicate GRANTED the floor for UE2 and UE3; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | floor priority | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED	6	P
33a	UE1 with permission for multi-talker FLOOR control. FLOOR control activated. Handled by the MCx_S1 Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
33b	UE1 with permission for multi-talker FLOOR control. FLOOR control activated. Handled by the MCx_S1 Check marked elements: MCx_S1 indicate GRANTED the floor for UE1; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | **floor priority** | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	-->	RTCP (MCPT) Floor GRANTED	6	P
33c	UE1 migrate from MCx_S1 to MCx_S2 while FLOOR control is activated Check marked elements: MCx_S2 indicate GRANTED the floor for UE1; following information included: Subtype **1 Floor Granted** {duration | audio SSRC of granted participant | **floor priority** | user ID | queue size | SSRC of queued user | queued user ID | queue ID | floor indicator | track info | floor indicator}	--	RTCP (MCPT) Floor GRANTED	6	P
33d	Compare elements in **floor priority**. Position of UE1 shall not be changed	--	No differences	-	-
34	Check: UE1 terminates the call by sending BYE	<--	SIP BYE	-	-
35	MCx Service system responds with STATUS message to the call termination request	-->	SIP STATUS (note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
36a	UE originating case. UE1 send INVITE	<--	SIP INVITE	-	-
37a	MCx_S1 responds to the INVITE request (note 1))	-->	100 (TRYING)	-	-
38a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
39a	Call established confirmation from MCx_S1	-->	200 (OK)	-	-
36b	UE terminating case. UE receive INVITE	-->	SIP INVITE	-	-
37b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
38b	Call established confirmation towards MCx_S1	<--	200 (OK)	-	-
40	UE2 & UE3 are granted for multi-talker FLOOR control. MCx_S1 allows max. two multi-talker at the same time and override is activated Check MCx_S1 indicate GRANTED the floor for UE2 and UE3; following information included: Subtype **1 Floor Granted**	--	RTCP (MCPT) Floor GRANTED	7	P
41	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set equal to UE2 & UE3. FLOOR control activated Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
42	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set equal to UE2 & UE3. FLOOR control activated Check: MCx_S1 indicate TAKEN the floor for UE1; the following information included: Subtype **2 Floor Taken**	-->	RTCP (MCPT) Floor TAKEN	7	P
43	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set higher than UE2 or UE3 (override). FLOOR control activated	<--	RTCP (MCPT) Floor REQUEST		
44	UE1 with permission for multi-talker send FLOOR request message. Floor priority is set higher than UE2 or UE3 (override). FLOOR control activated Check: MCx_S1 indicate GRANTED the floor for UE1; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED	7	P
45	UE2 or UE3 override Check: MCx_S1 indicate TAKEN the floor for UE2 or UE3; the following information included: Subtype **2 Floor Taken**	-->	RTCP (MCPT) Floor TAKEN	7	P
46	Check: UE1 terminates the call by sending BYE	<--	SIP BYE	-	-
47	MCx Service system responds with STATUS message to the call termination request	-->	SIP STATUS (note 1) 200 (OK)	-	-
	EXCEPTION: Procedure 'MCX CT call release' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.12	--		-	-
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
48a	UE originating case. UE1 send INVITE	<--	SIP INVITE	-	-
49a	MCx_S1 responds to the INVITE request (note 1))	-->	100 (TRYING)	-	-
50a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
51a	Call established confirmation from MCx_S1	-->	200 (OK)	-	-
48b	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
49b	UE terminating case. UE receive INVITE	-->	SIP INVITE	-	-
50b	Call established confirmation towards MCx_S1	<--	200 (OK)	-	-
52	UE1 with permission for multi-talker send FLOOR request message. FLOOR control activated. Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
53	Check: UE1 receive floor GRANTED message; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED		P
54	UE2 with permission for multi-talker send FLOOR request message while UE1 is still GRANTED. FLOOR control activated Check: UE2 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
55	UE2 with permission for multi-talker send FLOOR request message. FLOOR control activated Check: MCx_S1 indicate GRANTED the floor for UE2; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED	8	P
56	UE3 with permission for multi-talker send FLOOR request message while UE1 and UE2 still GRANTED. FLOOR control activated Check: Check: UE3 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
57	UE3 with permission for multi-talker send FLOOR request message. FLOOR control activated Check: MCx_S1 indicate GRANTED the floor for UE3; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED	8	P
58	Check: UE1 terminates the call by sending BYE	-->	SIP BYE		
59	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (note 1) 200 (OK)		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
60a	EXCEPTION: UE originating case. UE1 send INVITE for ad hoc group communication	<--	SIP INVITE	-	-
61a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (note 1)	-->	100 (TRYING)	-	-
62a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	-->	180 (RINGING)	-	-
63a	Call established confirmation from MCx_S1	-->	200 (OK)	-	-
60b	EXCEPTION: UE terminating case. UE1 receive INVITE for ad hoc group communication	-->	SIP INVITE	-	-
61b	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
62b	Call established confirmation from UE1	<--	200 (OK)	-	-
64	UE2 and UE3 have requested the floor, MCx_S1 has granted for both. UE1 with permission for multi-talker send FLOOR request message. Maximum numbers for simultaneous talkers exceed. FLOOR control activated Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST		
65	Check: UE1 receives TAKEN message by MCx_S1; the following information included: Subtype **2 Floor Taken**	-->	RTCP (MCPT) Floor TAKEN	8	P
66	MCx_S1 has permission to change the maximum number of simultaneous talkers. Number increased	--		8	P
67	Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST		

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
68	UE1 with permission for multi-talker send FLOOR request message. FLOOR control activated Check: UE1 GRANTED the floor; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED	8	P
69	Check: UE1 terminates the call by sending BYE	-->	SIP BYE		
70	MCx Service system responds with STATUS message to the call termination request	<--	SIP STATUS (note 1) 200 (OK)		
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2				
71a	Pre-arranged group communication with implicit floor control EXCEPTION: UE-Con send INVITE for pre-arranged group communication	<--	SIP INVITE	-	-
72a	Incoming response to the INVITE; MCx_S1 responds to the INVITE request (note 1)	-->	100 (TRYING)	-	P
73a	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
74a	Call established confirmation from MCx_S1	-->	200 (OK)	-	P
71b	UE1 receive INVITE for pre-arranged group communication	-->	SIP INVITE	-	P
72b	OPTIONAL: incoming response to TRYING if no automatic answering mode is activated	<--	180 (RINGING)	-	-
73b	Call established confirmation from UE1	<--	200 (OK)	-	-
75	FLOOR control activated. UE-Con receive immediately floor granted message without explicit request Check: UE-Con receive GRANTED; following information included: Subtype **1 Floor Granted**	-->	RTCP (MCPT) Floor GRANTED		P
76	Check: Encapsulated multipart session description protocol in the INVITE message. It contains following information: Media attribute (a): MCPTT implicit request – Receiver expects the information about (MCPTT) FLOOR control messages in the real time control protocol part	--	Media format specific parameters: fmtp: mc_implicit_request	-	-
77	Check: RTP data (voice) from UE-Con can be received at all involved UE RTP stream contains payload type information	<--	RTP stream	-	-
78	Check: UE1 terminates the call by sending BYE	<--	SIP BYE	-	-
79	MCx Service system responds with STATUS message to the call termination request	-->	SIP STATUS (note 1) 200 (OK)	-	-

NOTE 1: When the UE is confirming the successful resource reservation using an UPDATE request (or a PRACK request) and the UE receives a 180 (Ringing) response or a 200 (OK) response to the initial INVITE request before receiving a 200 (OK) response to the UPDATE request (or a 200 (OK) response to the PRACK request), the UE does not treat this as an error case and does not release the session.

NOTE 2: The application shall indicate the detected status of the floor parameter on the graphical user interface as icon and text:

a. Status is IDLE, REVOKE or RELEASED: green; text: e.g. "push PTT to talk!"
b. Status REQUESTED, DENY or TAKEN by other user, no multi-talker: red; text: none.
c. Status REQUESTED, DENY or TAKEN by other user, multi-talker: blue; text: e.g. "push PTT to talk!"
d. Status is GRANTED to own UE: yellow; text: e.g. "please talk!"

NOTE 3: Reject Causes are given in ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.2.

5.2.3.3 Specific message contents

Table 5.2.3.3-1: SIP INVITE from the UE
(Table 5.2.3.2-1; step 1a, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.5.1-1)

Derivation Path: Table 5.1.3.3-1: SIP INVITE from the UE

Table 5.2.3.3-2: SIP STATUS 100 from the MCx Service system
(step 2a, Table 5.1.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.16-1)

Derivation Path: Table 5.1.3.3-4: SIP STATUS from the MCx Service system

Table 5.2.3.3-3: SIP STATUS 180 from the UE
(step 3a, Table 5.2.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.16-1)

Derivation Path: Table 5.1.3.3-5: SIP STATUS from the UE

Table 5.2.3.3-4: SIP STATUS 200
(step 4a, Table 5.2.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.17-1)

Derivation Path: Table 5.1.3.3-6: SIP STATUS

Table 5.2.3.3-5: SDP message
(step 7, Table 5.2.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.3.1.0-1)

Derivation Path: IETF RFC 3556 [18]; condition SDP-ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subtype	0 - Floor Request	UE request the floor		SDP
	1 - Floor Granted	MCx Service system grants		
	2 - Floor Taken	Other UE has priority to speak		
	3 - Floor Deny			
	4 - Floor Release	UE or MCx Service system release the floor usage		
	5 - Floor Idle	No party has active floor		
	6 - Floor Revoke			
	10 - Floor Ack			
Media attribute			ETSI TS 124 229 [8] (3GPP TS 24.229)	MCPTT
Mission Critical Push To Talk(MCPTT)	<list of parameter>	Relevant parameter given in the test step		

Table 5.2.3.3-6: RTP (step 5, Table 5.2.3.2-1; IETF RFC 3550 [19])

Derivation Path: IETF RFC 3550 [19]; condition STREAM-INFORMATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	SDP			
Payload-type	Used codec e.g. ITU-T G.711 [i.6] PCMU			

Table 5.2.3.3-7: SIP BYE from the UE
(step 9, Table 5.2.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.5.2.2.1-1)

Derivation Path: Table 5.1.3.3-12: SIP BYE from the UE

5.3 MCPTT Groups configured for multi-talker control

5.3.1 Test Purpose (TP)

(1)

```
with { multi-Talker configuration by MCPTT Administrator }
ensure that {
    when { MCPTT Group with activated multi-talker floor control is configured on the MCx Service
system }
    then { multiple MCPTT service users can activate the floor control simultaneously and transmit
voice streams and all affiliated MCPTT service users receive the transmitted audio }
}
```

(2)

```
with { maximum number of simultaneous talkers configuration }
ensure that {
    when { an authorized MCPTT service user changes the maximum number of simultaneous talkers
during an ongoing MCPTT Group call configured for multi-talker }
    then { the MCx Service system enforces the new limit, revokes talk permission until the maximum
number of simultaneous talkers is reached and MCx Service system provides notification of the
revocation to all affected participants }
}
```

(3)

```
with { floor control request with priority hierarchy }
ensure that {
    when { several MCPTT service users request permission to transmit simultaneously in an MCPTT
Group configured for multi-talker floor control }
    then { the MCx Service system determines the transmitting participants based on the configured
priority hierarchy, grants permission to the higher-priority MCPTT users and rejects lower-priority
requests with notification of rejection }
}
```

(4)

```
with { override of active transmission in multi-Talker floor control }
ensure that {
    when { an authorized MCPTT service user with higher priority request to override an ongoing
MCPTT transmission in MCPTT Group communication which is configured for multi-talker }
    then { the MCx Service system revokes the active transmission, grants transmit permission to the
overriding MCPTT user and overridden MCPTT user(s) receive a notification that their transmission
has been terminated }
}
```

(5)

```
with { notification of denied transmission request }
ensure that {
    when { MCPTT service user request permission to transmit in a multi-talker configured MCPTT
Group and the request is denied}
    then { the MCPTT Service provide explicit notification to the requesting MCPTT service user
including the reason for denial (e.g., insufficient priority or limit exceeded) }
}
```

(6)

```
with { queue handling of floor control requests }
ensure that {
    when { multiple MCPTT service users request permission to transmit in a multi-talker group and
the requests are queued }
    then { the MCx Service system maintains the queue, provide the queued MCPTT service user with
their position at the time of request and MCx Service system allows them to query their position on
demand until the floor is granted, or request is withdrawn }
}
```

(7)

```
with { call termination on hang time expiry }
ensure that {
    when { MCPTT Group communication is ongoing and no further activity occurs until the configured
hang time expires }
    then { the MCx Service system terminates the call, release all resources and provide indication
of termination due to hang time expiry to all participants }
}
```

(8)

```
with { call time limit expiry handling }
ensure that {
    when { MCPTT Group communication approach the configured call time limit }
    then { the MCx Service system provides a pre-expiry indication to all participants before the
time limit is reached and upon reaching the limit, the MCx Service system release the call and all
participants receive notification, that the call was terminated due to time limit }
}
```

(9)

```
with { participant pre-emption handling due to insufficient capacity }
ensure that {
    when { one or more MCPTT service users are pre-empted from an ongoing MCPTT Group communication
due to insufficient capacity }
    then { MCx Service system ensures that the pre-empted MCPTT users receive a notification that
they were removed from the call due to capacity limitations and the remaining participants continue
the call with the updated participant list }
}
```

5.3.2 Conformance requirements

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-001]]

"An MCPTT Group shall be configurable to allow multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-002]]

"The MCPTT Service shall provide a mechanism for multiple MCPTT Users to talk simultaneously in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-003]]

"The MCPTT Service shall determine which Participant(s) are allowed to transmit to all other Participant(s) in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-004]]

"The MCPTT Service shall support all Participant(s) to receive audio from all other Participant(s) that are transmitting in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-005]]

"The MCPTT Service shall provide a mechanism for the MCPTT Administrator to configure the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-006]]

"The MCPTT Service shall allow an authorised MCPTT User to change the maximum number of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-007]]

"The MCPTT Service shall allow an authorised MCPTT User to change the priority hierarchy of simultaneous talkers at any time during a group call in an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.2-008]]

"In case the maximum number of simultaneous talkers is reduced by an authorised MCPTT User, the permission to talk will be revoked, until the new maximum number of simultaneous talkers is met.

NOTE: The selection of the participants for revoking the permission to talk is left for implementation."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-001]]

"The MCPTT Service shall enable authorised Participants to request to transmit to an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-002]]

"At call setup the MCPTT Service shall provide a notification, for example audio and/or visual, to the MCPTT Group Member attempting to transmit that there are no other Group Members who have affiliated to the MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-003]]

"The MCPTT Service shall determine the transmitting Participant(s) when there are simultaneous requests for permission to transmit within the same call for an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was granted the request an indication of being allowed to transmit."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004a]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was denied the request, the reason for denial."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-004b]]

"Following an MCPTT Request for permission to transmit on the Selected MCPTT Group configured for multi-talker control the MCPTT Service shall provide an Affiliated MCPTT Group Member that made and was queued the request, the position in the queue.

NOTE: The queueing mechanism is left for implementation."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.3-005]]

"Once in the queue, the MCPTT service shall provide to an Affiliated MCPTT Group Member, the position in the queue on demand."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-001]]

"If the number of MCPTT Users requesting the permission to talk exceeds the maximum number of simultaneous talkers in an MCPTT Group configured for multi-talker control, the MCPTT Service shall apply the override mechanism."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-002]]

"The MCPTT Service shall enable MCPTT Administrators to create a priority hierarchy for determining what Participants, Participant types (e.g. dispatcher) and urgent transmission types shall be granted a request to override an active MCPTT transmission on an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-003]]

"The priority hierarchy used for granting a request to override an active MCPTT transmission on a group configured for multi-talker control shall contain at least four (4) levels."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-004]]

"The transmitting Participant on an MCPTT Group a group configured for multi-talker control shall be determined by the relative priorities of the Participants and Call type based on priority (e.g. MCPTT Emergency)."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-005]]

"Transmission requests of Participants with insufficient relative priority shall be rejected."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-006]]

"The MCPTT Service shall provide a mechanism for Participants, to override an active MCPTT transmission of a transmitting Participant when the priority level of the overriding Participant or Call type based on priority (e.g. MCPTT Emergency) are ranked higher than the priority level of the transmitting Participant or Call type based on priority for an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-007]]

"If an authorised Participant overrides an MCPTT transmission, the MCPTT Service shall provide a means of notifying the overridden Participant(s) that the transmission has been overridden for an MCPTT Group configured for multi-talker control.'

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.3.7.4.1-008]]

"The MCPTT Service shall revoke the transmit permission of the overridden transmitting Participant on an MCPTT Group configured for multi-talker control."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-001]]

"If a Participant of an MCPTT Group call is pre-empted, the MCPTT Service shall terminate the call or continue the call with an indication to the transmitting Participant that one or more receiving Participants was pre-empted."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-002]]

"If MCPTT User(s) are pre-empted from an ongoing MCPTT call as there is insufficient capacity to support their ongoing participation, the MCPTT Service shall ensure that the MCPTT User(s) receive a notification that they have been removed from the call for reasons of lack of capacity."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-003]]

"The MCPTT Service shall terminate a call after the Hang Time expires."

[ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-005]]

"The MCPTT Service shall provide an indication to the Participants that the call is within a configurable amount of time before the call time limit is reached."

ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-006]]

"The MCPTT Service shall release the call when the call time limit has been reached."

ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-007]]

"The MCPTT Service shall provide an indication to the Participants that the call time limit has been reached."

ETSI TS 122 179 [1] (3GPP TS 22.179), clause [R-6.2.4-008]]

"The MCPTT Service shall release an MCPTT Group call if any of the termination conditions are met (e.g. last Participant leaving, second last Participant leaving, initiator leaving) or the minimum number of Affiliated MCPTT Group Members are not present."

5.3.3 Test description

5.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx service server is configured to support the following features:
  - Pre-arranged groups containing minimum UE2 and UE-Con.
  - Ad-hoc Group Communication functionality.
  - Implicit and explicit floor control.
  - Multiple user floor control.
  - Multi-talker floor control.
  - Prioritization of floor control requests.
  - In case the MCx Service system detects floor inactivity for a configurated period of time, the call will be terminated to unblock resources. The inactivity timer for any group communication type is configured to 240 seconds.
  - In case floor control will be used by the participants, the MCX service system provide an overall duration for an ongoing communication. This call time limit timer is configured to 500 seconds.
- Multimedia Broadcast Multicast Service (MBMS) is supported by MCx_S1 and UE1, UE2 and UE-Con. According to ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6.1 the floor DENY message is only supported in unicast bearer which is specified in the MBMS context. Unicast figures an end-to-end connection.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT (MCx_S1) is the acting user.

Preamble:

- The UE1and UE2 are connected to PLMN of the primary 5GS.
- The MCx Service server (MCx_S1) is connected to the IUT (UE1, MCx client), UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- MCx service SIP registration and affiliation are executed on MCx_S1.

5.3.3.2 Test procedure sequence

Table 5.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
1	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"=2	-->	HTTPS link to configuration interface	1	P
2	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=2	-->	XML access (e.g. via SCP or SFTP connection)	1	P
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	--		-	-
3a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	<--	SIP INVITE	-	-
3b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call.	<--	SIP INVITE	-	-
4	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	-->	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	-->	SIP INVITE	-	-
5	UE2 and UE-Con return STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
6	UE1 receives STATUS information: group communication established	-->	SIP STATUS 200 (OK)	-	-
7	Check: UE1 and UE2 send floor REQUEST messages at the same time	<--	RTCP (MCPT) Floor REQUEST	-	-
8a	Check: MCx_S1 sends floor GRANTED message without sending explicit floor request message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor Granted	1	P
8a1	Check: UE1 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
8b	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor Granted	1	P
8b1	Check: UE2 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
9a	Check: RTP data (voice) from UE1 can be received at all involved UE; RTP stream contains payload type information	<--	RTP stream	-	-
9b	Check: RTP data (voice) from UE2 can be received at all involved UE; RTP stream contains payload type information	<--	RTP stream	-	-
10	Check: UE1 and UE-Con sends floor request message at the same time	<--	RTCP (MCPT) Floor REQUEST	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
11a	Check: MCx_S1 sends floor GRANTED message without sending explicit floor request message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor Granted	1	P
11a1	Check: UE1 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
11b	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor Granted	1	P
11b1	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
12a	Check: RTP data (voice) from UE1 can be received at all involved UE; RTP stream contains payload type information	<--	RTP stream	-	-
12b	Check: RTP data (voice) from UE-Con can be received at all involved UE; RTP stream contains payload type information	<--	RTP stream	-	-
13	UE1 release the floor	<--	RTCP (MCPT) Floor RELEASE	-	-
14	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	1	P
15	UE2 send floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
16	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	1	P
17	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
18	Check: UE1 send floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
19	Check: MCx_S1 sends TAKEN message by MCx_S1; the following information included: Subtype 2 Floor Taken {RTCP SSRC of sending floor | granted party ID | permission to request the floor | user ID | message sequence | track info | floor indicator | audio SSRC of granted user | FA | list of granted users | list of audio SSRC | list of FA | location info | list of locations of granted users}	-->	RTCP (MCPT) Floor TAKEN with { floor indicator= 1 for multi-talker | list of granted users } According to ETSI TS 124 380 [12] (3GPP TS 24.380), Figure 6.2.4.1-1	2	P
20	Check: UE1 send floor ACK message to confirm the received Taken message	<--	RTCP (MCPT) Floor ACK	-	-
21a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"=3	-->	HTTPS link to configuration interface	2	P
21b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=3	-->	XML access (e.g. via SCP or SFTP connection)	2	P
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
22	MCx_S1 send REVOKE information to all involved participants; the following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (note 2) | track info | floor indicator	-->	RTCP (MCPT) Floor REVOKE message containing Cause #6 - No resources available	2	P
23	Check: involved UE send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
24a	Check: UE1 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
25a	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	2	P
25a1	Check: UE1 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
24b	Check: UE2 sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
25b	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	2	P
25b1	Check: UE2 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
24c	Check: UE-Con sends floor request message	<--	RTCP (MCPT) Floor REQUEST	-	-
25c	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	2	P
25c1	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
26a	Check: UE1 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	-	-
26a1	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	2	P
26b	UE2 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	-	-
26b1	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	2	P
26c	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	-	-
26c1	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	2	P
27a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 1 and the following priorities (note 3): UE1=2, UE2=4, UE-Con=0	-->	HTTPS link to configuration interface	3	P
27b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=1 and the following priorities (note 3): UE1=2, UE2=4, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	3	P
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
28a	Check: UE1 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	-	-
29a	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	3	P
29a1	Check: UE1 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
28b	Check: UE2 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=4}	-	-
29b	Check: MCx_S1 sends floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (note 2) | User granted the floor | floor indicator}	-->	RTCP (MCPT) Floor DENY; message containing field {reject cause=#1 (Another MCPTT client has permission)}	3	P
29b1	Check: UE2 send floor ACK message to confirm the received Deny message	<--	RTCP (MCPT) Floor ACK	-	-
28c	Check: UE-Con send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
29c	Check: MCx_S1 sends floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (note 2) | User granted the floor | floor indicator}	-->	RTCP (MCPT) Floor DENY; message containing field {reject cause=#4 (Media Burst pre-empted)}	3	P
29c1	Check: UE1 send floor ACK message to confirm the received Deny message	<--	RTCP (MCPT) Floor ACK	3	-
30	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	3	-
31	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	3	P
32a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 2 and the following priorities (note 3): UE1=2, UE2=2 UE-Con=0	-->	HTTPS link to configuration interface	4	P
32b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=2 and the following priorities (note 3): UE1=2, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	4	P
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
33a	Check: UE1 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	4	-
34a	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	4	P
34a1	Check: UE1 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	4	-
33b	Check: UE2 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	4	-
34b	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted (note 1)	-->	RTCP (MCPT) Floor GRANTED	4	P
34b1	Check: UE2 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
35	Check: UE-Con send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	-
36a	Check: MCx_S1 sends floor REVOKE message to the active transmission. The following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (note 2) | track info | floor indicator}	-->	RTCP (MCPT) Floor REVOKE (note 4) with Reject Cause #1 - Another MCPTT client has permission	4	P
36a1	Check: UE1 send floor ACK message to confirm the received Revoke message	<--	RTCP (MCPT) Floor ACK	-	-
36b	Check: MCx_S1 sends REVOKE message to the active transmission. The following information included: Subtype 6 Floor Revoke {SSRC of floor control server | reject cause (note 2) | track info | floor indicator}	-->	RTCP (MCPT) Floor REVOKE (note 4) with Reject Cause #1 - Another MCPTT client has permission	4	P
36b1	Check: UE2 send floor ACK message to confirm the received Revoke message	<--	RTCP (MCPT) Floor ACK	-	-
37	Check: MCx_S1 sends floor GRANTED	-->	RTCP (MCPT) Floor GRANTED	4	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
38	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
39a	Check: RTP data (voice) from UE-Con can be received at all involved UE; no other RTP source; outgoing condition RTP stream contains payload type information	<--	RTP stream	4	-
39b	Check: RTP data (voice) from UE-Con can be received at all involved UE; incoming condition RTP stream contains payload type information	<--	RTP stream	4	-
40	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	4	-
41	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	4	P
41a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 2 and the following priorities (note 3): UE1=4, UE2=2 UE-Con=0	-->	HTTPS link to configuration interface	5	P
41b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"=2 and the following priorities (note 3): UE1=4, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	5	P
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
42a	Check: UE2 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	5	-
43a	Check: MCx_S1 sends GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor GRANTED	5	P
43a1	Check: UE2 send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
42b	Check: UE-Con send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	-
43b	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor GRANTED	5	P
43b1	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	-	-
42c	Check: UE1 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=4}	-	-
43c	Check: MCx_S1 sends floor DENY message; following information included: Subtype 3 Floor Deny {Reject Cause (note 2) | User granted the floor | floor indicator}	-->	RTCP (MCPT) Floor DENY; message containing field {reject cause #7 – Queue full	5	P
43c1	Check: UE1 responds with floor ACK message to confirm the received Deny message	<--	RTCP (MCPT) Floor ACK	-	-
44a	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via HTML web interface; maximum number of simultaneous talker set to "MaxTalkers"= 1, "MaxQueueCount"=2 and the following priorities (note 3): UE1=2, UE2=2 UE-Con=2	-->	HTTPS link to configuration interface	6	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
44b	Check: MCPTT Administrator configures MCPTT Group for multi-talker floor control on the MCx Service system; access via XML configuration file interface; ; maximum number of simultaneous talker set to "MaxTalkers"= 1, "MaxQueueCount"=2 and the following priorities (note 3): UE1=2, UE2=2, UE-Con=0	-->	XML access (e.g. via SCP or SFTP connection)	6	P
	EXCEPTION: group communication establishment shall not terminate while changing MaxTalkers parameter	--		-	-
45a	Check: UE-Con send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	6	-
46a	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor GRANTED	6	P
46a1	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Floor ACK	6	-
45b	Check: UE2 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	6	-
46b	Check: MCx_S1 sends floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	-->	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 1 and the ID of the queued user	6	P
46b1	Check: UE2 send floor ACK message to confirm the received Floor Queue Position Info message	<--	RTCP (MCPT) Floor ACK	6	-
45c	Check: UE1 send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=2}	6	-
46c	Check: MCx_S1 sends floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	-->	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 2 and the ID of the queued user	6	P
46c1	Check: UE1 send floor ACK message to confirm the received Floor Queue Position Info message	<--	RTCP (MCPT) Floor ACK	6	-
47	Check: UE1 send periodically floor QUEUE POSITION REQUEST message; following information included: Subtype 9 Floor Queue Position Info Request	<--	RTCP (MCPT) Floor QUEUE POSITION REQUEST	6	-
48	Check: MCx_S1 sends floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	-->	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position index 2 and the ID of the queued user	6	P
49	Check: UE1 send floor ACK message to confirm the received Floor Queue Position Info message	<--	RTCP (MCPT) Floor ACK	6	P
50	UE2 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	6	-
51	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	6	P
52	Check: UE1 send periodically floor QUEUE POSITION REQUEST message; following information included: Subtype 9 Floor Queue Position Info Request	<--	RTCP (MCPT) Floor QUEUE POSITION REQUEST	6	-
53	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	6	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
54	Check: MCx_S1 sends floor QUEUE POSITION INFO message; following information included: Subtype 8 Floor Queue Position Info	-->	RTCP (MCPT) Floor QUEUE POSITION INFO containing the queue info field with the position (index 2 has changed to index 1) and the ID of the queued user	6	P
55a	UE1 send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	6	-
56a	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	6	P
55b	UE-Con send floor RELEASE message; the following information included: Subtype 4 Floor Release {user ID | track info | floor indicator}	<--	RTCP (MCPT) Floor RELEASE	6	-
56b	Check: MCx_S1 send floor ACK	-->	RTCP (MCPT) Floor ACK	6	P
	EXCEPTION: no action until hang timer (240s) expires	-		7	-
57a	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	-->	SIP BYE	7	P
58a	UE1 responds with STATUS message to the call termination request	<--	SIP STATUS1) 200 (OK)	7	-
57b	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	-->	SIP BYE	7	P
58b	UE2 responds with STATUS message to the call termination request	<--	SIP STATUS1) 200 (OK)	7	-
57c	Check: Hang time expired, MCx_s1 send termination request to all participants; UE 1 receive SIP BYE	-->	SIP BYE	7	P
58c	UE-Con responds with STATUS message to the call termination request	<--	SIP STATUS1) 200 (OK)	7	-
59a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	<--	SIP INVITE	8	-
59b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	<--	SIP INVITE	8	-
60	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	-->	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	-->	SIP INVITE	-	-
61	UE2 and UE-Con return STATUS information: group communication established	<--	SIP STATUS 200 (OK)	8	-
62	UE1 receives STATUS information: group communication established	-->	SIP STATUS 200 (OK)	-	-
63	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE send floor REQUEST message	<--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	8	-
64	Check: MCx_S1 sends floor GRANTED message; following information included: Subtype 1 Floor Granted	-->	RTCP (MCPT) Floor GRANTED	8	P
65	Check: UE-Con send floor ACK message to confirm the received Granted message	<--	RTCP (MCPT) Flo or ACK	8	-
	None. (Wait until call time limit expires)	--		-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
66	Call time limit timer expires after 300 s. Check: MCx_S1 disconnect UE1 from group communication. UE1 receive DISCONNECT message	-->	RTCP (MCPT) DISCONNECT with reason clause 7 according to ETSI TS 124.380 [12] (3GPP TS 24.380) clauses 8.3.5 and 8.3.3.11	8	P
67	Ceck: UE1 acknowledge the DISCONNECT message	<--	SIP STATUS 200 (OK)	8	-
68a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	-->	SIP INVITE	9	-
68b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	-->	SIP INVITE	9	-
69	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	-->	SIP STATUS 100 (Trying)	-	-
	EXCEPTION: UE2 and UE-Con receive INVITE to the group communication in step 3a/3b provided by MCx_S1	-->	SIP INVITE	-	-
70	UE2 and UE-Con return STATUS information: group communication established	<--	SIP STATUS 200 (OK)	9	-
71	UE1 receives STATUS information: group communication established	-->	SIP STATUS 200 (OK)	-	-
72	MCx_S1 removes UE1 due to capacity limitations Check: UE receive DISCONNECT message	-->	RTCP (MPTC) DISCONNECT with reason clause 10 according to ETSI TS 124.380 [12] (3GPP TS 24.380) clauses 8.3.5 and 8.3.3.11	9	P
73	MCx_S1 sends group dynamic notification to all remaining participants Check: UE2 and UE-Con receive group dynamic notification	-->	RTCP (MPTC) Group Dynamic NOTIFY	9	P
74	MCx_S1 terminates the group communication call using BYE Check: UE2 and UE-Con receive BYE	-->	SIP BYE	-	-
75	UE2 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	-
76	UE-Con confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	-

NOTE 1: UE2 and UE-Con are in status floor GRANTED and MaxTalkers exceeds with a new request. These states the precondition to check that the change of the MaxTalker counter will be possible during an ongoing group communication (pre-arranged or ad hoc).
NOTE 2: Reject Causes are given in [12] clause 8.2.6.2.
NOTE 3: Priorities depending on the MCx Service server configuration. For this test consider 4=low priority, 2=mid priority and 0=high priority.
NOTE 4: According to ETSI TS 122 179 [1] (3GPP TS 22.179) clause R-6.2.3.7.2-008 the selection of the participants for revoking the permission to talk is left for implementation.

5.3.3.3 Specific message contents

Table 5.3.3.3-1: SIP INVITE
(Table 5.3.3.2-1; step 3a, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.5.1-1)

Derivation Path: Table 5.1.3.3-1: SIP INVITE from the UE

Table 5.3.3.3-2: SIP STATUS 100 from the MCx Service system
(Table 5.3.3.2-1; step 4, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.16-1)

Derivation Path: Table 5.1.3.3-4: SIP STATUS from the MCx Service system

Table 5.3.3.3-3: SIP STATUS 200
(step 5, Table 5.3.3.2-1; step 5, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.17-1)

Derivation Path: Table 5.1.3.3-6: SIP STATUS

Table 5.3.3.3-4: SDP message
(step 7, Table 5.3.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.3.1.0-1)

Derivation Path: IETF RFC 3556 [18]; condition SDP-ANSWER				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Subtype	0 – Floor Request	UE request the floor		SDP
	1 – Floor Granted	MCx Service system grants		
	2 – Floor Taken	Other UE has priority to speak		
	3 – Floor Deny			
	4 – Floor Release	UE or MCx Service system release the floor usage		
	5 – Floor Idle	No party has active floor		
	6 – Floor Revoke			
	10 – Floor Ack			
Media attribute			ETSI TS 124 229 [8] (3GPP TS 24.229)	MCPTT
Mission Critical Push To Talk(MCPTT)	<list of parameter>	Relevant parameter given in the test step		

Table 5.3.3.3-5: RTP DISCONNECT
(step 66, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.3.5)

Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5; condition REASON-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DISCONNECT			
Reason Cause	7	Group timer expired	ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.3.3.11	
	10	Group participant disconnected		

Table 5.3.3.3-6: RTP DENY
(step 29b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.6)

Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.6; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DENY			
Reject Cause	1	Another MCPTT client has permission	ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.3.3.11	
	4	Media Burst pre-empted		
	7	Queue full		

Table 5.3.3.3-7: RTP REVOKE
(step 29b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5)

Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.3.5; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	DENY			
Reject Cause	1	Another MCPTT client has permission	ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.6.2	
	4	Media Burst pre-empted		

Table 5.3.3.3-8: RTP QUEUE POSITION INFO
(step 46b, Table 5.3.3.2-1; ETSI TS 124 380 [12] (3GPP TS 24.380), clause 8.2.12)

Derivation Path: ETSI TS 124 380 [12] (3GPP TS 24.380) clause 8.2.12; condition REJECT-CAUSE-FIELD				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	QUEUE POSITION INFO			
Queue info field	<index>	Position	ETSI TS 124 380 [12] (3GPP TS 24.380) Table 8.2.12-1	
ID of the queued user	<ID>	Requesting user ID		

Table 5.3.3.3-9: RTP
(step 9a, Table 5.3.3.2-1; IETF RFC 3550 [19])

Derivation Path: ETSI TS 124 380 [12]; condition STREAM-INFORMATION				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Stream setup				
Method	SDP			
Payload-type	Used codec e.g. ITU-T G.711 [i.6] PCMU			

Table 5.3.3.3-10: SIP BYE from the UE
(step 57a, Table 5.3.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.5.2.2.1-1)

Derivation Path: Table 5.1.3.3-12: SIP BYE from the UE

6 Mission critical ad hoc group communications

6.1 Interception interface on MCX service system

6.1.1 Test Purpose (TP)

(1)

```
with {MCX service system providing discrete listen capabilities}
ensure that {
    when {MCX service system provides discrete listening capabilities to a target MCX service user
which requires spec authorizations}
    then {No noticeable impact, information, facility or impact can be detected by non-interception-
target MCX service users and all information from or to the ad hoc group communication MCX service
users are transmitted to the authorized interception-target MCX service user including all session
audit information (e.g. regrouping, leaving of participants, voice durations, position information)}
            }
```

6.1.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.1-001a]]

"The MCX Service shall provide discreet listening capabilities without noticeable impact on or knowledge of the target MCX User, or the members of the target MCX Service Group including Ad hoc groups, and all other unauthorised MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.1-002a]]

"The MCX Service shall provide a mechanism for an authorised MCX User to receive MCX Service Ad hoc Group Communication transmissions to and from a specific target MCX User that is within the authority of the authorised MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-011]]

"The MCX Service shall provide recording and audit functions for all types of MCX Service group communication (e.g. normal group, regroup group, Ad hoc group)."

6.1.3 Test description

6.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- The configuration for recording, audition and discrete listening is prepared according to Figure 4.1-3.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx service server is configured to support the following features:
    - Configuration options for discrete listening, recording and session audition.
    - UEa1 shall be configured as discrete listening user.
    - Pre-arranged groups containing minimum UE2 and UE-Con.
    - Ad-hoc Group Communication functionality.
    - Multiple user floor control.
    - Multi-talker floor control.
    - Prioritization of floor control requests.
    - In case the MCx Service system detects floor inactivity for a configurated period of time, the call will be terminated to unblock resources. The inactivity timer for any group communication type is configured to 240 seconds.
    - In case floor control will be used by the participants, the MCx service system provide an overall duration for an ongoing communication. This call time limit timer is configured to 500 seconds.
    - Time period for keeping recording and session details shall be configurable between 0 (off) and 30 days.

IUT:

- IUT is MCx_S1 (MCx Service system 1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT (MCx_S1), UE1, UEa1 and UE2 are connected to PLMN of the primary 5GS.
- The MCx Service server (MCx_S1) is connected to the IUT (UEa1, MCx client, discrete listener), UE1, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- MCx service SIP registration and affiliation are executed on MCx_S1.

6.1.3.2 Test procedure sequence

Table 6.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), clause 5.1.3				
	MCx Service system provides a configuration interface for feature activation in respect to the FA	--			
1a	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. With correct credentials the access to the discrete listening configuration management will be granted (note 1)	-->	HTTPS link to configuration interface	1	P
1b	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. With correct credentials the access to the discrete listening configuration management will be granted (note 1)	-->	XML access (e.g. via SCP or SFTP connection)	1	P
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), clause 5.4.3 (note 2)	--		-	-
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), clause 5.3.2 (note 2)	--		-	-
	EXCEPTION: Procedure 'Functional aliasing client side' as described ETSI TS 104 069-1 [26], clause 5.1.3 (note 2)	--		-	-
	EXCEPTION: Procedure 'Floor control in general' as described in ETSI TS 104 069-1 [26], clause 5.2.3 (note 2)	--		-	-
	EXCEPTION: Procedure 'MCPTT Groups configured for multi-talker control' as described in ETSI TS 104 069-1 [26], clause 5.3.3 (note 2)	--		-	-
	EXCEPTION: Procedure 'Arbitration function' as described in ETSI TS 104 069-1 [26], clause 5.4.3 (note 2)	--		-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
2a	Check: Open the MCx Service server GUI and login as "Replay Client". Session audition log will be presented on the GU	-->	The session audition log contains all information regarding {PTT activities | regrouping | leaving of participants | voice durations | position information | termination causes | optional information}	1	P
2b	Check: Open the MCx Service server GUI and login as "user" or "admin"	-->	The session audition log is not available	1	P
3a	Pre-arranged group communication case Check: UE1 initiates pre-arranged group communication call	<--	SIP INVITE	-	-
3b	Ad-hoc Group Communication case Check: UE1 initiates ad-hoc group communication call	<--	SIP INVITE	-	-
4	UE1 receives STATUS information from MCx_S1 regarding the establishment procedure	-->	SIP STATUS 100 (Trying)	-	-
5	EXCEPTION: UEa1, UE2 and UE-Con receive INVITE to the group communication in step 2a/2b provided by MCx_S1	-->	SIP INVITE	-	-
6	UEa1, UE2 and UE-Con return STATUS information: group communication established	<--	SIP STATUS 200 (OK)	-	-
7	UE1 receives STATUS information: group communication established	-->	SIP STATUS 200 (OK)	-	-
8a	Check for pre-arranged group communication: UE1 does not see any information about UE1a in the group communication participants list	<--	RTCP (MCPT) Floor REQUEST	-	-
9a	MCx Service system replies with floor granted	-->	RTCP (MCPT) Floor GRANTED	-	-
10a	Check: UE1 can send RTP data - media stream is transmitted	<--	RTP stream	-	-
11a	RTP data from UE1 is provided to UE1a	-->	RTP stream	1	P
12a	Check: UE1a is not allowed to use the PTT. If floor request is sent by the UE1a, it is denied by the MCx Service system	<--	RTCP (MCPT) Floor REQUEST	-	-
13a	MCx Service system replies with floor deny	-->	RTCP (MCPT) Floor DENY	1	P
8b	Check for ad hoc group communication: UE1 does not see any information about UE1a in the ad hoc group communication participants list	<--	RTCP (MCPT) Floor REQUEST	-	-
9b	MCx Service system replies with floor granted	-->	RTCP (MCPT) Floor GRANTED	1	P
10b	Check: UE1 can send RTP data - media stream is transmitted	<--	RTP stream	-	-
11b	RTP data from UE1 is provided to UE1a	-->	RTP stream	1	P
12b	Check: UE1a is not allowed to use the PTT. If floor request is sent by the UE1a, it is denied by the MCx Service system	<--	RTCP (MCPT) Floor REQUEST	-	-
13b	MCx Service system replies with floor deny	-->	RTCP (MCPT) Floor DENY	1	P
14	MCx_S1 terminates the group communication call using BYE Check: UE1, UE1a and UE-Con receive BYE	-->	SIP BYE	-	-
15	UE1, UE1a confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	-

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
16	UE-Con confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	-
NOTE 1: The configuration section on the server shall be in available only with specific user rights to prevent misuse. The discrete listening function can be enabled or disabled. On enabling the settings for recording and user selection can be configured. The recording shall include SIP messaging as well as RTP audio stream data. The discrete listening UE terminal shall be excluded in all lists. PTT function shall be disabled; UE is in listening only mode. NOTE 2: The exceptional procedures are required as input for the recording and session audition function.					

6.2 Administrative and configurational parameters required for Ad hoc group communications

6.2.1 Test Purpose (TP)

(1)

```
with {Options for adopting the termination of an ad hoc group service by a MCX Service
administrator}
ensure that {
    when {A configuration interface is available (e.g. web interface or XML configuration)}
    then {MCX Service administrator can configure the ad hoc group communication termination on last
participant or second last participant is leaving}
            }
```

(2)

```
with {Options for adopting the individual permissions of an ad hoc group service by a MCX Service
administrator}
ensure that {
    when {A configuration interface is available (e.g. web interface or XML configuration) and
individual permissions of the MCX Users are set}
    then {The MCX Service administrator can define the maximum number of participants for an ad hoc
group communication and if a MCX Service user has the permission to participate the ad hoc group
communication and the MCX Service administrator can configure the standard value set of the ad hoc
group communication (e.g. priority, hang time, broadcast mode) and the MCX Service administrator can
override the authorizations of individual MCX users}
            }
```

6.2.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-004]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure additional conditions under which MCX Service Ad hoc Group Communication shall be terminated (e.g. last Participant leaving, second last Participant leaving, initiator leaving)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.3-001]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure which MCX Users, within their authority, are authorised to initiate a MCX Service Ad hoc Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.3-002]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure the maximum number of MCX Users who can participate in a MCX Service Ad hoc Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.3-003]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure which MCX Users are authorised to participate in a MCX Service Ad hoc Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.3-004]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to define the default parameters for MCX Service Ad hoc Group communication (e.g. priority, hang time, broadcast mode)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.3-005]]

"The MCX Service shall provide a mechanism for an MCX Service Administrator to configure whether MCX Service Ad hoc Group Communication is allowed on the MCX system regardless of individual MCX User authorizations."

6.2.3 Test description

6.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- Parameter interface is accessible.
- MCx service server is supports configurable parameter values for the following features:
  - Configuration options for discrete listening, recording and session audition.
  - UEa1 shall be configured as discrete listening user.
  - Pre-arranged groups containing minimum UE2 and UE-Con.
  - Ad-hoc Group Communication functionality.
  - Multiple user floor control.
  - Multi-talker floor control.
  - Prioritization of floor control requests.
  - In case the MCx Service system detects floor inactivity for a configurated period of time, the call will be terminated to unblock resources. The inactivity timer for any group communication type is configured to 240 seconds.
  - In case floor control will be used by the participants, the MCx service system provide an overall duration for an ongoing communication. This call time limit timer is configured to 500 seconds.
  - Time period for keeping recording and session details shall be configurable between 0 (off) and 30 days.

IUT:

- IUT is MCx_S1 (MCx Service system 1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The MCx_S1 is the acting user.

Preamble:

- The IUT (MCx_S1) provides a configuration interface.

6.2.3.2 Test procedure sequence

Table 6.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), clause 5.1.3				
1	MCX Service Administrator opens configuration interface (Web/XML API).	-->	HTTPS POST {ConfigRequest: TerminationCondition = "Last / Second Last Participant Leaves"}		P
2	MCX Service validates request and stores configuration	<--	HTTPS 200 OK {ConfigApplied}		
3	Ad hoc group call is established with multiple participants	<--	SIP INVITE → 200 OK → ACK		
4	Last or second last participant leaves group (depending on admin config)	<--	SIP BYE → 200 OK		
5	MCX Service terminates the ad hoc group communication automatically	-->	SIP NOTIFY {Group Termination Event} sent to all relevant participants / administrator		P
6	MCX Service Administrator configures individual permissions for ad hoc group communication	-->	HTTPS POST {ConfigRequest: MaxParticipants, Priority, HangTime, BroadcastMode, OverrideRules}		P
7	MCX Service validates configuration	<--	HTTPS 200 OK {ConfigApplied}		
8	MCX Service User attempts to join ad hoc group	<--	SIP INVITE {UserID, GroupID}		
9	MCX Service applies permission policy	-->	If unauthorized → SIP 403 Forbidden. If authorized → SIP 200 OK		P
10	During ongoing session, low-priority user requests floor	<--	RTCP Floor Request		
11	Administrator override is applied (higher-priority user)	<--	RTCP Floor Revoke (low priority), RTCP Floor Granted (higher priority)		
12	MCX Service applies broadcast mode or hang time as configured	<--	SIP re-INVITE / UPDATE {Session-Expires, BroadcastMode} → 200 OK		
13	Ongoing participants remain unaffected	<--	RTP streams continue seamlessly		

6.3 Ad hoc group communication functionality for MCX service user involving one MCX system

6.3.1 Test Purpose (TP)

(1)

```
with {Establishment of an ad hoc group communication by an authorized user} service user}
ensure that {
    when {MCX service user is allowed to initiate ad hoc communication (initiator) manually or based
on automatic criterion (e.g. location) and Multiple MCX service users using FA are available for ad
hoc group communication and participants can be added or removed by initiator or authorized MCX
server user}
    then {MCX service user initiates an ad hoc group communication manually or based on an automatic
criterion (e.g. location) and MCX service user with multiple FA receives only one ad hoc group
communication based on the unique MCX service ID and the FA can be determined by the MCX service
user and the initiator or an authorized MCX service user is able to add or remove participants from
the ad hoc group communication and the list of participants will be updated by the MCX service
system and information will be sent to the initiator of the ad hoc group communication}
    else when {MCX service user has no authorisation to initiate an ad hoc group communication}
    then {MCX service user receive a reason for denial from the MCX service system}}
        }
```

(2)

```
with {Ad hoc group communication processing}}
ensure that {
    when {Ad hoc group communication is in preparation or ad hoc group communication is active}
    then {MCX service user receive information about the initialisation and transmit request queue
and hang time and mode (e.g. broadcast mode)}
            }
```

(3)

```
with {Termination of an ad hoc group communication}
ensure that {
    when {MCX ad hoc group communication service is configured, that ad hoc group communication can
be released by initiator or by timer expiration only or termination by MCX service system. Timer is
set to 300 seconds}
    then {Initiator of an ad hoc group communication releases the ad hoc communication or timer
expires, and ad hoc group communication no longer exists after release}
    else when {MCX ad hoc group communication service is configured, that ad hoc group communication
can be released by initiator or by timer expiration after 300 seconds only and ad hoc group call
participant which is not initiator release the group communication}
    then {MCX service user leaves the ad hoc group call communication, and the list of participants
will be updated by the MCX service system and information will be sent to the initiator of the ad
hoc group communication}
            }
```

(4)

```
with {Suppression of the participants list}
ensure that {
    when {MCX service system suppress the list of participants for MCX service users which are not
the initiator of the ad hoc group communication}
    then {The MCX service user, which is participant in an ad hoc group communication has no option
to request the list of participants or the MCX service system reject the request}
            }
```

(5)

```
with {Generating a permanent group from an ad hoc group communication participants list}
ensure that {
    when {MCS service user is able to initiate an ad hoc group communication by selecting different
participants}
    then {The MCX service user application provides options to save the ad hoc group configuration
as permanent and the MCX service user application provides options to manipulate the saved list
(e.g. add or remove participants, export or delete the list)}
            }
```

(6)

```
with {Joining an ongoing ad hoc group communication}
ensure that {
    when {Ongoing ad hoc group communication and} MCX Service user added to the ad hoc group
communication list}
    then {MCX service user is able to join an ongoing ad hoc group communication while entering the
location or on request to the initiator or an authorized MCX service user or on invitation from the
initiator or an authorized MCX service user}
            }
```

(7)

```
with {MCX service user provide location information}
ensure that {
    when {MCX service user is initiator or participant of an ad hoc group communication}
    then {MCX service user provides location information}
            }
```

(8)

```
with {Ad hoc group communication parameter selection and usage}
ensure that {
    when {MCX service user prepares an ad hoc group communication}
    then {MCX service user can select additional parameters for hang time and priority (e.g. general
group, emergency, imminent peril) and operational mode (e.g. broadcast or non-broadcast mode)}
```

6.3.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001]]

"The MCX Service shall provide a mechanism for an authorised MCX User to combine an ad hoc multiplicity of MCX Users into a MCX Service Ad hoc Group Communication.

NOTE: Selection of the list of MCX Users can be manual, or automatic based on certain criteria (e.g. location). This is left for implementation."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001a]]

"The MCX Service shall provide the reason for denial to an MCX user who was not authorised to initiate an MCX Service Ad hoc Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001b]]

"The participant of an MCX Ad hoc Group Communication who has activated several Functional Alias(es) shall receive only one communication based on the unique MCX Service ID."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-001c]]

"The MCX Service shall provide a mechanism for the participant of an Ad hoc Group Communication to determine the Functional Alias, through which this participant is addressed by the communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-002]]

"An MCX Service Ad hoc Group Communication is a type of MCX Service Group communication and shall support MCX Service Group Communication mechanisms for call processing (e.g. transmit request queuing, hang time, broadcast mode)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-003]]

"MCX Service Ad hoc Group Communications shall be terminated by an authorised user or using the same mechanisms as MCX Service Group communications (e.g. initiator release, time expiration, server release)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-005]]

"When an MCX Service Ad hoc Group Communication is terminated the group shall not persist."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-007]]

"The MCX Service shall provide a mechanism for an authorised initiator of a MCX Service Ad hoc Group Communication to define the communication parameters for the Ad hoc Group Communication (e.g. priority, hang time, broadcast/non-broadcast)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-008]]

"MCX Service Ad hoc Group Communications shall be able to support the same urgency as MCX Service Group communication (e.g. general group, emergency, imminent peril)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-010]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Communication to request that the list of participants is suppressed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-011]]

"The MCX Service shall provide a mechanism for authorised MCX Users to create a permanent MCX Service Group from the members of the MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-012]]

"The MCX Service shall provide a mechanism for the initiator and/or an authorised user to add or remove participants during an in progress MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-013]]

"The MCX Service shall provide a mechanism for a participant to join an in progress MCX Service Ad hoc Group communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-014]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Communication to request that the list of participants be determined and updated by the MCX Service system using pre-defined criteria."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-014a]]

"When the list of participants is determined or updated by the MCX Service system, the MCX Service shall provide a mechanism that monitors and ensures that the participants list is applied for an MCX Service Ad hoc Group communication, performing retries when needed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-015]]

"The MCX Service shall provide a mechanism for a participant of an MCX Service Ad hoc Group Communication to provide his location information to an authorised participant of the same group."

6.3.3 Test description

6.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad-hoc Group Communication supporting criterion-based functionality:
    - Functional Alias (optional).
    - Location Information (optional).
    - Criteria for determining the participants (mandatory).
  - In case the MCx Service system detects floor inactivity for a configurated period, the call will be terminated to unblock resources. The inactivity timer T4 for any ad hoc group communication is configured to T4=300 seconds.
- Supporting user UE2 has permission for using ad hoc functionality and has activated multiple FA (minimum two).
- Supporting user UEa1 user has following configuration:
  - Local parameter set allows the usage of ad hoc functionality.
  - MMCx_S1 has a parameter set with no permission for UEa1 to use ad hoc functionality.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The UE1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS according to Figure 4.1-1.
- The MCx Service system (MCx_S1) is connected to the IUT, UEa1 (MCx client with restrictions), UE2 (MCx client with multiple FA) and UE-Con (MCx client with controller/dispatcher functionality).

6.3.3.2 Test procedure sequence

Table 6.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3. No FA is active at the MCx Service system	<--	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	-	--
1	Check: MCx_S1 confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
2	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	<--	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	-	--
3	Check: MCx_S1 responds with a STATUS (note 1) message	-->	SIP STATUS (note 1) 200 OK	1	P
4	UE1 send PUBLISH message containing an automatic FA information	<--	SIP PUBLISH	-	--
5	Check: MCx_S1 responds with a STATUS (note 1) message	-->	SIP STATUS (note 1) 200 OK	1	P
6	UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	--	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 6.2.8.1.24 is set to "true"	-	--
7a	UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup, ParticipantList}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
8a	System validates authorization of UE1	-->	DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	1	P
9a	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	1	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
7b	UE1 initiates ad hoc group communication based (automatic; criterion based: functional alias) {InitiateGroup, ParticipantList, FA ID}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
8b	System validates authorization of UE1		DB/AAA check → VALID: ACK	1	P
9b	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	1	P
7c	UE1 initiates ad hoc group communication based (automatic criterion: location information) {InitiateGroup, ParticipantList, location information}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
8c	System validates authorization of UE1.		DB/AAA check → VALID: ACK	1	P
9c	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	1	P
10	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
11	MCx_S1 invites UE-con into the ongoing ad hoc group communication	<--	SIP INVITE	-	--
12	UE-con responds to the INVITE request (note 2)	<--	SIP STATUS 100 (TRYING)	-	--
13	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
14	UE-con call established confirmation towards MCx_S1	<--	SIP ACK	-	--
15	Chenck: Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
16	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
17	Check: MCx_S1 invites UE2 (multiple FA) into the ongoing ad hoc group communication	-->	SIP INVITE (note 3)	1	P
18	UE2 responds to the INVITE request (note 2)	<--	SIP STATUS 100 (TRYING)	-	--
19	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
20	UE2 call established confirmation towards MCx_S1	<--	SIP STATUS 200 (OK) {Group Established, GroupID} → ACK	-	--
21	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
22	UEa1 initiates ad hoc group communication based (automatic criterion: functional alias) {InitiateGroup, ParticipantList, FA ID}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
23	Check: System validates authorization of UEa1		DB/AAA check → NOT VALID: → 40x (note 4) FORBIDDEN or UNAUTHORIZED {supressed}	1	P
24	Check: MCx_S1 sends confirmation to UEa1	-->	SIP STATUS 40x (note 4)	1	P
25b	Criterion based on FA UE1 change FA to a value which is not part of the ad hoc group communication		none		
26b	UE1 sends PUBLISH message for subscribe functional alias request to MCx_S1	<--	SIP PUBLISH Message content according to ETSI TS 123 280 [4] (3GPP TS 23.280) clause 10.13.10.2	-	--
27b	Check: MCx Service system responds with a STATUS (note 4) message	-->	SIP STATUS 200 OK	1	P
25c	Criterion based on location information UE1 receive new location information via internal interface and change to a location which is not part of the ad hoc group communication		none		

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
26c	UE1 sends PUBLISH message containing Location Information Report to MCx_S1	<--	SIP PUBLISH Message content according to ETSI TS 123 280 [4] (3GPP TS 23.280) clause 10.9.3.2	-	--
27c	Check: MCx Service system responds with a STATUS (note 4) message	-->	SIP STATUS 200 OK	1	P
28	System validates authorization of UE1		DB/AAA check → VALID but not part of the current ad hoc: → BYE	1	P
29	Check: MCx_S1 disconnect UE1 from ad hoc group communication. UE1 receive BYE message	-->	SIP BYE	1	P
30	UE1 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
31	Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
32	Check: ad hoc group communication activity timer T4 expires. Termination by system timer (300 s)		Internal Timer Expired → SIP NOTIFY {Group Terminated by Timer} → Participants receive BYE	2	P
33a	Check: MCx_S1 disconnect UE2 from ad hoc group communication. UE2 receive BYE message	-->	SIP BYE	3	P
34a	UE2 confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
33b	Check: MCx_S1 disconnect UE-con from ad hoc group communication. UE-con receive BYE message	-->	SIP BYE	3	P
34b	UE-con confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3 Check: UE1 publish the basic POC settings	<--	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	-	--
35	Check: MCx_S1 confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
36	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	<--	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	-	--
37	Check: MCx_S1 responds with a STATUS (note 1) message	-->	SIP STATUS (note 1) 200 OK	1	P
38	UE1 send PUBLISH message containing an automatic FA information	<--	SIP PUBLISH	-	--
39	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS (note 1) 200 OK	1	P
40	UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	<--	- none - Client checks the ruleset configuration according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 6.2.8.1.24 is set to "true"	-	--
41	Check: UE1 initiates ad hoc group communication based (criterion: manually) {InitiateGroup, ParticipantList}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
42	System validates authorization of UE1		DB/AAA check → VALID: ACK {Group Established, GroupID} → ACK	1	P
43	UE1 receives confirmation from MCx_S1	-->	SIP ACK	2	P
44	Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
45	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	<--	SIP INVITE	-	--
46	Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
47	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message		RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
48	Check: MCx_S1 invites UE2 into the ongoing ad hoc group communication	-->	SIP INVITE	5	P
49	UE2 responds to the INVITE request	<--	SIP STATUS 100 (TRYING)	-	--
50	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
51	UE2 call established confirmation towards MCx_S1	-->	SIP ACK	-	--
52	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	4	P
53	UE2 (participant) leaves ad hoc group communication sending BYE	<--	SIP BYE	-	--
54	Check: MCx_S1 confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P
55	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {Updated Participant List} → Remaining UEs	3	P
60	Check: UE1 (initiator) terminates ad hoc group communication sending BYE	<--	SIP BYE	-	--
61	MCx_S1 confirms the received BYE message	-->	SIP STATUS 200 (OK)	3	P
62	Check: MCx_S1 terminates the ad hoc group communication; UE-con receive BYE message	-->	SIP BYE	3	P
63	UE-con confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--

NOTE 1: May notify the answer state to the user (i.e. "Unconfirmed" or "Confirmed") if received in the P-Answer-State header field.

NOTE 2: The application shall indicate the detected status of the floor parameter on the graphical user interface as icon and text:

a. Status is IDLE, REVOKE or RELEASED: green; text: e.g. "push PTT to talk!".
b. Status REQUESTED, DENY or TAKEN by other user, no multi-talker: red; text: none.
c. Status REQUESTED, DENY or TAKEN by other user, multi-talker: blue; text: e.g. "push PTT to talk!".
d. Status is GRANTED to own UE: yellow; text: e.g. "please talk!".

NOTE 3: User with multiple FA receives only one ad hoc group communication based on the unique MCX service ID and the FA can be determined by the MCX service user and the initiator.

NOTE 4: According to ETSI TS 124 229 [8] (3GPP TS 24.229) a request should be rejected by sending a suitable 4xx response.

NOTE 5: The application on the UE GUI shall indicate a menu for editing the participant lists via user action. This may include:

a. Adding, renaming and remove a list.
b. Adding, renaming, repositioning and removing of participants in a list.
c. Save an ad hoc participants list during ongoing ad hoc group communication call.

6.3.3.3 Specific message contents

Table 6.3.3.3-1: SIP INVITE (criteria based) from MCx_S1
(step 48, Table 6.3.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	INVITE SIP/2.0	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
MIME Multipart Media Encapsulation		Type: multipart/mixed		
application/sdp		Session description parameter set		
Application/vnd.3gpp.mcptt-info+xml	<mcptt-Params>			
<session-type>	Ad hoc	Type: ad hoc group communication		
<mcptt-request-uri>				
<mcptt-client-id>				
<any-ext>/<call-participants-criterias>	active	Criteria based		

Table 6.3.3.3-2: SIP INVITE (list based) from UE1
(step 7b, Table 6.3.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.5.2.5.1-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	INVITE SIP/2.0	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16]	
MIME Multipart Media Encapsulation		Type: multipart/mixed		
application/sdp		Session description parameter set		
application/vnd.3gpp.mcptt-info+xml	<mcptt-Params>			
<session-type>	Ad hoc	Type: ad hoc group communication		
<mcptt-request-uri>				
<mcptt-client-id>				
Application/resource-lists+xml	<resource-lists>			
<lists>	<entry>	List entries		

6.4 Ad hoc group communication security

6.4.1 Test Purpose (TP)

(1)

with {Ad hoc group communication security compliance}
ensure that {
    when {MCX Ad hoc group communication is initiated by a MCX service user}
    then {Ensure that the MCX Service user initiating the session using secure encryption mechanisms on the application layer for signalling **and** the encryption is end-to-end for all transmitted media between all involved MCX UE **and** traffic and MCX service supports information exchange for regulatory constraints to support confidentiality, message integrity, and source authentication for some information exchanges (e.g., MCX Service User Profile management, kill commands)}
}

(2)

with {Identities for MCX Service Users}
ensure that {
    when {MCX Service users are registered as identities on the MCX Service}
    then {Each MCX User shall register to a unique identity and the MCX Service system shall assign the MCX Service user to this globally unique identifier and the identifier is independent of the IMSI, provided by the 3GPP system and the unique identity shall use as basis for the MCX Service application security layer}
}

(3)

with {Mechanism to perform a single authentication}
ensure that {
when {MCX Service user request the authentication for a set of unique identifiers}
then {The MCX Service provides a mechanism for a single authentication procedure to the requesting MCX Service user to access all authorized features and with authentication the MCX Service system grants full access to all authorized features connected to the requested identity and authorized features can vary between the different requested identities}
    else when {MCX Service user not performing authentication procedure or not all required identities are authenticated by the MCX Service user}
    then {MCX Service user has no access or only limited access to the authorized features on the MCX Service system depending on the requested set of identities}
}

(4)

with {Limited feature access Prior to authentication}
ensure that {
    when {MCX Service system provides access for selected features for a MCX UE prior the MCX Service user authentication}
then {An authorized MCX UE shall have access to selected features prior the authentication of the MCX Service user}
Only predefined low risk features are accessible before the full authentication, and all emulate the access levels for unauthenticated users}
}

(5)

with {End-to-End security for media traffic and cryptographic key management}
ensure that {
    when {Media traffic is sent between MCX UEs}
then {The MCX Service system provides a management for cryptographic key services and ensures that the cryptographic keys are confidentiality protected, integrity protected and authenticated when delivered over air **and** media traffic remains end-to-end encrypted throughout transmission **and** security mechanisms persist regardless of network connectivity and secure direct peer to peer communication is established when out of network coverage and the cryptographic key management service support pre-provisioning and over-the air provisioning **and** the MCX Service provide end-to-end confidentiality and integrity protection to the MCX User Profile when transferred to and/or from and while stored on an MCX Server, an MCX UE or both}
}

6.4.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.5.2-009]]

"MCX Service Ad hoc Group Communications shall support the applicable security requirements as identified in clause 5.12."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-001]]

"The MCX Service shall provide a means to support the confidentiality and integrity of all user traffic and signalling at the application layer."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-002]]

"The MCX Service shall support MCX User with globally unique identities, independent of the mobile subscriber identity (IMSI) assigned by a 3GPP network operator to UEs."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-003]]

"The MCX Service identities shall be part of the MCX Service application service domain."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-004]]

"The MCX Service identities shall form the basis of the MCX Service application layer security for the MCX Service."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-005]]

"The MCX Service shall provide the MCX User with a mechanism to perform a single authentication for access to all authorised features."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-006]]

"The MCX Service shall provide a means for an authorised MCX UE to access selected MCX Service features prior to MCX User authentication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-007]]

"The MCX Service shall require authentication of the MCX User before service access to all authorised MCX Service features is granted.

NOTE: The MCX Service features available are based on the authenticated user identity(s)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-008]]

"Subject to regulatory constraints, the MCX Service shall provide a means to support confidentiality, message integrity, and source authentication for some information exchanges (e.g. MCX Service User Profile management, kill commands) that have the potential to disrupt the operation of the target MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-009]]

"The MCX Service shall provide a means to support end-to-end security for all media traffic transmitted between MCX UEs."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-010]]

"End-to-end security shall be supported both within and without network coverage and regardless of whether the traffic is transmitted directly or via the network infrastructure."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-011]]

"Subject to regulatory constraints, the MCX Service shall provide a cryptographic key management service(s)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-012]]

"The cryptographic key management service(s) shall support both pre-provisioning and over-the-air provisioning of cryptographic keys."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-013]]

"The cryptographic key management service(s) shall ensure that cryptographic keys are confidentiality protected, integrity protected and authenticated when delivered over-the-air."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.12-014]]

"The MCX Service shall provide end-to-end confidentiality and integrity protection to the MCX User Profile when transferred to and/or from and while stored on an MCX Server, an MCX UE or both."

6.4.3 Test description

6.4.3.1 Pre-test conditions

System Configuration:

- The system is configured as shown in Figure 4.1-4.
- Security Policy Database (SPD) is defined and all required values for Authentication Header (AH) and Encapsulating Payload (ESP). Pre Shared Keys (PSK) are agreed between UE (MCx client) and MCx Service system.
- For generation of the keys the Internet Key Exchange (IKE) shall be used. Keys and required passwords are specified.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The MCx_S1 is the acting user.

Preamble:

- The MCx_S1 responds to an ICMP request of the UE1 and vice versa to ensure that the IP layer is available.

6.4.3.2 Test procedure sequence

Table 6.4.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
1	UE1 initiate IPSec Phase 1 for providing policy proposal (start ISAKMP SA negotiation)	<--	UE sends ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [24]	-	--
2	Check: SIP core on MCx_S1 responds Identity Protection (Main Mode) request from UE with its own policy proposals (respond ISAKMP SA negotiation)	-->	SIP core responds ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [24]	1	P
3	UE1 starts key exchange procedure (Phase 2); UDP message format Identity Protection (Main Mode); init other SA negotiation	<--	UE sends ISAKMP message with proposal; Identity Protection (Main Mode) regarding IETF RFC 2408 [24]	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
4	Check: SIP core on MCx_S1 responds Identity Protection (Main Mode) request from UE (respond other SA negotiation)	-->	SIP core responds ISAKMP message with proposal; Identity Protection (Main Mode)	1	P
5	UE1 starts authentication request; UDP message format Identity Protection (Main Mode); Other (KE, ID, etc.)	<--	UE sends ISAKMP message with authentication request; Identity Protection (Quick Mode) regarding IETF RFC 2408 [24]	-	--
6	Check: SIP core on MCx_S1 confirms authentication request with message format Identity Protection (Main Mode)	-->	SIP core confirms ISAKMP message with own ID; Identity Protection (Quick Mode)	1	P
7	UE1 start tunnel protected by ESP or AH; establishment via Quick Mode	<--	UE sends ISAKMP message with tunnel protection mechanism; (Main Mode) regarding IETF RFC 2408 [24]	-	--
8	Check: SIP core on MCx_S1 confirms tunnel establishment request with message format (Quick Mode)	-->	SIP core sends ISAKMP message for tunnel establishment (Quick Mode)	1	P
9	Check: MCx_S1 starts TLS request via TLSv1.3 handshake; sending "Client Hello"	-->	MCx_S1 sends TLS handshake with Cipher Suite proposal according to IETF RFC 8446 [25] including supported ciphers, random number and session ID		P
10	UE1 responds with TLS message "Server Hello"	<--	Client sends TLS handshake "Server Hello" including certificate and "Hello Done" information including selected ciphers, random number and session ID		
11	Check: MCx_S1 initiates client key exchange procedure	-->	MCx_S1 sends request for "Cipher Key Exchange" with encrypted premaster secret		P
12	UE confirms cipher change	<--	UE1 confirms the cipher key exchange procedure with Cipher specification exchange finished.		

6.4.3.3 Specific message contents

Table 6.4.3.3-1: ISAKMP from MCx_S1
(step 2, Table 6.4.3.2-1; IETF RFC 2408 [24])

Derivation Path: IETF RFC 2408 [24]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**			
Payload				
Security Association (1)	IPSEC			
Proposal (2)	ISAKMP			
Transform (3)/ IKE Attribute	KEY_IKE			
Encryption-Algorithm	AES-CBC			
Key-Length	128			
Hash-Algorithm	SHA2-256			
Group-Description	1024 MODP			
Authentication-Method	Pre-shared key			
Life-Type	Seconds			
Life-Duration	86 400			

Table 6.4.3.3-2: ISAKMP from MCx_S1
(step 4, Table 6.2.3.2-1; IETF RFC 2408 [24])

Derivation Path: IETF RFC 2408 [24]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Payload				
Key Exchange (4)	IPSEC			
Key Exchange Data	<key>			
Payload				
Vendor ID (13)	Dead Peer Detection		IETF RFC 3706 [35]	
Vendor ID (13)	XAUTH			

Table 6.4.3.3-3: ISAKMP from MCx_S1
(step 6, Table 6.2.3.2-1; IETF 2408 [24])

Derivation Path: IETF RFC 2408 [24]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Payload				
Key Exchange (4)	IPSEC			
Key Exchange Data	<key>			
Payload				
Vendor ID (13)	Dead Peer Detection		IETF RFC 3706 [35]	
Vendor ID (13)	XAUTH			

Table 6.4.3.3-4: ISAKMP from MCx_S1
(step 8, Table 6.2.3.2-1; IETF RFC 2408 [24])

Derivation Path: IETF RFC 2408 [24]; condition ISAKMP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Security and Key Management Protocol	**Payload**	**Main Mode**		
Next Payload: Identification (5)	HASH Data			

Table 6.4.3.3-5: TLS from MCx_S1
(step 9, Table 6.2.3.2-1; IETF RFC 8446 [25])

Derivation Path: IETF RFC 8446 [25]; condition TLS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Secure Socket Layer	**Payload**			
TLSv1.3 Record Layer				
Content Type	Handshake (22)			
Handshake Protocol	Client Hello			
Handshake Type	Client Hello (1)			
Cipher Suites (No. of suites)	TLS_RSA_WITH_ <type>	Supported cipher suites		

Table 6.4.3.3-6: TLS from UE1
(step 11, Table 6.2.3.2-1; IETF RFC 8446 [25])

Derivation Path: IETF RFC 8446 [25]; condition TLS				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Secure Socket Layer	**Payload**			
TLSv1.3 Record Layer	**Handshake Protocol**			
Content Type	Handshake (22)			
Handshake Protocol	Client Key Exchange			
Handshake Type	Client Key Exchange (16)			
Diffie-Hellman Client Params	Pubkey	<key>		
TLSv1.3 Record Layer	**Change Cipher Specification Protocol**			
Content Type	Change Cipher Spec (20)			
Change Cipher Spec Message				
TLSv1.3 Record Layer	**Handshake Protocol**			
Content Type	Handshake (22)			
Handshake Protocol	Encrypted Handshake Message			

6.5 Modifying the participants list of on-going ad hoc group communication

6.5.1 Test Purpose (TP)

(1)

```
with {MCPTT user clients on different partner MCx Service systems meet the criteria to participate
an ad hoc group communication}
ensure that {
when {MCPTT client meets the criteria for adding to an ongoing ad hoc group communication}
then {MCPTT client will be successfully added to the ongoing ad hoc group communication
independently to which partner MCx Service system it belongs}
else when {MCPTT client is in ongoing ad hoc group communication and does not meet the criteria
successfully}
then {MCPTT client will be removed by the originating MCx Service system}
```

6.5.2 Conformance requirements

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.4]

"MCPTT server modifying the ad hoc group call participants procedure involving multiple MCPTT systems."

6.5.3 Test description

6.5.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.

- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).

- MCx Service system is configured to support the following features:

  - Ad-hoc Group Communication supporting criterion-based functionality; UE1 is allowed to change the list of participants (mandatory).

  - UE2 is not part of the initial list based ad hoc group communication.

IUT:

- IUT (MCx_S1).

- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.

- The MCx_S1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.

- The MCx Service system (MCx_S1) is connected to the IUT, UEa1 (MCx client with restrictions), UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).

6.5.3.2 Test procedure sequence

Table 6.5.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
1	UE1 publish the basic POC settings	<--	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	-	--
2	Check: MCx_S1 confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
3	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	<--	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	-	--
4	Check: MCx_S1 responds with a STATUS (note 1) message	-->	SIP STATUS (note 1) 200 OK	1	P
5	UE1 send PUBLISH message containing an automatic FA information	<--	SIP PUBLISH	-	--
6	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS (note 1) 200 OK	1	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
7	Check: UE1 is allowed to participate on ad hoc group communication based on criterion <allow-emergency-adhoc-group-call> in the <ruleset> of the client profile	--	- none – Client checks the ruleset configuration according to ETSI TS 124 379 [20] (3GPP TS 24.379), clause 6.2.8.1.24 is set to "true"	-	--
8	Check: UE1 initiates ad hoc group communication based (criterion: manually) {InitiateGroup, ParticipantList}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379), clause 17.2.2.1.1	-	--
9	System validates authorization of UE1.	-->	DB/AAA check → VALID: ACK {Group Established, GroupID} → ACK	1	P
10	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	1	P
11	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	1	P
12	Check: MCx_S1 invites UE-con into the ongoing ad hoc group communication	-->	SIP INVITE	1	P
13	UE-con responds to the INVITE request	<--	SIP STATUS 100 (TRYING)	-	--
14	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
15	UE-con call established confirmation towards MCx_S1	<--	SIP ACK	-	--
16	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
17	Floor activation by one participant to ensure the usage of the call time limit expiry. UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
18	UE1 add participant UE2 to the resource list (note 2). UE send re-INVITE	<--	SIP re-INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379), clause 17.6.2	-	--
19	Check: MCx_S1 sends ACK response to UE1	-->	SIP ACK	1	P
20	Check: MCx_S1 sends INVITE message to UE2	-->	SIP INVITE	1	P
21	UE2 responds to the INVITE request	<--	SIP STATUS 100 (TRYING)	-	--
22	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
23	UE-con call established confirmation towards MCx_S1	<--	SIP ACK	-	-
24	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
25	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
26	Check: UE1 remove participant UE2 from the resource list (note 2). UE send re-INVITE	-->	SIP re-INVITE	-	--
27	UE1 receive ACK response from MCx_S1	-->	SIP ACK	1	P
28	UE2 receive BYE message from MCx_S1	-->	SIP BYE	1	P
29	UE2 responds to the BYE request	<--	SIP STATUS 200 (OK)	-	--
30	{Group Established, GroupID} → ACK Session preparation phase. MCx_S1 sends NOTIFICATION	--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
31	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	--	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
32	Check: UE1 (initiator) terminates ad hoc group communication sending BYE	<--	SIP BYE	-	--
33	MCx_S1 confirms the received BYE message	-->	SIP STATUS 200 (OK)	1	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
34	Check: MCx_S1 terminates the ad hoc group communication; UE-con receive BYE message	-->	SIP BYE	1	P
35	UE-con confirms the received BYE message	<--	SIP STATUS 200 (OK)	-	--
NOTE 1: May notify the answer state to the user (i.e. "Unconfirmed" or "Confirmed") if received in the P-Answer-State header field. NOTE 2: The application on the UE GUI shall indicate a menu for adding or removing participants via user action.					

6.6 Ad hoc group emergency alert support involving one MC system

6.6.1 Test Purpose (TP)

(1)

```
with {Number of parallel allowed MCX Service ad hoc group emergency alerts}
ensure that {
    when {MCX Service administrator configures the number of parallel ad hoc group emergency alerts}
    then {The MCX service system supports the permission settings for MCX Service users
participating ad hoc emergency group alerts: (none or one or maximum number)}
            }
```

(2)

```
with {MCX Service ad hoc group emergency alert termination}
ensure that {
    when {MCX Service emergency ad hoc group communication or ad hoc group emergency alert is
terminated by authorized MCX Service user}
    then {In-progress Emergency condition for ad hoc group communication shall be cancelled or ad
hoc group emergency alert is terminated and if no other ad hoc group call is active the list of
participants is not persistent}
            }
```

(3)

```
with {MCX Service user can initiate an MCX Service ad hoc Emergency Group alert}
ensure that {
    when {MCX Service user is enabled for initiating an ad hoc group emergency alert}
    then {MCX UE send MCX Service ad hoc emergency alert and the responsible user change state to
MCX Service emergency and authorized MCX Service user can set the ad hoc group emergency alert into
an ad hoc group communication and MCX Service ad hoc group emergency alert is distributed to the
determined list of participants and MCX Service emergency alert shall contain location information,
MCX Service user ID, functional alias, criteria for determining list of participants, available MCX
Services, additional information related to the alert, and the user's Mission Critical organization
name}
    else when {MCX Service user is disabled for initiating an ad hoc group emergency alert}
    then {MCX user receive the reason for denial}
            }
```

(4)

```
with {MCX Service system shall provide the list of participants of an ad hoc group emergency alert}
ensure that {
    when {The MCX Service administrator or the MCX Service system changes the list of participants
of an ad hoc group emergency alert (e.g. on request or on location information provided by a MCX
Service user}
    then {The authorized MCX Service user requests the determined list of participants of an ad hoc
group emergency alert and the MCX Service system monitors the changes on the list (e.g. periodically
on timer or on specific trigger) and provide an update on any change and if necessary, the MCX
Service system performs retrials to involve changed list participants into the ad hoc group
emergency alert}
            }
```

6.6.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.4.2-007a]]

"Depending on meaningful elements of a functional alias the MCX Service shall be able to restrict or allow MCX Users to be involved in more than one MCX Service Emergency Group Communication or more than one MCX Service Ad hoc Group Emergency Alert at a time."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-8.8.8.5-001]]

"The In-progress Emergency condition for a MCX Service Ad hoc Group Communication shall be cancelled when the communication is terminated."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-001]]

"The MCX Service shall support an MCX Service Ad hoc Group Emergency Alert capability, which on initiation by an MCX User causes that MCX UE to send an MCX Service Ad hoc Group Emergency Alert and may put that MCX User into the MCX Service Emergency State."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-002]]

"The MCX Service shall provide a means for an authorised user to be able to activate the MCX Service Ad hoc Group Emergency Alert capability."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-002a]]

"The MCX Service shall provide the reason for denial to an MCX user who was not authorised to activate the MCX Service Ad hoc Group Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-003]]

"The MCX Service Emergency Alert shall contain the following information: Location information, MCX Service User ID, Functional Alias, Criteria for determining list of participants, available MCX Services, additional information related to the alert, and the user's Mission Critical Organization name."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-004]]

"The MCX Service shall provide a mechanism for the initiator of an MCX Service Ad hoc Group Emergency Alert to request that the list of participants is to be determined and updated by the MCX Service system using pre-defined criteria."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-004a]]

"The MCX Service shall provide a mechanism for an authorised user to update the pre-defined criteria during an on-going Ad hoc Group Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005]]

"The MCX Service Ad hoc Group Emergency Alert shall be distributed to the list of participants determined by the MCX Service system."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005a]]

"When the list of participants is determined and updated by the MCX Service system, the MCX Service shall provide notifications to the relevant participants and authorised users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005b]]

"When the list of participants is determined or updated by the MCX Service system, the MCX Service shall provide a mechanism that monitors and ensures that the participants list is applied for MCX Service Ad hoc Group Emergency Alert, performing retries when needed."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-006]]

"The MCX Service shall support MCX Service Ad hoc Group Emergency Alert cancellation by authorised MCX Users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-007]]

"The MCX Service shall provide a mechanism for an authorised user which is participant of an active MCX Service Ad hoc Group Emergency Alert to set up group communications using the ad hoc group."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-008]]

"When an MCX Service Ad hoc Group Emergency Alert is cancelled and ongoing calls in the ad hoc group are terminated the ad hoc group shall not persist."

[ETSI TS 123 280 [4] (3GPP TS 23.280), clause 10.10.3]

"The following sub clauses specify the procedures for ad hoc group emergency alert initiation, ad hoc group emergency alert cancel, entering an ongoing ad hoc group emergency alert and leaving an ad hoc group emergency alert that are utilised by the following MC services:

- *MCPTT*
- *MCVideo*
- *MCData"*

6.6.3 Test description

6.6.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad-hoc Group Emergency Alert supporting functionality.
- Supporting user UEa1, UEb1, UEc1, UE2 and UE-con.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The MCx_S1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system (MCx_S1) is connected to UE1, UEa1 (MCx client with restrictions), UEa1, UEb1, UEc1, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).
- UEa1 and UEb1 are not part of the initial ad hoc emergency alert but can be added.

6.6.3.2 Test procedure sequence

Table 6.6.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
	Check: configuration (as file or user settings entry) with multiple FA is available for the UE1. The configuration contains numerical and organizational arrangements of FA and flags for expiration timer, priority and location. MCx Service user accepts the configuration	--		-	--
1a	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via HTML web interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=0	-->	HTTPS link to configuration interface	1	P
2a	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via XML configuration file interface; ; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=0	-->	XML access (e.g. via SCP or SFTP connection)	1	P
1b	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via HTML web interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=3 (note 1)	-->	HTTPS link to configuration interface	1	P
2b	Check: MCPTT Administrator configures permissions for the number of parallel ad hoc group emergency alerts for UE1 on the MCx_S1; access via XML configuration file interface; maximum number of parallel ad hoc group emergency alerts set to "MaxAhGEA"=3 (note 1)	-->	XML access (e.g. via SCP or SFTP connection)	1	P
	EXCEPTION: Procedure 'MCX CO communication' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.4.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.2	-		-	--
3	Execute MCX Group Affiliation Status Change as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. Check: UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
4	Check: MCx_S1 confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
5	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	<--	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	-	--
6	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS 200 (OK)	1	P
7	UE1 send PUBLISH message containing an automatic FA information	<--	SIP PUBLISH	-	--

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
8	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS 200 (OK)	1	P
9	UE1 is allowed to participate on ad hoc group emergency alert based on criterion <allow-adhoc-emergency-alert> in the <ruleset> of the client profile	<--	- none – Client checks the ruleset configuration according to ETSI TS 124 379 [20] (3GPP TS 24.379), clause 6.2.8.1.24 is set to "true"	-	--
10	UE1 activates ad hoc group emergency alert (note 2) #1	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379), clause 17.2.2.1.1	-	--
11	Check: System validates authorization of UE1.	-	check → VALID: 200	1	P
12	Check: UE1 receives confirmation from MCx_S1	-->	SIP STATUS 200 (OK)	1	P
13	Check: For the initiator the ad hoc group emergency alert automatically is confirmed. Other participants need to confirm automatically or by manually action		- none -		
14a	UEa1 activates ad hoc group emergency alert (note 2) #2	<--	SIP MESSAGE according to ETSI TS 124 379 [20] 3GPP TS 24.379 clause 17.2.2.1.1	-	--
15a	System validates authorization of UEa1.	-	check → VALID: NOK; MaxAhGEA=0 (no parallel ad hoc group emergency alert)	-	--
16a	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 403 (FORBIDDEN)	1	P
14b	UEa1 activates ad hoc group emergency alert (note 2) #2	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
15b	Check: System validates authorization of UEa1.	-	check → VALID: 200	1	P
16b	Check: MCx_S1 sends confirmation to UEa1	-->	SIP STATUS 200 (OK)	1	P
17b	UEb1 activates ad hoc group emergency alert (note 2) #3	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	-
18b	Check: System validates authorization of UEb1.	-	check → VALID: 200	1	P
19b	Check: MCx_S1 sends confirmation to UEb1	-->	SIP STATUS 200 (OK)	1	P
17b	UEb1 activates ad hoc group emergency alert (note 2) #4	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
18b	Check: System validates authorization of UEc1.	-	check → VALID: NOK; MaxAhGEA=3 (no parallel ad hoc group emergency alert)	1	P
19b	Check: MCx_S1 sends confirmation to UEc1	-->	SIP STATUS 403 (FORBIDDEN)	1	P
20b	UE1 shows three different ad hoc group emergency alerts	<--	Visual check on the UE1 GUI	-	--
21	UE1 deactivates ad hoc group emergency alert (note 2) #1	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	-
22	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	2	P
23	Check: UE-con deactivates ad hoc group emergency alerts #2 and #3				
24	Check: MCx_S1 sends confirmation to UE-Coc for each, ad hoc group emergency alert #2 and #3	-->	SIP STATUS 200 (OK)	2	P

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
25	No list of participants is available on UE1 after ad hoc group emergency alert was terminated.	<--	Visual check on the UE1 GUI	-	--
26	UE-con activates ad hoc group emergency alert (note 2)	<--	SIP MESSAGE	-	--
27	Check: MCx_S1 sends confirmation to UE-con	-->	SIP STATUS 200 (OK)	2	P
28	Check: MCx_S1 send ad hoc group emergency alert activation to UE1	-->	SIP MESSAGE	2	P
29	UE1 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
30	UE1 confirm ad hoc group emergency alert activation (note 3)	<--	SIP MESSAGE	-	--
31	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	2	P
32	Check: MCx_S1 sends ad hoc group emergency alert activation to UE2	-->	SIP MESSAGE	2	P
33	UE2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
34	UE2 confirm ad hoc group emergency alert activation	<--	SIP MESSAGE	-	--
35	Check: MCx_S1 sends confirmation to UE2	-->	SIP STATUS 200 (OK)	2	P
36	UE-con deactivates the ad hoc group emergency alert for				

NOTE 1: MaxAhGEA=3 simulates the maximum value for parallel ad hoc emergency alerts.
NOTE 2: The application on the UE GUI shall indicate a menu for editing the participant lists via user action. This may include:
a. Adding, renaming and remove a list.
b. Adding, renaming, repositioning and removing of participants in a list.
c. Save an ad hoc participants list during ongoing ad hoc group communication call.
NOTE 3: The user GUI shall change the visualization between active, unconfirmed alert and confirmed alert e.g. like:
a. Unconfirmed: alert tone + blinking indication on main screen.
b. Confirmed: static information in the status field.

6.6.3.3 Specific message contents

Table 6.6.3.3-1: SIP MESSAGE from MCx_S1
(step 28, Table 6.6.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/mcptt-info+xml		Type: multipart/mixed		
<mcptt-request-uri>				
<mcptt-calling-user-id>				
<mcptt-calling-group-id>				
<any-ext>				
<adhoc-emergency-alert-ind>	true			
<location-info>	<report>			
<call-participants-criterias>	active			
<adhoc-alert-participant-list>	<entry>			
<adhoc-alert-participant-not-ack-list>	<entry>			

6.7 Configuration parameters required for Ad hoc group emergency alert support

6.7.1 Test Purpose (TP)

(1)

```
with {MCX Service administrator configuration options on ad hoc emergency group alert}
ensure that {
    when {MCX Service administrator has access to the MCX Service system}
    then {Enable or Disable a MCX Service user for specific ad hoc group emergency capabilities and
change the pre-defined criteria for an ad hoc group emergency alert and change the determined list
of participants and a notification sent to all relevant participants and authorized MCX Service
users}
            }
```

6.7.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-002]]

"The MCX Service shall provide a means for an authorised user to be able to activate the MCX Service Ad hoc Group Emergency Alert capability."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-004a]]

"The MCX Service shall provide a mechanism for an authorised user to update the pre-defined criteria during an on-going Ad hoc Group Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005a]]

"When the list of participants is determined and updated by the MCX Service system, the MCX Service shall provide notifications to the relevant participants and authorised users."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.6.2-005b]]

"When the list of participants is determined or updated by the MCX Service system, the MCX Service shall provide a mechanism that monitors and ensures that the participants list is applied for MCX Service Ad hoc Group Emergency Alert, performing retries when needed."

6.7.3 Test description

6.7.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system provides an interface for changing the configuration.

IUT:

- IUT (MCx_S1).

- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The MCx_S1 is the acting user.

Preamble:

- No UE required.
- The MCx Service system (MCx_S1) is connected to the IUT.

6.7.3.2 Test procedure sequence

Table 6.7.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
1	Precondition: MCX Service administrator is authenticated and has management access to the MCX Service system.	-->	Authentication procedure (HTTP/TLS or Web portal login) → 200 OK (Admin access granted).	1	P
2	Check: Administrator enables or disables a specific MCX Service User for ad hoc emergency group alert participation.	-->	Admin → Service system: Configuration Request {UserID, EmergencyAlert=Enable/Disable} → Service system → Admin: 200 OK {Confirmation}.	1	P
3	Check: Administrator modifies pre-defined criteria for triggering ad hoc emergency group alert (e.g. location, role, priority).	-->	Admin → Service system: Config Update {Criteria=Location/Role/Priority} → Service system: 200 OK {Criteria Updated}.	1	P
4	Check: Administrator updates the participant list for the ad hoc emergency group.	-->	Admin → Service system: ParticipantListUpdate {AddUser/RemoveUser, GroupID} → Service system: 200 OK {Updated List, VersionID}.	1	P
5	Check: MCX Service system automatically propagates the updated configuration to all authorized entities.	-->	Service system → Participants: NOTIFY {EmergencyGroup Updated, ParticipantsList, Criteria} Service system → Authorized users: NOTIFY {Emergency group configuration change event}.	1	P
6	Negative Test: Unauthorized administrator attempts to modify configuration.	-->	Admin → Service system: Config Request → Service system: 403 Forbidden {Insufficient Rights}.	1	P
7	Negative Test: MCX Service user without authorization attempts to trigger emergency alert after being disabled by administrator.	-->	User → Service system: Emergency Alert Request → Service system: 403 Forbidden {User Not Authorized}.	1	P

6.8 Interworking between MCX Service systems

6.8.1 Test Purpose (TP)

(1)

```
with {Establishment of an MCX Service Ad hoc Group Emergency Alert and MCX Ad hoc Group
Communication in interconnection case}
ensure that {
    when {MCX Service supports interconnection with multiple MC Systems. MCX Service user initiate
an Ad hoc Group Emergency Alert or an MCX Ad hoc Group Communication in interconnection context}
    then {MCX Service shall establish the Ad hoc Group Emergency Alert and Ad hoc Group
communication and ensure that participants connected to different MC Systems also be involved in the
Ad hoc group emergency alert and Ad hoc group communication}
            }
```

6.8.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.17.2-008]]

"The MCX Service shall enable to establish an MCX Service Ad hoc Group Emergency Alert and MCX Ad hoc Group Communication with Participants of multiple MC Systems, which are interconnected."

6.9 Ad hoc group call and ad hoc group emergency alert involving multiple MC systems

6.9.1 Test Purpose (TP)

(1)

```
with {Information provided by the initiator of an Ad hoc group communication}
ensure that {
    when {Initiation of an ad hoc group communication call or  ad hoc group emergency alert by a MCX
Service user}
    then {The MCX Service user provide participants list for an Ad hoc group communication call or
ad hoc group emergency alert on procedures for call setup and call release and MCPTT server can
determine the participants list during Ad hoc group call setup and MCPTT server or an authorised
user can modify the participants list and authorized user can modify the ad hoc group communication
criteria}
            }
```

(2)

```
with {Releasing an Ad hoc group call by partner MCPTT system}
ensure that {
    when {The ad hoc group communication is active with participants involving different MCPTT
systems and the ad hoc group call shall be released by partner MCPTT server}
    then {Ad hoc group communication shall be terminated and the partner MCPTT system terminates the
designation of the participant, and a termination notification is received from all active
participants and MCPTT server determines the participants list}
```

6.9.2 Conformance requirements

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.1]

"Procedure for ad hoc group call setup - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.1]

"Procedure for ad hoc group call setup - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.2]

"Procedure for ad hoc group call release by MCPTT server - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.1]

"Procedure for ad hoc group call setup - Participants list provided by the Initiator."

[**ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.3]**

"Ad hoc group call setup - Participants list determined by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.1]

"Procedure for ad hoc group call setup - Participants list provided by the Initiator."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.4]

"Modification of ad hoc group call participants by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.5]

"Release ad hoc group call and stop determining the ad hoc group call participants by partner MCPTT system - Participants list determined by the MCPTT server."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.6]

"Modification of ad hoc group call participants by an authorised user."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 10.19.3.2.7]

"Modification of ad hoc group call criteria by an authorised user."

6.9.3 Test description

6.9.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Ad hoc Group Emergency Alert supporting functionality.
  - Ad hoc group communication call supporting functionality.
  - MCx Service system 1 (MCx_S1) and MCx Service system 2 (MCx_S2) are connected according to Figure 4.1-2.
- Supporting UE2, UE3 and UE-con:
  - UE2 is connected to MCx_S1.
  - UE3 is connected to MCx_S2 with permission to cancel or terminate ad hoc group emergency alerts or ad hoc group communication calls. UE3 can request list of participants.

IUT:

- IUT (MCx_S1, MCx_S2).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT has the permission to cancel or terminate ad hoc group communication requests.
- MCx_S1 is the acting user in Table 6.9.3.2-1.
- MCx_S1 is the acting user in Table 6.9.3.2-2.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the UE1, UE2 (MCx client) and UE-Con (MCx client with controller/dispatcher functionality).

- The MCx Service system 2 (MCx_S2) is connected to UE3.

6.9.3.2 Test procedure sequence

Table 6.9.3.2-1: Main behaviour for MCx_S1

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	<--		-	--
	No FA is active at the MCx Service system. UE1 publish the basic POC settings	-->	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	1	P
1	Check: MCx_S1 confirm the POC settings	<--	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
2	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	-->	SIP SUBSCRIBE according ETSI TS 124 229 [9] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	1	P
3	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS200 OK	-	--
4	UE1 send PUBLISH message containing an automatic FA information	-->	SIP PUBLISH	1	P
5	Check: MCx_S1 responds with a STATUS message	<--	SIP STATUS 200 OK	-	--
6	UE1 activates ad hoc group emergency alert involving UE2, UE3 and UE-con as list-based group initiation	--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
7	System validates authorization of UE1.	-->	check → VALID: 200	1	P
8	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	-	--
9	UE-con receives ad hoc group emergency alert activation from MCx_S1	<--	SIP MESSAGE	-	--
10	UE-con confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	1	P
11	Check: MCx_S1 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S2, MCx_S2 provide MESSAGE to UE3	-->	SIP MESSAGE	-	--
12	Check: MCx_S2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
13	Check: UE3 receives ad hoc group emergency alert activation from MCx_S1 via MCx_S2	<--	SIP MESSAGE	-	--
14	UE3 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)		SIP STATUS 200 (OK)		P
16	Check: UE3 change list of participants and includes UE2 into the ad hoc group emergency alert	-->	SIP MESSAGE	-	--
17	Check: MCx_S2 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	1	P
18	Check: MCx_S1 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	1	P
19	Check: MCx_S1 sends ad hoc group emergency alert activation to UE2	<--	SIP MESSAGE	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
20	UE2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
21	UE1 confirm ad hoc group emergency alert activation	-->	SIP MESSAGE	1	P
22	Check: MCx_S1 sends confirmation to UE1	<--	SIP STATUS 200 (OK)	-	--
23	UE1 deactivates ad hoc group emergency alert	-->	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	1	P
24	Check: MCx_S1 sends confirmation to UE1 MCx_S1	-->	SIP STATUS 200 (OK)	-	--
25	Check: MCx_S2 provides SIP MESSAGE for deactivating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	1	P
26	Check: MCx_S1 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
27	Check: UE3 receives ad hoc group emergency alert deactivation from MCx_S1 via MCx_S2	<--	SIP MESSAGE	-	--
28	UE3 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	--	SIP STATUS 200 (OK)	-	--
29	Check: all involved participants have changed back to idle mode				
30	Check: UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup, ParticipantList};	-->	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	2	P
31	System validates authorization of UE1	-->	DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	2	P
32	Check: MCx_S1 sends confirmation to UE1	<--	SIP ACK	-	--
33	MCx_S1 invites UE-con into the ongoing ad hoc group communication	<--	SIP INVITE	-	--
34 35	UE-con responds to the INVITE request	<--	SIP STATUS 100 (TRYING)	-	--
36	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
37	Check: UE-con call established confirmation towards MCx_S1	-->	SIP ACK	2	P
38	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
39	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	-->	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	2	P
40	Check: MCx_S1 forwards the INVITE for the ad hoc group communication to MCx_S2	-->	SIP INVITE	-	--
41	Check: MCx_S2 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	-	--
42	Check: UE3 receive INVITE from MCx_S2 for the ongoing ad hoc group communication	<--	SIP INVITE	-	--
43	UE3 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	-	--
44	Check: MCx_S2 updates the INVITE request from MCx_S1	<--	SIP STATUS 200 (OK)	-	--
45	OPTIONAL: UE3 sends outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
46	UE3 call established confirmation towards MCx_S2	-->	SIP STATUS 200 (OK) {Group Established, GroupID} → ACK	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
47	UE3 call established confirmation towards MCx_S2	-->	SIP ACK	2	P
48	Check: Session preparation phase. MCx_S2 sends NOTIFICATION to MCx_S1 and UE3	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
49	Check: Session preparation phase. MCx_S2 sends NOTIFICATION	<--	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
50a	UE3 terminates ad hoc group communication, sending BYE	-->	SIP BYE	-	--
51a	Check: MCx_S2 send termination request confirmation to UE3	-->	SIP STATUS 200 (OK)	-	--
52a	Check: Session termination phase. MCx_S2 sends BYE to MCx_S1	-->	SIP BYE	2	P
53a	Check: MCx_S1 confirms the receipt of the termination request to MCx_S2	-->	SIP STATUS 200 (OK)	2	P
54a	Check: MCx_S1 sends BYE to UE1 and UE-con	<--	SIP BYE	-	--
55a	UE1 and UE-con confirm the receipt of the termination request to MCx_S1	<--	SIP STATUS 200 (OK)	-	--
50b	UE1 terminates ad hoc group communication, sending BYE	-->	SIP BYE	2	P
51b	Check: MCx_S1 sends termination request confirmation to UE1	-->	SIP STATUS 200 (OK)	2	P
52b	Session termination phase. MCx_S1 sends BYE to MCx_S2	-->	SIP BYE	-	--
53b	Check: MCx_S2 confirms the receipt of the termination request to MCx_S1	-->	SIP STATUS 200 (OK)	-	--
54b	Check: MCx_S2 sends BYE to UE3	-->	SIP BYE	2	P
55b	Check: MCx_S1 sends BYE to UE-con	<--	SIP BYE	-	--
56b	Check: UE3 and UE-con confirm the receipt of the termination request	<--	SIP STATUS 200 (OK)	-	--

Table 6.9.3.2-2: Main behaviour for MCx_S2

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
	No FA is active at the MCx Service system. UE1 publish the basic POC settings	<--	SIP PUBLISH according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3903 [17]	-	--
1	Check: MCx_S1 confirm the POC settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	-	--
2	UE1 announce interest for a specific FA to the MCx_S1 (Interrogation, entrance to a specific list); UE1 is not allowed to use interrogation	<--	SIP SUBSCRIBE according ETSI TS 124 229 [8] (3GPP TS 24.229), IETF RFC 3856 [9] and IETF RFC 6665 [10]	-	--
3	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS200 OK	-	--
4	UE1 send PUBLISH message containing an automatic FA information	<--	SIP PUBLISH	-	--
5	Check: MCx_S1 responds with a STATUS message	-->	SIP STATUS 200 OK	-	--
6	UE1 activates ad hoc group emergency alert involving UE2, UE3 and UE-con as list-based group initiation	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
7	System validates authorization of UE1	--	check → VALID: 200	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
8	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	-	--
9	UE-con receives ad hoc group emergency alert activation from MCx_S1	-->	SIP MESSAGE	-	--
10	UE-con confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
11	Check: MCx_S1 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S2, MCx_S2 provide MESSAGE to UE3	-->	SIP MESSAGE	-	--
12	Check: MCx_S2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	1	P
13	Check: UE3 receives ad hoc group emergency alert activation from MCx_S1 via MCx_S2	-->	SIP MESSAGE	1	P
14	UE3 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
16	Check: UE3 change list of participants and includes UE2 into the ad hoc group emergency alert	<--	SIP MESSAGE	-	--
17	Check: MCx_S2 provides SIP MESSAGE for activating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	1	P
18	Check: MCx_S1 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
19	Check: MCx_S1 sends ad hoc group emergency alert activation to UE2	-->	SIP MESSAGE	-	--
20	UE2 confirms the receipt of the SIP MESSAGE for activation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
21	UE1 confirm ad hoc group emergency alert activation	<--	SIP MESSAGE	-	--
22	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	-	--
23	UE1 deactivates ad hoc group emergency alert	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
24	Check: MCx_S1 sends confirmation to UE1 MCx_S1	-->	SIP STATUS 200 (OK)	-	--
25	Check: MCx_S2 provides SIP MESSAGE for deactivating ad hoc group emergency alert to MCx_S1, MCx_S1 provide MESSAGE to UE2	-->	SIP MESSAGE	1	P
26	Check: MCx_S1 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	-->	SIP STATUS 200 (OK)	-	--
27	Check: UE3 receives ad hoc group emergency alert deactivation from MCx_S1 via MCx_S2	-->	SIP MESSAGE	1	(P)
28	UE3 confirms the receipt of the SIP MESSAGE for deactivation of the ad hoc group emergency alert (no alert confirmation)	<--	SIP STATUS 200 (OK)	-	--
29	Check: all involved participants have changed back to idle mode	--		-	--
30	Check: UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup, ParticipantList}	<--	SIP INVITE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	--
31	System validates authorization of UE1	-->	DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
32	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	-	--
33	MCx_S1 invites UE-con into the ongoing ad hoc group communication	<--	SIP INVITE	-	--
34 35	UE-con responds to the INVITE request (note 2)	<--	SIP STATUS 100 (TRYING)	-	--
36	OPTIONAL: outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
37	Check: UE-con call established confirmation towards MCx_S1	<--	SIP ACK	-	--
38	Check: Session preparation phase. MCx_S1 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
39	Floor activation by one participant to ensure the usage of the call time limit expiry. Check: UE1 send floor REQUEST message	-	RTCP (MCPT) Floor REQUEST; message containing field {Floor priority=0}	-	--
40	Check: MCx_S1 forwards the INVITE for the ad hoc group communication to MCx_S2	-->	SIP INVITE	-	--
41	Check: MCx_S2 responds to the INVITE request	-->	SIP STATUS 100 (TRYING)	2	P
42	Check: UE3 receive INVITE from MCx_S2 for the ongoing ad hoc group communication	-->	SIP INVITE	2	P
43	UE3 responds to the INVITE request	<--	SIP STATUS 100 (TRYING)	-	--
44	Check: MCx_S2 updates the INVITE request from MCx_S1	-->	SIP STATUS 200 (OK)	2	P
45	OPTIONAL: UE3 sends outgoing response to TRYING if no automatic answering mode is activated	<--	SIP STATUS 180 (RINGING)	-	--
46	UE3 call established confirmation towards MCx_S2	<--	SIP STATUS 200 (OK) {Group Established, GroupID} → ACK	-	--
47	UE3 call established confirmation towards MCx_S2	-->	SIP ACK	-	--
48	Check: Session preparation phase. MCx_S2 sends NOTIFICATION to MCx_S1 and UE3	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	-	--
49	Check: Session preparation phase. MCx_S2 sends NOTIFICATION	-->	RTCP (MPTC) Group Dynamic NOTIFY {InitStatus, QueuePolicy, HangTime, Mode} → All UEs	2	P
50a	UE3 terminates ad hoc group communication, sending BYE	<--	SIP BYE	-	--
51a	Check: MCx_S2 send termination request confirmation to UE3	-->	SIP STATUS 200 (OK)	2	P
52a	Check: Session termination phase. MCx_S2 sends BYE to MCx_S1	-->	SIP BYE	2	P
53a	Check: MCx_S1 confirms the receipt of the termination request to MCx_S2	-->	SIP STATUS 200 (OK)	-	--
54a	Check: MCx_S1 sends BYE to UE1 and UE-con	-->	SIP BYE	-	--
55a	UE1 and UE-con confirm the receipt of the termination request to MCx_S1	<--	SIP STATUS 200 (OK)	-	--
50b	UE1 terminates ad hoc group communication, sending BYE	<--	SIP BYE	-	--
51b	Check: MCx_S1 sends termination request confirmation to UE1	-->	SIP STATUS 200 (OK)	-	--
52b	Session termination phase. MCx_S1 sends BYE to MCx_S2	-->	SIP BYE	-	--
53b	Check: MCx_S2 confirms the receipt of the termination request to MCx_S1	-->	SIP STATUS 200 (OK)	2	P
54b	Check: MCx_S2 sends BYE to UE3	-->	SIP BYE	2	P
55b	Check: MCx_S1 sends BYE to UE-con	-->	SIP BYE	-	--
56b	Check: UE3 and UE-con confirm the receipt of the termination request	<--	SIP STATUS 200 (OK)	-	--

6.10 MCData subservices SDS and FD support involving one MC system

6.10.1 Test Purpose (TP)

(1)

```
with {MCX Service system supporting MCData functionality SDS}
ensure that {
    when {SDS is initiated by an MCX Service user}
    then {The MCX Service user can enter the content of the message and size for the message is
limited to a content payload of at least [1000] characters of 8 bit text or [500] characters of 16
bit text or [250] characters of 32 bit text and the necessary character encoding information or at
least [1000] characters of hyperlink or interleaved text and hyperlink(s) to allow subsequent access
to linked content (which may be a large file) or at least [1000] bytes of binary data to be used by
a local running application and the necessary addressing detail to identify the intended application
and SDS is provided to a MCX service user with policy assertion capabilities (e.g. policy to limit
certain types of message or content to certain users due, for example, to location or user
privilege)}
}
```

(2)

```
with {MCX Service system supporting MCData functionality SDS to a group of service affiliated
members}
ensure that {
    when {SDS is initiated by an MCX Service user to a group}
    then {SDS is provided to a group service to affiliated members with policy assertion
capabilities (e.g. certain types of messages or content may only be relevant to certain members of a
group due, for example, to location)}
}
```

(3)

```
with {MCX Service system supporting MCData message thread indication functionality}
ensure that {
    when {MCData SDS functionality}
    then {The MCX Service user provides a message thread indication function so that multiple
message flows can be managed independently and on reply to a message or any message coupled with a
previous message (sent or received) the SDS message thread indication function use the same
indication mechanism as for previous messages}
}
```

(4)

```
with {Additionally MCData SDS functionality configurable for the sender of an SDS}
ensure that {
    when {MCData SDS is configured for read confirmation active}
    then {The MCX Service user initiating a SDS can select a read confirmation tag and the
activation includes automatically the presentation of the delivery status for the specific SDS add a
delivery history can requested by an authorized user and the MCX Service user initiating a SDS
containing no information or payload according (1) can select a location tag and system add the
current location of the MCX UE and send the SDS}
else when {MCData SDS is configured for read confirmation not active}
then {The MCX Service user initiating a SDS is not able to select a read confirmation tag and the
presentation of the delivery status for the specific SDS will not be delivered and the MCX Service
user initiating a SDS containing no information or payload according (1) can select a location tag
and system add the current location of the MCX UE and send the SDS}
}
```

(5)

```
with {MCX Service system provides file distribution functionality}
ensure that {
    when {MCX Service user distributes files to the MCData System}
    then {MCData file distribution capability provide a service to allow a user to send a file to
any combination of individual users and/or affiliated groups and it provides an option for each
recipient to reject or choose to receive the file or not (e.g. by storing the file and sending a
link (URL) to all relevant members and the user can choose when to receive the file}
}
```

(6)

```
with {MCX Service system provides file distribution additionally functionality}
ensure that {
    when {MCX Service user distributes files to the MCData System}
    then {MCData file distribution capability provide a service to allow a user to send a file to
any combination of individual users and/or affiliated groups and it provides an option for each
recipient to reject or choose to receive the file or not (e.g. by storing the file and sending a
link (URL) to all relevant members and the user can choose when to receive the file and the sending
MCX Service user is able to indicate the file as immediately to send or mandatory for download and a
system level capability on the receiving UE invoke the download automatically by need or by
selection of a user and on download complete the MCData file distribution capability provides an
indication to each recipient successfully downloading the file and on availability of system
delivery efficiencies the MCData file distribution capability is able to distribute common
information to users within a specific geographical area at the same time and the user (sender) or
an authorized user is able to cancel the distribution of files which waiting to be sent or have been
sent but not have been delivered}
}
```

6.10.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-001]]

"The MCData Service shall provide an SDS feature for conveyance of limited size, variable content, messages."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-002]]

"The MCData SDS shall provide a group service to affiliated members with policy assertion capabilities (e.g. certain types of messages or content may only be relevant to certain members of a group due, for example, to location)."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-003]]

"The MCData SDS shall provide a one-to-one service with policy assertion capabilities (e.g. policy to limit certain types of messages or content to certain users due, for example, to location or user privilege)."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-004]]

"The MCData SDS shall provide the option to include a content payload of at least [1 000] characters of 8-bit text or [500] characters of 16-bit text or [250] characte32-bit text and the necessary character encoding information (for example to identify alphabet used)."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-005]]

"The MCData SDS shall provide the option to include a content payload of at least [1 000] characters of hyperlink or interleaved text and hyperlink(s) to allow subsequent access to linked content (which may be a large file)."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-006]]

"The MCData SDS shall provide the option to include a content payload of at least [1 000] bytes of binary data to be used by a local running application and the necessary addressing detail to identify the intended application."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-007]]

"The MCData SDS shall provide a message thread indication so that multiple message flows can be managed independently."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-008]]

"When replying to a message on the MCData SDS or sending any message which should be coupled with previously sent or received messages or message flows; the message thread indication shall use the same indication as was used for those previous messages."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-009]]

"The MCData SDS shall provide a selectable read receipt indication. When requested, the receiving entity shall provide receipt indication for delivered and read messages as appropriate.

NOTE 1: The read indication will implicitly indicate delivered as well."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-010]]

"The MCData SDS shall provide a configurable read receipt indication. When configured, the receiving entity shall provide receipt indication addressed to the application for delivered and read messages as appropriate.

NOTE 2: The read indication will implicitly indicate delivered as well."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-011]]

"The MCData SDS shall permit delivery history interrogation for suitably authorised users."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-012]]

"The MCData SDS shall provide the option to add a field indicating location of the sending user/UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.2.2-013]]

"The MCData SDS shall allow empty messages including only a field indicating location of the sending user/UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-001]]

"The MCData Service shall provide a file distribution capability."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-002]]

"The MCData file distribution capability shall provide a service to allow a user to send a file to any combination of individual users and/or affiliated groups."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-003]]

"The MCData file distribution capability shall provide an option for each recipient to choose to receive the file or not (e.g. by storing the file and sending a link (URL) to all relevant members).

NOTE 3: By using this capability, a user can choose when to receive the file."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-004]]

"The MCData file distribution capability shall allow a user to reject to receive the file where appropriate."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-005]]

"The MCData file distribution capability shall provide a sending user selectable indication for mandatory download so that the UE, for all relevant receiving members, will automatically download the file.

NOTE 4: The mandatory indication provides a system level capability and could be invoked automatically by need of the application or might be provided for selection by the user.

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-006]]

"The MCData file distribution capability shall provide download complete indications for each recipient successfully downloading the file."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-007]]

"The MCData file distribution capability shall allow the sender to select to send the file immediately to all chosen users."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-008]]

"The MCData file distribution capability shall make use of available system delivery efficiencies for distribution of common information to users within a specific geographic area and able to receive at the same time."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-009]]

"The MCData file distribution capability shall allow a user to cancel distribution of files they have sent but have not been delivered."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.3.2-010]]

"The MCData file distribution capability shall allow an authorised user to cancel distribution of files being sent or waiting to be sent."

6.10.3 Test description

6.10.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - MCData services SDS and File Distribution.
- Supporting MCx Service user UE2 and UE-con:
  - UE2 is connected to MCx_S1.
  - UE-con is able to receive and display SDS and FD.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports MCData client functionality.
- MCx_S1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT (MCData client) and UE-Con (MCData client with controller/dispatcher functionality).

6.10.3.2 Test procedure sequence

Table 6.10.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
1	No FA is active at the MCx Service system. Check: UE1 publish the basic settings	<--	SIP PUBLISH according ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), IETF RFC 3903 [17]	-	--
2	Check: MCx_S1 confirm the basic settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
4a	UE1 provides a user GUI for entering a message in 8 bit character format (note 1)	<--	The input of 1 000 characters is possible	-	--
4b	UE1 provides a user GUI for entering a message in 16 bit character format (note 1)	<--	The input of 500 characters is possible	-	--
4c	UE1 provides a user GUI for entering a message in 32 bit character format (note 1)	<--	The input of 250 characters is possible	-	--
4d	UE1 provides a user GUI for adding a hyperlink to connected content (note 1)	<--	The input of the hyperlink is possible.	-	--
4e	UE1 provides a user GUI for providing abstract binary data (note 2)	<--	The input of the abstract binary data up to 1000 bytes is possible.	-	--
5a1	UE1 (MCData client) send a SIP MESSAGE request with the content generated in test steps 4a, 4b, 4c, 4d and 4e to a unique user UE-con	<--	SIP MESSAGE according to ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), clause 5.3C.1.3	-	--
5a2	Check: MCx_S1 sends (Accepted) response to UE1	-->	SIP STATUS 202 (ACCEPTED)	1	P
5a3	MCx_S1 checks the policy rules for the targeting user	-->	Services authorized	1	P
5a4	Check: MCx_S1 sends SDS to UE-con; UE-con receive the SDS message	-->	SIP MESSAGE	1	P
5a5	UE-con confirms reception with (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
5b1	Check: UE1 (MCData client) send a SIP MESSAGE request with the content generated in test steps 4a, 4b, 4c, 4d and 4e to a user group	<--	SIP MESSAGE according to ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), clause 5.3C.1.3	-	--
5b2	Check: MCx_S1 sends (Accepted) response to UE1	-->	SIP STATUS 202 (ACCEPTED)	2	P
5b3	Check: MCx_S1 checks the policy rules for the targeting user group	-->	Services authorized	2	P
6	UE1 provides an option for activating a read confirmation (note 4)	<--	Read confirmation can be activated on the GUI	-	--
7	UE1 (MCData client) send a SIP MESSAGE with SDS NOTIFICATION request to a unique user UE-con	<--	SIP MESSAGE	-	--
8	Check: MCx_S1 sends (Accepted) response to UE1	-->	SIP STATUS 202 (ACCEPTED)	4	P
9	Check: MCx_S1 sends SDS to UE-con; UE-con receive the SDS message	-->	SIP MESSAGE	4	P
10	UE-con confirms reception with (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
11	SDS read confirmation: UE-con read the SDS and transmit the confirmation reply	<--	SIP MESSAGE	-	--
12	UE1 confirms reception with (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
13	Repeat test steps 7, 8, 9, 10 and 11 with UE-con as acting user (UE1 receives SDS with read confirmation request)			-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
14	UE-con confirms reception with (Accepted) response	<--	SIP STATUS 202 (ACCEPTED)	-	--
15	Check: MCx_S1 provides an option for activating a read confirmation (note 4)	-->	Read confirmation can be activated on the GUI	4	P
16	EXCEPTION: MCx_S1 supports FD according to the procedure described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.3C.10.3			5	P

NOTE 1: The GUI shall provide an information for the user of characters left based on the configuration setting for the format (8 bit, 16 bit or 32 bit).
NOTE 2: Local application provides an option for providing binary information (e.g. abstract control data).
NOTE 3: The GUI shall provide functions for organizing and filtering to optimize the presentation of the message threads on the GUI.
NOTE 4: GUI shall provide the following options regarding to the creation of an SDS or FD:
a. [SDS] Boolean option active/inactive for read confirmation.
b. [FD] Create an URL instead of adding a file on upload.
c. [FD] Provide an option for the user to receive the file or not.

6.10.3.3 Specific message contents

Table 6.10.3.3-1: SIP MESSAGE from the MCx_S1
(Table 6.10.3.2-1; step 5a1, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3), Table 5.5.2.7.1-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.7.1; SIP MESSAGE				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-body			IETF RFC 3261 [16] IETF RFC 3909 [17]	
MIME -part-body	MCData-Info as described in Table 6.7.3.3-2			
Application/				
vnd3.3gpp.mcdata-info+xml	<mcdata-Params>	one-to-one-sds group-sds		
resource-list+xml	<list>			
vnd3.3gpp.mcdata-signalling				
vnd3.3gpp.mcdata-payload		Payload		

Table 6.10.3.3-2: SIP MESSAGE from the MCx_S1
(Table 6.10.3.2-1; step 15 ETSI TS 124 282 [33] (3GPP TS 24.282), Table 15.1.5.1-1)

Derivation Path: ETSI TS 124 282 [33] (3GPP TS 24.282) clause 15.1.5.-1; condition SDS NOTIFICATION message content

6.11 MCData subservices SDS and FD support involving multiple MC system

6.11.1 Test Purpose (TP)

(1)

```
with {MCData service system supporting Short Data Service (SDS) and File Distribution (FD)
subservices across multiple MC systems}
ensure that {
when {MCData supports SDS or FD message or file is exchanged between users in different MC systems}
then {MCx Service system correctly handles the intra- and inter-system signalling, routes data and
files, performs necessary security and authorization checks, and delivers acknowledgments or failure
reports to the originator and MCx Service system use specific request and response messages in
respect to the specified message and file semantics during the exchange}
}
```

(2)

```
with {MCData supporting exchange between different users}
ensure that {
    when {MCData service users are connected to the same or to different MCData Service systems and
MCData users exchange SDS or FD}
    then {MCData services (SDS and FD) are distributed through the different MCData Service systems
including a primary MC system and file transfer mechanisms as initiation and segmentation and
completion are supported and MCData Service system implements control mechanisms to ensure
confidentiality, integrity and authentication by applying the MC security architecture}
}
```

(3)

```
with {MCData supporting SDS exchange between different users and ongoing MCData distribution}
ensure that {
    when {SDS messages are distributed using MSRP}
    then {MCData Service system uses inter-working signalling functions including routing,
forwarding and delivery confirmation and it provides additional content information for
acknowledgement and delivery reports}
}
```

(4)

```
with {MCData supporting FD exchange between different users and ongoing MCData distribution}
ensure that {
    when {FD transfer negotiated the media plane using MSRP}
    then {MCData Service system use inter-working signalling functions including session setup, file
metadata exchange and reporting of success or failure transmission and it uses mechanisms for
partial transfer, error handling and meta data exchange}
}
```

(5)

```
with {Mechanisms for security and trust}
ensure that {
    when {SDS exchange or FD transfer}
    then {MCData Service system use mechanisms to ensure that partner MC Service systems are
confidentiality, integrity and maintained for using specified key management and protection schemes}
}
```

(6)

```
with {different domains or home servers}
ensure that {
    when {SDS exchange or FD transfer}
    then {MCx Service system acting as originating system support SDS exchange and FD to recipients
homed in a different domain or home server}
}
```

6.11.2 Conformance requirements

[ETSI TS 123 282 [36] (3GPP TS 23.282), clause 8.7.2.3]

"The MCData service shall support sending and receiving SDS messages between MCData users belonging to different MC systems, using the inter-MC system functional model defined in clause 8.7.2.3."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clause 4.2.2.6]

"The MCData service shall support the exchange of Short Data Service (SDS) and File Distribution (FD) messages and files between users in different MC systems, including a primary MC system and one or more partner MC systems."

[ETSI TS 123 282 [36] (3GPP TS 23.282), clause 8.7.2.4]

"The MCData service shall support File Distribution (FD) across multiple MC systems, ensuring that the file transfer procedures (initiation, segmentation, completion) follow the requirements of clause 8.7.2.4."

[ETSI TS 124 582 [21] (3GPP TS 24.582), clauses 6.3 and 7.3]

"MCData SDS message exchange across MC system boundaries shall use the media plane control protocols specified in TS 24.582 (MSRP for SDS content transport), and shall support acknowledgements / delivery reports as defined."

[ETSI TS 124 582 [21] (3GPP TS 24.582), clauses 7.4 and 8.4]

"MCData FD transfer between MC systems shall use the negotiated media plane (MSRP or equivalent) with support for partial transfers, error handling, and metadata exchange, as specified in TS 24.582."

[ETSI TS 129 582 [22] (3GPP TS 29.582), clause 5.2]

"The MCData service shall provide inter-working signalling functions for SDS messages across MC systems, including routing, forwarding, and delivery confirmation via the MCData server-to-server interface."

[ETSI TS 129 582 [22] (3GPP TS 29.582), clause 5.3]

"The MCData service shall provide inter-working signalling functions for FD across MC systems, including session setup, file metadata exchange, and reporting of success or failure."

[ETSI TS 133 180 [15] (3GPP TS 33.180), clause 6.6]

"The MCData service shall ensure confidentiality, integrity, and authentication of SDS and FD messages across MC systems by applying the mission critical security architecture defined in TS 33.180."

[ETSI TS 123 379 [5] (3GPP TS 23.379), clauses 10.19.2.20 and 10.19.2.21]

"The MCData service shall support the use of specific request and response messages (e.g. MCData-FD-Request/Response, MCData-SDS-Request/Response) for both intra-system and inter-system communications."

[ETSI TS 124 582 [21] (3GPP TS 24.582), clause 10]

"The MCData service shall enforce security and trust requirements, including ensuring that partner MC systems are trusted and that confidentiality and integrity are maintained for SDS and FD messages/files as they cross system boundaries using the specified key management and protection schemes."

[ETSI TS 129 582 [22] (3GPP TS 29.582), clauses 9.2 and 10]

"The system shall support MCData SDS messages and FD files between users in different domains or home servers, allowing an originator in one system to send to recipients homed in another."

6.12 Configuration parameters required for MCData subservices SDS and FD support

6.12.1 Test Purpose (TP)

(1)

```
with {MCData service system supporting variable configuration parameter for Short Data Service (SDS)
and File Distribution (FD) subservices}
ensure that {
when {MCx Service systems provides a configuration interface}
then {Settings for SDS and FD regarding group communication and broadcast communication and FA
management and recording can be changed}
}
```

6.12.2 Conformance requirements

5.1.2 Group configuration

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.1.3-002]] (server side; off network)

"The MCX Service shall enable a properly provisioned and authorised MCX UE operating on the network to receive its application layer level parameters (e.g. MCX Service Group ID, group keys) necessary for initiating and participating in Selected MCX Service Group and Private Communications at a future time, while off the network.

NOTE: This is a "run-time" requirement applicable to an already configured MCX UE, when MCX Service Groups and/or MCX Users, in addition to what was already configured, need to participate in future off-network communication."

5.2.1 General Broadcast Group Communication

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.2.1-002]]

"The MCX Service shall only allow the initiating MCX Service Group Member to transmit on a Broadcast Group Communication, unless overridden (e.g. by a supervisor)."

5.16.2 General requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.16.2-001]] (server side)

"The MCX Service shall be able to support multiple Mission Critical Organizations, each with their own MCX Users and MCX Service Groups, on the same MCX Service system."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.16.2-002]] (server side)

"The MCX Service shall provide a mechanism by which Mission Critical Organizations designate and manage (i.e. add, delete, change authorizations, etc.) MCX Service Administrators with authority to manage users, groups, other MCX Service Administrators, security controls, and other mission affecting parameters (e.g. authorizations and priorities) of the MCX Service."

5.17 General administrative - groups and users

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.17-004]] (server side)

"The MCX Service shall provide mechanisms for an MCX Service Administrator to assign and amend the identifying information of an MCX Service Group (e.g. name, alias)."

6.4 General MCX Service Group Communications

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.4.3-001]] (server side)

"The MCX Service shall provide the MCX Service User ID, associated MCX Service User ID alias(es), MCX Service Group ID, group aliases and, if available, the identity of the Mission Critical Organization name of the transmitting Participant to the receiving MCX UEs unless the transmitting Participant's identity is restricted."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause R-6.4.3-002]] (server side)

"The MCX Service shall present users with human readable identifiers (with a minimum length of Nc3) for MCX Users (i.e. MCX Service User ID alias(es)) and for the MCX Service Groups (i.e. group alias(es))."

6.9 IDs and aliases

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.9-004]] (server side)

"The MCX Service shall provide the MCX Service User ID and /or associated aliases, the identity of the Selected MCX Service Group, and, if available, the identity of the Mission Critical Organization name of the transmitting MCX User to all MCX UEs that are receiving for display by each MCX UE."

6.15.4 Recording and audit requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-001]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log the metadata of the MCX Service Group Communications and MCX Service Private Communications under the organization's authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-002]] (server side)

"Metadata shall be logged for both the transmitting Participant and the receiving Participant(s)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-003]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to record the transmissions of the Group Communications and Private Communications under the organization's authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-004]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log at least the following metadata per communication: depending on service this may include; start time, date, MCX User ID, functional alias(es), MCX Group ID, Location information of the transmitting Participant, end time or duration, end reason, type of communication (e.g. MCX Service Emergency, regroup, private) and success/failure indication."

ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-005]] (server side)

"If an MCX Service Group Communication or MCX Service Private Communication uses end-to-end confidentiality, the MCX Service shall provide a mechanism for a Mission Critical Organization to maintain the end-to-end confidentiality when the MCX Service Group Communication or MCX Service Private Communication is logged."

ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-006]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log the metadata of non-communication related user activities under the agency's authority."

ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-007]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log at least the following non-communication activity types: MCX Service Emergency Alert, MCX Service Emergency Alert cancellation, In-progress Emergency cancellation, registration state change, overridden event, user remote logout, changing another user's affiliations, affiliation change, change of Selected MCX Service Group, (de)activation of functional alias(es) and User and Group Regroup operations."

ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-008]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log at least the following metadata per non-communication activity: time, date, MCX Service User identity, activity type and pertinent information about the activity (e.g. MCX User/Group ID(s) included in the temporary group created by the User/Group Regroup operation). The following metadata should be logged if applicable to the activity type: MCX Service Group ID, Location information of the MCX User, affiliation list, target MCX Service User ID and success/failure indication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-009]] (server side)

"The MCX Service shall provide a mechanism for a Mission Critical Organization to log metadata for all failed authorization attempts (e.g. invalid login password) by an MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-6.15.4-010]] (server side)

"The MCX Service shall provide a mechanism to collect metadata for network access events (e.g. pre-emption of the 3GPP system bearer services, loss of signal, failed registration attempts)."

7 Enhanced Mission Critical Push-to-talk architecture phase 4

7.1 Mission critical location enhancements for on-network and off-network

7.1.1 Test Purpose (TP)

(1)

```
with {MCx Service obtaining and conveying high accuracy location information}
ensure that {
    when {MCx Service user detects its position}
    then {MCx Service system can describe exact position of the MCx Service user and MCx Service has access to multiple external sources (e.g. magnetometers, orientation sensors, GNSS) and }
            }
```

(2)

```
with {Manage the privacy of Location information for MCx Users}
ensure that {
    when {MCx Service user can provide location information}
    then {The MCx Service system provides the Location information only to authorised participants (e.g. dispatcher) and the Location information is provided by the MCx Service user at the time of conveying }
            }
```

(3)

```
with {Manage the Location information update trigger}
ensure that {
    when {MCx Service system can configure several Location information update triggers}
    then {Based on identified conditions the MCx Service system can provide several Location information update triggers (e.g. initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change or MCX communication initiation) and the MCx Service system is able to modify the triggers on the MCx Service user while connected to the network and the MCx Service administrator can connect the triggers with a geographical area}
            }
```

(4)

```
with {Location information on ad hoc group emergency communication or alert}
ensure that {
    when {MCx Service user initiates ad hoc group emergency alert or ad hoc group communication}
    then {The MCx Service user is providing Location information independent from the duration of the ongoing alert or communication and in any case, a defined trigger condition will be satisfied}
            }
```

7.1.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-001]] <server>

"The MCX Service shall support obtaining and conveying Location information describing the position of the MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-002]]

"The MCX Service should support obtaining and conveying high accuracy Location information describing the position of the MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-002a]]

"The MCX Service shall be able to provide a mechanism for obtaining high accuracy Location information by integrating position information from multiple external sources (e.g. magnetometers, orientation sensors, GNSS)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-003]]

"The MCX Service shall provide for the flexibility to convey future formats of Location information."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-004]]

"The MCX Service shall provide a means for MCX Service Administrators to manage the privacy of Location information for MCX Users within their authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-005]]

"An authorised MCX User shall be able to control the supplying of Location information by the MCX UE for MCX Service communications."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-006]]

"The conveyed Location information shall be the most recently obtained information about the position of the MCX UE at the time of the Location information conveyance."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-007]]

"The MCX Service shall be capable of configuring and re-configuring one or more Location information update triggers (i.e. identified conditions that, when satisfied, cause the MCX UE to report its current Location information)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-008]]

"The MCX Service shall be able to modify Location information update triggers of an MCX User while the MCX User is on the network."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-009]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever a trigger condition is satisfied (e.g. initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation)."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-010]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever the MCX User initiates an MCX Service Emergency Alert."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-011]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever the MCX User initiates an MCX Service Emergency Group Communication."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-012]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update if the MCX User is in an MCX Service Emergency State and a configured amount of time has passed since the previous location information update."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-013]]

"The MCX Service shall provide a means for an MCX UE to send a Location information update whenever a trigger condition is satisfied while the MCX User is in MCX Service Emergency State (e.g. initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation).

NOTE 1: The Location information update triggers for an MCX User in an MCX Service Emergency State might be different than the Location information update triggers used when the MCX User is not in an MCX Service Emergency State."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-014]]

"The MCX Service shall provide a means for an MCX Service Administrator to define geographical areas to be used for Location information update triggers for MCX Users within their authority."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.11-015]]

"The MCX Service shall provide a means for an MCX UE in a predefined area to send a Location information update whenever a trigger condition configured in an MCX User's active MCX Service User Profile is satisfied (e.g. initial registration, distance travelled, elapsed time, cell change, tracking area change, PLMN change, MCX Service communication initiation).

NOTE 2: The Location information update triggers for an MCX User in a predefined area might be different than the Location information update triggers used when the MCX User is not in a predefined area."

7.1.3 Test description

7.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Handling of geographical information data.
  - Configuration of different triggers is possible. The following triggers shall be supported:
    - initial registration;
    - distance travelled;
    - elapsed time;
    - cell change;
    - tracking area change;
    - PLMN change;
    - MCX communication initiation.
- Supporting MCx Service user UE2, UE-con:
  - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports the detection of the geographical position and authorized to provide the Location information to the MCx Service.

- UE1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).
- UE2 has no authorization to receive Location information from UE1.
- UE-con is authorized to fetch Location information from UE1. UE-con provides a graphical interface to visualize the position of UE1 on a map.

7.1.3.2 Test procedure sequence

Table 7.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Configure the different Location information trigger for UE1	-->	HTTPS link to configuration interface	1	P
1	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Configure the different Location information trigger for UE1	-->	XML access (e.g. via SCP or SFTP connection)	1	P
2	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
3	No FA is active at the MCx Service system Check: UE1 publish the basic settings	<--	SIP PUBLISH according ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), IETF RFC 3903 [17]	-	--
4	Check: MCx_S1 confirm the basic settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
5	Check: MCx_S1 provides location information configuration to UE1; receive SIP MESSAGE	-->	SIP MESSAGE	1	P
6	UE1 respond with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
7	UE1 provides location information (notes 1 and 2) on MCPTT signalling event: INVITE UE-con	<--	SIP INVITE	-	--
8	Check: MCx_S1 confirm INVITE	-->	SIP STATUS 100 (TRYING)	1	P
9	UE-con receive incoming call	<--	SIP INVITE	-	--
10	UE-con responds with SIP STATUS 200	<--	SIP STATUS 200 (OK)	-	--
11	Check: MCx_S1 confirms call setup towards UE1	-->	SIP STATUS 200 (OK)	1	P
12a	UE1 provides location information (notes 1 and 2) on cell change event: MESSAGE	<--	SIP MESSAGE	-	--
13a	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P
12b	Check: UE1 provides location information (notes 1 and 2) on tracking area change event: MESSAGE	<--	SIP MESSAGE	-	--
13b	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P
12c	UE1 provides location information (notes 1 and 2) on PLMN change event: MESSAGE	<--	SIP MESSAGE	-	--
13c	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
12d	UE1 provides location information (notes 1 and 2) on periodic event: MESSAGE	<--	SIP MESSAGE	-	--
13d	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P
12e	UE1 provides location information (notes 1 and 2) on travel distance event: MESSAGE	<--	SIP MESSAGE	-	--
13e	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P
12f	UE1 provides location information (notes 1 and 2) on geographical area change event: MESSAGE	<--	SIP MESSAGE	-	--
13f	Check: MCx_S1 responds with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	2	P
14	Check: MCx_S1 provides updated location information configuration to UE1; receive SIP MESSAGE	-->	SIP MESSAGE	3	P
15	UE1 respond with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
16	Check: MCx_S1 terminates the call; UE1 receive BYE	-->	SIP BYE	-	--
17	UE1 respond with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
18a	UE1 initiates ad hoc group communication based (criterion: manual; list based) {InitiateGroup | ParticipantList | **Location Information**};	<--	SIP INVITE	-	--
19a	Check: System validates authorization of UE1	-->	DB/AAA check → VALID: 200 OK {Group Established, GroupID} → ACK	-	--
20a	Check: MCx_S1 sends confirmation to UE1	-->	SIP ACK	4	P
18b	UE1 activates ad hoc group emergency alert {Location Information}	<--	SIP MESSAGE	-	--
19b	Check: System validates authorization of UE1	-->	check → VALID: 200	4	P
20b	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	4	P

NOTE 1: MCx application of the user shall determinate the geographical position, velocity, time, travel distance, etc. based on external sensors or supporting applications, which are not on focus for this conformance test. The MCx application further shall authorized to merge all the available information into the MCx stack.
NOTE 2: MCx_S1 forwards the SIP MESSAGE with the Location information to the UE-con. UE-con updates the position of UE1 on the underlaying map on its graphical user interface.

7.1.3.3 Specific message contents

Table 7.1.3.3-1: SIP MESSAGE from MCx_S1
(step 5, Table 7.1.3.2-1; ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.7.2; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
application/vnd.3gpp.mcptt-location-info+xml		Type: multipart/mixed		
<location-info>				
<mcpttloc:configuration>	<mcpttlog:Triggering Criteria>	CellChange TrackingAreaChange PlmnChange MbmsSaChange MbsfnAreaChange PeriodicReport TravelDistance McpttSignalingEvent GeographicalAreaChange		

7.2 Preconfigured group enhancements

7.2.1 Test Purpose (TP)

(1)

```
with {MCx Service system handles preconfigured user regroup procedures}
ensure that {
    when {Authorised MCx Service user request regroup of preconfigured user groups}
    then {MCx Service system can handle the procedures request, remove request and remove response}
        }
```

7.2.2 Conformance requirements

[ETSI TS 123 280 [4] (3GPP TS 23.280), clause 10.15.2.15]

"Preconfigured user regroup remove request (MC service server - MC service client)."

[ETSI TS 123 280 [4] (3GPP TS 23.280), clause 10.15.2.16]

"Preconfigured user regroup remove response (MC service client - MC service server)."

[ETSI TS 123 280 [4] (3GPP TS 23.280), clause 10.15.2.17]

"Preconfigured regroup request return (MC service server - MC service client)."

7.2.3 Test description

7.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Re-grouping of predefined user groups.
- Supporting MCx Service user UE2, UE-con:
  - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx Service system including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user:
  - Can request a removal from the preconfigured list.

Preamble:

- The UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).

7.2.3.2 Test procedure sequence

Table 7.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-			
	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Authorize UE1 for re-grouping functionality	-->	HTTPS link to configuration interface	1	P
	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Authorize UE1 for re-grouping functionality	-->	XML access (e.g. via SCP or SFTP connection)	1	P
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.37	-			
1	UE1 send a SIP MESSAGE request for removal from the preconfigured list (note 1)	<--	SIP MESSAGE with group regroup notification request according to ETSI TS 123 280 [4] (3GPP TS 23.280)	1	P
2	Check: MCx_S1 confirms the reception with SIP STATUS	-->	SIP STATUS 200 (OK) (note 2)		
3	Check: The group management part of the MCx_S1 changes the preconfigured groups and confirm the change towards UE1	-->	SIP MESSAGE MCx_S1 responds with group regroup notification response	-	--
4	Check: UE1 confirm with SIP STATUS	<--	SIP STATUS 200 (OK) (note 2)	1	P
NOTE 1: UE transmit a changed MC service group ID list. The request can be triggered manually or coupled to an automation mechanism. NOTE 2: According to ETSI TS 124 229 [8] (3GPP TS 24.229) a request should be rejected by sending a suitable 4xx response.					

7.2.3.3 Specific message contents

Table 7.2.3.3-1: SIP MESSAGE from MCx_S1
(step 1, Table 7.2.3.2-1; ETSI TS 124 379 [20] (3GPP TS 24.379) Table 5.5.2.7.2-1)

Derivation Path: ETSI TS 124 379 [20] (3GPP TS 24.379) clause 16.2.2; condition client procedures				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"MESSAGE"**			
Message-body			IETF RFC 3261 [16]	
Extensible Markup Language				
application/vnd.3gpp.mcptt-regroup+xml	<mcptt-regroup-URI>	MCPTT user identity		MCPTT
<groups-for-regroup>	<list>			
<users-for-regroup>	<list>			

7.3 Remotely initiated call request to support additional call setup parameters and implicit affiliation to the groups

7.3.1 Test Purpose (TP)

(1)

```
with {Remotely initiated and requested MCx services}
ensure that {
    when {MCx Service user is authorised to initiate remote controlled communications}
    then {Authorised MCx Service user (e.g. Dispatcher) can request or initiate remotely a private
communication or a group communication from another MCx Service user UE and MCx Service user can
accept or deny the request and Authorised MCx Service user (e.g. Dispatcher) can terminate remotely
the connection}
            }
```

(2)

```
with {MCx Service user can pull MCx communication data towards an authorised MCx Service user (e.g.
dispatcher)}
ensure that {
    when {MCX User to request an authorized MCX User (e.g., a dispatcher) to send an MCX
Communication (e.g., video or data) to the MCX UE (downlink pull)}
    then {Data received by the authorised MCx Service user}
            }
```

7.3.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-001]]

"The MCX Service shall provide a mechanism for an authorised MCX User (e.g. dispatch operator) to remotely initiate an MCX Service Private Communication or MCX Service Group Communication from another user's MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-002]]

"The MCX Service shall provide a mechanism for an authorised MCX User (e.g. dispatch operator) to request remote activation of an MCX Service Group Communication from another user's (e.g. officer in the field with wearable camera) MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-003]]

"The MCX Service shall provide a mechanism for an authorised MCX User (e.g. officer in the field with wearable camera) to accept or deny access to an MCX Service Group Communication from their MCX UE."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.1.2-004]]

"The MCX Service shall provide a mechanism for an MCX User to request an authorised MCX User (e.g. a dispatcher) to send an MCX Communication (e.g. video or data) to the MCX UE (downlink pull).

NOTE: A video could be stored on the MCX UE or taken online by the camera."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.21.2.1-001]]

"The MCX Service shall provide a mechanism for an authorised MCX User (e.g. dispatch operator) to remotely terminate an MCX Service Private Communication or MCX Service Group Communication from another user's MCX UE."

7.3.3 Test description

7.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Remote initiation of association and call activation.
- Supporting MCx Service user UE2, UE-con:
  - UE2 and UE-con are connected to MCx_S1.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- MCx_S1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT, UE2 (MCx Service client) and UE-Con (MCx Service client with controller/dispatcher functionality).

7.3.3.2 Test procedure sequence

Table 7.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
1	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Authorize UE1 for acting as remotely controlled user entity	-->	HTTPS link to configuration interface	1	P
2	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Authorize UE1 for acting as remotely controlled user entity	-->	XML access (e.g. via SCP or SFTP connection)	1	P
3	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
4	No FA is active at the MCx Service system. Check: UE1 publish the basic settings	<--	SIP PUBLISH according ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), IETF RFC 3903 [17]	-	--

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
5	Check: MCx_S1 confirm the basic settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
6a	UE-con initiates the request for remote initiated private communication by SIP MESSAGE	<--	SIP MESSAGE containing <remotely-initiated-private-call-request>	-	--
7a	Check: MCx_S1 checks the authorization for UE-con to initiate remote connections and confirms with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
8a	Check: MCx_S1 forwards the request for remote initiated private communication by SIP MESSAGE to UE1	-->		1	P
9a	UE1 confirm the receipt with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
10a	UE1 initiates group communication; sending SIP INVITE	<--	SIP INVITE <remotely-initiated-private-call-response>	-	--
6b	UE-con initiates the request for remote initiated group communication by SIP MESSAGE	<--	SIP MESSAGE containing <remotely-initiated-group-call-request>	-	--
7b	Check: MCx_S1 checks the authorization for UE-con to initiate remote connections and confirms with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
8b	Check: MCx_S1 forwards the request for remote initiated group communication by SIP MESSAGE to UE1	-->		1	P
9b	UE1 confirm the receipt with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
10b	UE1 initiates group communication, sending SIP INVITE	<--	SIP INVITE <remotely-initiated-group-call-response>	-	--
6c	UE1 initiates the request for remote initiated private communication by SIP MESSAGE	<--	SIP MESSAGE containing <remotely-initiated-private-call-request>	-	--
7c	Check: MCx_S1 checks the authorization for UE1 to initiate remote connections and confirms with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
8c	Check: MCx_S1 forwards the request for remote initiated private communication by SIP MESSAGE to UE2	-->		1	P
9c	UE2 confirm the receipt with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
10c	UE2 initiates group communication, sending SIP INVITE	<--	SIP INVITE <remotely-initiated-private-call-response>	-	--
6d	UE1 initiates the request for remote initiated group communication by SIP MESSAGE	<--	SIP MESSAGE containing <remotely-initiated-group-call-request>	-	--
7d	Check: MCx_S1 checks the authorization for UE1 to initiate remote connections and confirms with SIP STATUS 200 (OK)	-->	SIP STATUS 200 (OK)	1	P
8d	Check: MCx_S1 forwards the request for remote initiated group communication by SIP MESSAGE to UE2	-->		1	P
9d	UE2 confirm the receipt with SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
10d	UE2 initiates group communication, sending SIP INVITE	<--	SIP INVITE <remotely-initiated-group-call-response>	-	--

7.3.3.3 Specific message contents

7.4 Geographical affiliation/de-affiliation

7.4.1 Test Purpose (TP)

(1)

```
with {Notification to an MCx Service user on entering a geographical area}
ensure that {
    when {MCx Service system informs MCx Service user about the need of association regarding
entering or leaving a geographical area}
    then {On entering a geographical area which is connected to a pre-defined group the MCx Service
system triggers the MCx Service user to affiliate or de-affiliate on leaving}
```

7.4.2 Conformance requirements

[ETSI TS 124 379 [20] (3GPP TS 24.379), clause 6.3.2.4.2]

"Generating a SIP MESSAGE request for notification of entry into or exit from a group geographic area. This clause describes the procedures for generating a SIP MESSAGE request to notify an MCPTT client that it has entered a pre-defined group geographic area or exited from a pre-defined group geographic area requiring affiliation to or de-affiliation from a group. The procedure is initiated by the participating MCPTT function when the participating MCPTT function determines that the MCPTT client has entered a pre-defined group geographic area or exited from a pre-defined group geographic area."

[ETSI TS 124 379 [20] (3GPP TS 24.379), clause 12.1.1.4]

"MCPTT client receives notification of entry into or exit from a group geographic area. Upon receipt of a "SIP MESSAGE request for notification of entry into or exit from a group geographic area", the MCPTT client."

7.4.3 Test description

7.4.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support Location information.
  - Location information is connected to a pre-defined group communication.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.

- The IUT can measure its geographical position based on e.g. GNSS or GPS and provide Location information to the MCx Service system.
- UE1 is the acting user.

Preamble:

- The IUT MCx is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT and UE-Con (MCx client with controller/dispatcher functionality).

7.4.3.2 Test procedure sequence

Table 7.4.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	--
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.3.37	-		-	--
1	UE1 provides location information on cell change event: MESSAGE	<--	SIP MESSAGE according to test case 7.1	-	--
2	Check: MCx_S1 responds with STATUS	-->	SIP STATUS 200 (OK)	1	P
3	Check: MCx_S1 indicates to UE1 that de-affiliation is required; event: MESSAGE	-->	SIP MESSAGE	1	P
4	UE1 confirms with STATUS	<--	SIP STATUS 200 (OK)	-	--
5	UE1 starts de-affiliation with PUBLISH	<--	SIP PUBLISH	-	--
6	Check: MCx_S1 responds with STATUS	-->	SIP STATUS 200 (OK)	1	P
7	UE1 de-affiliated. (note)	--	- none -	-	--
8	UE1 starts re-affiliation with PUBLISH triggered by manual action	<--	SIP PUBLISH	-	--
9	Check: MCx_S1 responds with STATUS	-->	SIP STATUS 200 (OK)	1	P
10	UE1 affiliated. (note)	--	- none -	-	--
NOTE: Affiliation status can be checked via graphical user interface.					

7.5 Geographical sending of an emergency alert feature

7.5.1 Test Purpose (TP)

(1)

```
with {MCx Service user providing a mechanism to provide geographical location information}
ensure that {
    when {MCx Service user is entering or leaving a specific geographical area}
    then {An ad hoc group emergency alert can be activated on entering or deactivated on leaving for
the MCx Service user depending on a specific geographical area and MCx Service system can authorise
an MCx Service user to activate a preconfigured group emergency alert for this specific area}
            }
```

7.5.2 Conformance requirements

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.1.9-001]]

"The MCX Service shall provide a mechanism for an MCX Service Emergency Alert to be triggered when an MCX UE moves into a predefined area."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.1.9-002]]

"The MCX Service shall provide a mechanism for an MCX Service Emergency Alert to be cancelled when the MCX UE moves out of a predefined area or to remain active until cancelled by the MCX User."

[ETSI TS 122 280 [2] (3GPP TS 22.280), clause [R-5.6.2.4.1-005]]

"The MCX Service shall provide a mechanism for an authorised MCX User to configure an MCX Service Emergency Alert to send a notification to MCX Users within a configurable geographic area of the MCX User entering the MCX Service Emergency State, independent of the MCX Service Group Membership."

7.5.3 Test description

7.5.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support Location information.
  - Location information is connected to ad-hoc group emergency alert.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- The IUT have access to a database to compare or visualize the geographical positions of the connected users based on the GNSS or GPS information provided by the MCx service user's Location information poll to configure the following functionalities correctly:
  - Ad hoc group emergency alert activation while a MCx Service user enters a specific geographical location.
  - Cancellation of an ad hoc group emergency alert while MCx Service user is leaving a specific geographical location.
  - Send ad hoc emergency group alert notification independent from the group membership.
- MCx_S1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT and UE-Con (MCx client with controller/dispatcher functionality).

7.5.3.2 Test procedure sequence

Table 7.5.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-		-	
	EXCEPTION: Procedure 'Initial MCX Authentication, Registration, Configuration and Subscription' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.3.37	-		-	
1	UE1 provides location information on cell change event: MESSAGE.	<--	SIP MESSAGE according to test case 7.1	-	--
2	Check: MCx_S1 responds with STATUS	-->	SIP STATUS 200 (OK)	1	P
3	Check: MCx_S1 indicates to UE1 that ad-hoc emergency group alert is required; event: MESSAGE	-->	SIP MESSAGE	1	P
4	UE1 confirms with STATUS	<--	SIP STATUS 200 (OK)	-	--
5	UE1 activates ad hoc group emergency alert	<--	SIP MESSAGE according to ETSI TS 124 379 [20] (3GPP TS 24.379) clause 17.2.2.1.1	-	-
6	Check: MCx_S1 validates authorization of UE1.	-	check → VALID: 200	1	P
7	Check: MCx_S1 send confirmation to UE1	-->	SIP STATUS 200 (OK)	1	P
8	Check: For the initiator the ad hoc group emergency alert automatically is confirmed. Other participants need to confirm automatically or by manually action		- none -		

8 MCData using IP connectivity

8.1 IP Connectivity

8.1.1 Test Purpose (TP)

(1)

```
with {MCData service capability for transmission of IP data towards a receiver in the network}
ensure that {
    when {MCData user initiates the IP data transport}
    then {The MCData service capability enables a MCData user for initiating IP data transport
towards a server in the network or another MCData user and the MCData Service enables the incoming
transport of IP data to another MCData user which is not the initiator and the MCData Service
supports different per packet priorities for incoming and outgoing IP data transport}
            }
```

(2)

```
with {MCData Service authorizations}
ensure that {
    when {The MCData Service capability provides different authorisation options}
    then {The MCData Service system provides preconfigured or authorized MCData user or personnel
access to authorize the initiation of incoming or outgoing transport of IP data for the MCData user
to specific destinations and authorization can be requested by initiating transport IP data to new
destination or can be revoked on demand by authorized access and the MCData system allows an
authorized remote access to allow the use of higher per packet priority for a particular MCData
user}
            }
```

(3)

```
with {Authorized control over outgoing IP data transport}
ensure that {
    when {MCData user tries to send IP data to new destination and authorized person has set
predefined transport rules or needs dynamically to approve the request}
    then {MCData service checking if preconfigured authorization exists for the destination. And in
case is not preconfigured, a real-time approval request is sent to the authorization person.}
            then {IP data transport is initiated upon approval, and the authorized person can revoke
the authorization at any time, preventing further transmissions.}
            }
```

(4)

```
with {Remote control over higher packet priority assignment}
ensure that {
    when {Authorized person attempts to remotely enable higher per packet priority for an MCData
user}
    then {MCData service is validating the authorization credentials of requesting entity, Higher
priority setting is applied to the designated MCData user, and the priority ca be revoked from the
authorized person at any time. The system logs record all authorization actions precisely}
            }
```

8.1.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-001]]

"The MCData Service shall enable a MCData user to initiate transport of IP data towards a server in the network or another MCData user."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-002]]

"The MCData Service shall enable incoming transport of IP data towards an MCData user, initiated by another MCData user."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-003]]

"The MCData Service shall enable an authorised person for an MCData user to authorise initiation of transport of IP data from that MCData user to specific destinations. This authorization may be preconfigured or may be provided by the authorised person when IP data transport to a new destination is initiated. Authorization may be revoked on demand by the authorised person."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-004]]

"The MCData Service shall enable a MCData user or authorised person for this MCData user to authorise initiation of incoming transport of IP data from specific other MCData users. This authorization may be preconfigured or may be provided by the authorised person when IP data transport from a new destination is initiated. Authorization may be revoked on demand by the authorised person."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-005]]

"The MCData Service shall support incoming and outgoing IP data transport for a MCData user with a higher per packet priority."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.5.2-006]]

"The MCData Service shall enable an authorised person to remotely authorise the use of higher per packet priority for a particular MCData user."

8.1.3 Test description

8.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, including the Group Management Server, the Configuration Management Server, the Key Management Server, the Identity Management Server, the HTTP Server, and the SIP AS. The IUT (UE1) also consists of all sub-systems of the MCx user service.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - IPConn authorization for several users.
- Supporting MCx Service user UE2/:
  - UE2 is connected to MCx_S1.
  - UE2 is authorized to use IPConn functionality.
- Figure 4.1-5 shows the principal setup.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- IUT supports IPConn client functionality.
- MCx_S1 is the acting user.

Preamble:

- The IUT MCx_S1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT (MCData client) and UE-Con (MCData client with controller/dispatcher functionality).

8.1.3.2 Test procedure sequence

Table 8.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
1	Check: MCx_S1 provides a configuration interface based on HTTPS request. Login required. Activate the IPConn functionality for UE1 and UE2 including prioritization	-->	HTTPS link to configuration interface	1	P
2	Check MCx_S1 provides a configuration interface based on XML file import. Login for access to the file transfer function required. Activate the IPConn functionality for UE1 and UE2 including prioritization	-->	XML access (e.g. via SCP or SFTP connection)	1	P
3	UE1 provides local configuration for using IPConn functionality	<--		-	--
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	-	<mcdata-client-id> <mcdata-access-token>	-	--
	Execute MCData group Affiliation Status Change as described in ETSI TS 137 579-1 [34] (3GPP TS 37.579-1) clause 5.3.34. No FA is active at the MCx Service system. UE1 publish the basic settings	<--	SIP PUBLISH according ETSI TS 137 579-1 [34] (3GPP TS 37.579-1), IETF RFC 3903 [17]	-	--
	Check: MCx_S1 confirm the basic settings	-->	STATUS 200 OK (PUBLISH) according ETSI TS 124 229 [8] (3GPP TS 24.229)	1	P
4	UE1 starts initiation of IPConn using SIP INVITE message	<--	SIP INVITE	-	--
	MCx_S1 checks the policy rules for the targeting users UE1 and UE2.		-none- (both users have authorization for using IPConn)	-	--
5	Check: MCx_S1 provides proceeding information to UE1	-->	SIP STATUS 100 (TRYING)	2	P
	MCx_S1 invites UE2 as second IPConn user	<--	SIP INVITE	-	--
	UE2 confirms invitation request, replying SIP STATUS 200 (OK)	<--	SIP STATUS 200 (OK)	-	--
6	Check: MCx_S1 sends confirmation to UE1	-->	SIP STATUS 200 (OK)	2	P
7	UE1 start establishment of the GRE tunnel sending IP protocol with generic routing encapsulation (GRE) using IPv6	<--	IETF RFC 2784 [37]	-	--
8	UE2 initiates GRE tunnel like UE1; MCx_S1 connects the tunnel	-	IETF RFC 2784 [37]	-	--
9	UE1 starts Ping test through the GRE tunnel to check the availability of UE2	-	ICMP request message according to IETF RFC 792 [38]	-	-

8.1.3.3 Specific message contents

Table 8.1.3.3-1: SIP INVITE from MCx_S1
(Table 8.1.3.2-1; step 4, ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) Table 5.2.5.1-1)

Derivation Path: ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.5.2.5.1; condition MCPTT-INFO-EVENT				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Method	**"INVITE"**			
Request-URI	MCPTT_PublicServiceID	MCPTT originating user public ID		MCPTT
Message-header				
P-asserted identity/SIP PAI Address	<SIP PAI User part>			
	<SIP PAI Host part>			
Route				
addr-spec[1]	SIP URI			
user-info and host	P-CSCF address of the SS	P-CSCF address as assigned to the UE via NAS signalling or P-CSCF discovery		
port	protected server port of the SS	as assigned during registration		
uri-parameters	List of parameters			
Message-body				
Application//vnd.3gpp.mcdata-info+xml				
<mcdata-Params>	<request-type>	one-to-one-ipcon		IPCon

8.2 MCData with different QoS requirements

8.2.1 Test Purpose (TP)

(1)

```
with {FRMCS UE attach to FRMCS network including 5QI activation}
ensure that {
    when {UE is powered on or returns from flight mode}
    then {RRC attaching procedure including context activation and IP address assignment in the
network}
            }
```

8.2.2 Conformance requirements

[ETSI TS 103 765-3 [27], clause 6.2]

"The SMF receives policy rules from the PCF via N7 and sends instructions for QoS enforcement, traffic steering and charging control via N4 to the UPF. The UPF enforces the rules via Packet Detection Rules (PDRs) and forwarding actions."

[ETSI TS 103 765-3 [27], clause 8.2]

"FRMCS Quality of Service Principles. The FRMCS Transport Stratum QoS framework is based on the following principles:

a) QoS is applied via PDU sessions (up to 15 per UE) and QoS flows (up to 63 per PDU session).

b) QoS is enforced at UE, gNodeB and UPF based on rules propagated from the UDM/PCF or pre-configured.

c) QoS policy is controlled from the FRMCS service stratum via messaging over reference points Rx or N5.

d) The gNodeB can map multiple QoS flows with similar characteristics to a single Data Radio Bearer (DRB).

e) QoS flows can be of type GBR (Guaranteed Bit Rate), non-GBR and delay critical GBR.

f) Each QoS flow is associated with a QoS profile, including:

i) A QFI (QoS Flow identifiers) based on 5QI.

ii) An ARP (Allocation & Retention Priority).

iii) Type specific & optional parameters.

IP data is mapped (filtered) to QoS flows at UE (based on QoS rules) and UPF utilizing packet filter sets. The FRMCS Application Function (AF), defined in ETSI TR 103 791 [40], shall perform mapping from a Session Description Information (SDI), received within the AF session signalling, to service information passed to the PCF over the N5 or Rx reference points as per ETSI TS 129 513 [29] (3GPP TS 29.513). The FRMCS AF shall signal the PCF to request resources for media bearers using one of the following options:

i) as per ETSI TS 123 289 [14] (3GPP TS 23.289), clause 7.2.2 (see step 3 in Figure 8.1-1) if the FRMCS SIP Core represents the

ii) FRMCS AF; or as per ETSI TS 123 289 [14] (3GPP TS 23.289), clause 7.2.3.2 if the MC Service Server represents the FRMCS AF (see step 6 in Figure 8.2-2).

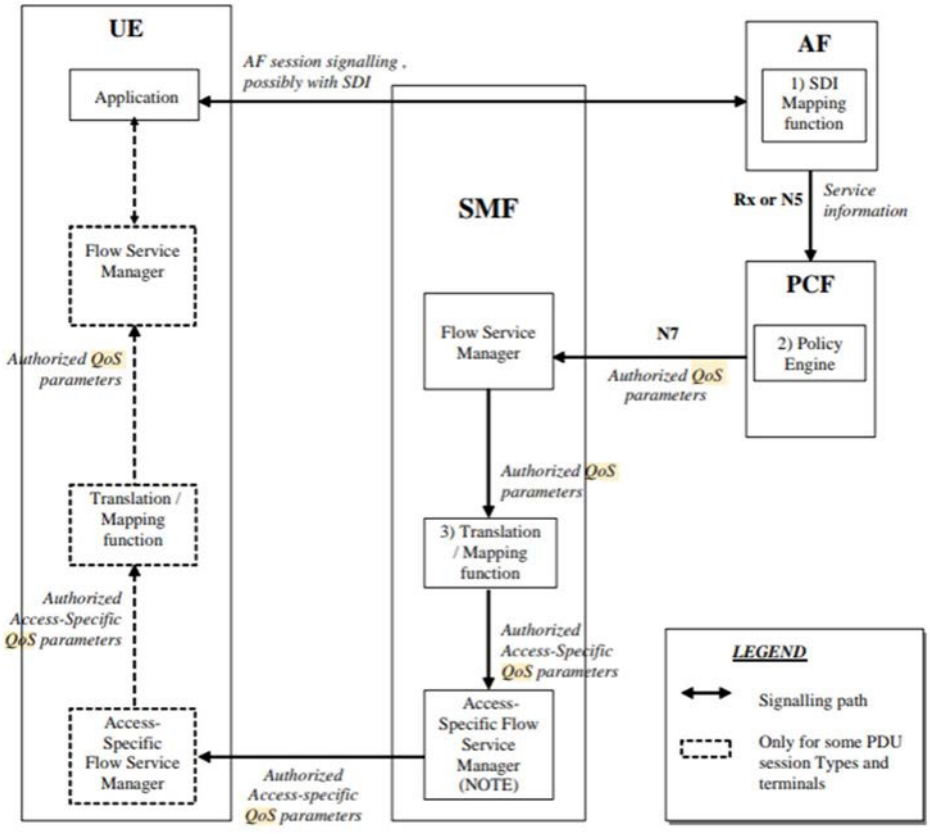


Figure 8.2-1: Communication Security

The MC Service Server shall utilize the Npcf_PolicyAuthorization Service for N5 or Rx procedures to request policies for the PDU session which the FRMCS MC session is bound to. The PCF shall perform mapping from the service information received over the Rx or N5 reference points to the Authorised QoS parameters that shall be passed to the SMF via the N7 reference point.

The Session binding function determines relevant PDU session and associates IP data flows to it. PCF shall consider any 5GC specific restrictions, the AF service information and other information available to the PCF. PCF associates a set of QoS parameters (5QI, ARP, etc.) to each PCC rule as per ETSI TS 123 289 [14] (3GPP TS 23.289).

The SMF shall perform mapping from the Authorised QoS parameters received from PCF to the access specific QoS parameters as per ETSI TS 123 289 [14] (3GPP TS 23.289).

FRMCS AF can use Npcf_EventExposure service to get notifications about the outcomes of UE policies delivery. The network sets up QoS flow(s) with a suitable QoS and indicates to the UE the QoS characteristics of those QoS flow(s) as per Figure 8.2-1 (reference to ETSI TS 123 289 [14] (3GPP TS 23.289), clause 7.1).

[ETSI TS 103 765-3 [27], clause 8.2]

"5QI and ARP Application per FRMCS Service. The FRMCS Transport Domain shall support at least the 5QI identified in table 8.2-1

Table 8.2-1: List of 5QI in FRMCS

5QI
4
5
8
65
69"

"

[ETSI TS 103 765-3 [27], clause 8.3]

"The QoS of FRMCS Transport Stratum traffic is classified using 5QI. Since the traffic flow to some endpoints (Trackside Gateway, VAS controller, etc.) traverses an IP/MPLS backbone the 5QI shall be mapped to a Traffic Class. For each QoS Flow characterized by its 5QI, the mapping of 5QI to DSCP values are done on different nodes depending on the traffic direction:

- *For UL traffic, the gNodeB maps the 5QI values to transport DSCP values. For DL traffic, the mapping is done by local configuration on the SMF, which provides UPF with the transport DSCP values. The UPF then marks DL traffic on a per QoS Flow basis.*

The DSCP mapping for mandatory FRMCS 5QIs from Table 8.2-1 is identified in Table 8.2-2.

Table 8.2-2: 5QI to DSCP Mapping

5QI	*DSCP*
4	*AF43 (DSCP 38)*
5	*CS5 (DSCP 40)*
8	*AF12 (DSCP 12)*
65	*EF (DSCP 46)*
69	*CS5 (DSCP 40)*

"

8.2.3 Test description

8.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system not required.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- Different bearers are available according to Table 8.2-2.

IUT:

- IUT (MCx_S1).

- The IUT (MCx_S1) consists of all sub-systems of the MCx Service system including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- MCx_S1 is the acting user.

Preamble:

- The UE1 is connected to PLMN of the primary 5GS.
- The IUT MCx Service system 1 (MCx_S1) is connected to the MCData client and UE-Con (MCData client with controller/dispatcher functionality).
- IUT is connected to all contexts presented by an individual APN and IPv4 or IPv6 IP address.

8.2.3.2 Test procedure sequence

Table 8.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	Exception: Switch on the UE1. UE1 starts radio resource allocation procedure (RRC). 5QI contexts are assigned to the SIM application permanently in the network and UE1 assign via PDU session initialization	--	RRC activation	-	--
	Network confirms the radio resource allocation and UE1 is attached to the network	--		-	--
	PDU contexts in the UE. For each 5QI there shall be a bearer available (e.g. check for IP address allocation)	--	5QI contexts are available	-	--
	Application A on UE1 request transmission 5QI=4. Bearer is accessible for UE1	--	5QI contexts are accessible and active	-	--
1	Check: MCx_S1 can access the application A (e.g. via PING) through the APN (5QI=4) (note)	-->	PING through the established network is possible	1	P
	Check: application B on UE1 request transmission 5QI=5. Bearer is accessible for UE1	--	5QI contexts are accessible and active	-	--
2	Check: MCx_S1 can access the application B (e.g. via PING) through the APN (5QI=5) (note)	-->	PING through the established network is possible	1	P
	Check: application C on UE1 request transmission 5QI=8. Bearer is accessible for UE1	--	5QI contexts are accessible and active	-	--
3	Check: MCx_S1 can access the application C (e.g. via PING) through the APN (5QI=8) (note)	--	PING through the established network is possible	1	P
	Check: application D on UE1 request transmission 5QI=65. Bearer is accessible for UE1	--	5QI contexts are accessible and active	-	--
4	Check: MCx_S1 can access the application D (e.g. via PING) through the APN (5QI=65) (note)	-->	PING through the established network is possible	1	P
	Check: application E on UE1 request transmission 5QI=69. Bearer is accessible for UE1	--	5QI contexts are accessible and active	-	--
5	Check: MCx_S1 can access the application E (e.g. via PING) through the APN (5QI=469 (note)	-->	PING through the established network is possible	1	P
NOTE:	This conformance tests should show the general path through the network while using the different 5QI. Performance is to be rated with further conformance test (e.g. part of GCF).				

8.3 MCData streaming and datagram mode

8.3.1 Test Purpose (TP)

(1)

```
with {MCData Service system provides a data streaming capability}
ensure that {
    when {MCData service user is authorized to use MCData streaming capability}
    then {Each recipient can decide to select the receipt of a data stream by using a provided link
or MCData service user can reject the receipt of a data stream}
            }
```

(2)

```
with {Automatic reception of streamed data for tracking delivery status per recipient}
ensure that {
    when {MCData user initiates a data stream and marks it for automatic reception by UEs}
    then {Recipient UEs automatically accept and start receiving the stream without user
intervention. MCData system generates and returns a start record for each recipient that has
successfully receiving the stream and when stream ends, MCData systems returns stop record to the
sender identifying, which users completed or terminated reception}
                        }
```

(3)

```
with {Optimize delivery of data streams to users in the same area}
ensure that {
    when {MCData is initiated for a group of users}
    then {MCData service detects the common delivery context, and the systems uses available network
and system efficiencies to deliver the stream using the least amount of transmission resources and
all the recipients within the area receive the stream in parallel with minimized delay and bandwidth
consumption.}
                        }
```

(4)

```
with {User or authorized initiated cancellation of active or buffered data streams}
ensure that {
    when {MCData user decides to terminate streaming that is currently active or cancel pending
initiated by another user}
    then {MCData service holds the transmission of ongoing data to all recipients. All buffered and
undelivered streams segments are purged from system and successful cancelation is returned to the
initiating entity. No further delivery attempts of the cancelled stream data are made}
                        }
```

(5)

```
with {Measurement of data communication setup times under different conditions}
ensure that {
    when {MCData user made request to initiate a data stream under normal and urgent conditions}
    then {MCData systems completes the setup and confirms establishment within less than 1 second
for immediate and for normal data the setup time does not exceed 3 seconds, and the measurement
includes the time from the initial request to successful establishment of communication as perceived
by the user and failures of those timing requirements are logged for systems analyses}
                        }
```

8.3.2 Conformance requirements

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-001]]

"The MCData Service shall provide a data streaming capability."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-002]]

"The MCData data streaming capability shall provide an option that allows each recipient to choose to receive the data stream or not (e.g. by sending a link (URL) to all relevant members)."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-003]]

"The MCData data streaming capability shall allow a user to reject to receive the data stream."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-004]]

"The MCData data streaming capability shall provide a sending user selectable indication for automatic reception by the UE."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-005]]

"The MCData data streaming capability shall provide start and stop records to the sender for each recipient successfully receiving the data stream."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-006]]

"The MCData data streaming capability shall make use of available system delivery efficiencies for streaming of common information to users within the same relevant area and able to receive at the same time."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-007]]

"The MCData data streaming capability shall allow a user to cancel streaming of data they have initiated including data remaining buffered in the system waiting to be streamed."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-008]]

"The MCData data streaming capability shall allow an authorised user to terminate streaming of data being sent and cancel streaming of data remaining buffered in the system waiting to be streamed."

[ETSI TS 122 282 [3] (3GPP TS 22.282), clause [R-5.4.2-009]]

"The MCData data streaming capability shall provide setup time of:

- *Less than 1 second for immediate setup of a data communication.*
- *Less than 3 second for normal setup of a data communication.*

NOTE 1: Setup time of a communication is essential because of their safety related character.

NOTE 2: Communication setup encompasses the value of the elapsed time between the communication establishment request and the indication of successful communication establishment."

8.3.3 Test description

8.3.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core, according to figure 4.1-6.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support QUIC according to IETF RFC 9000 [31].

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx Service system including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- MCx_S1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

8.3.3.2 Test procedure sequence

Table 8.3.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
1	Check: MCx_S1 initiates UDP Data connection with datagram mode according to IETF RFC 9897 [32] and IETF RFC 9000 [31]	-->	Connection on UDP established		P
2	UE1 confirms transmission after all datagrams are sent	<--	none		-
4	Check: MCx_S1 initiates UDP Data connection with streaming mode according to IETF RFC 9897 [32] and IETF RFC 9000 [31]	-->	Connection on UDP established		P
5	UE1 confirms transmission after all datagrams are sent	<--	none		-

8.3.3.3 Specific message contents

Table 8.3.3.3-1: UDP from UE1
(step 1, Table 8.3.3.2-1; IETF RFC 9897 [32])

Derivation Path IETF RFC 9897 [32]; condition: Stream Information				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Datagram Protocol	**Payload**			
Stream Index	Index	In stream mode stream index will be changed for each transmission	IETF RFC 9897 [32]	
Stream Packet Number	Number of packages			
QUIC IETF		Could be more than one QUIC IETF connection	IETF RFC 9000 [31]	

Annex A (normative): Additional Functionalities

A.1 Support of MP-QUIC and MP-TCP

A.1.1 Test Purpose (TP)

(1)

```
with { Support of MP-QUIC }
ensure that {
    when { FRMCS Service system open a MP-QUIC stream }
    then {FRMCS Service system can establish a data connection using QUIC IETF based on UDP }
            }
```

(2)

```
with { Support of MP-QUIC }
ensure that {
    when { FRMCS Service system open a MP-TCP stream }
    then {FRMCS Service system can establish a data connection using QUIC IETF based on TCP }
            }
```

A.1.2 Conformance requirements

[ETSI TS 103 765-1 [28], clause 5.3.3]

"Architecture. It is constituted of two functions, the Multipath Client and the Multipath Gateway. The Multipath Client and the Multipath Gateway are connected via two reference points:

- *FSMPM which enables the FRMCS Multipath control plane function, including Multipath Gateway discovery, information exchange on available Multipath Datapaths.*
- *FSMPU which enables the user plane transport of a Multipath Dataflow based on a Multipath Protocol over one or more Multipath Datapaths."*

A.1.3 Test description

A.1.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core according to figure 4.1-6.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- MCx Service system is configured to support the following features:
  - Support of MP-QUIC according IETF RFC 9000 [31].
  - Support of MP-TCP according to IETF RFC 9897 [32].
  - Multipath Control Function (MPF) to control the parameter for the switching of the uniflows.

IUT:

- IUT (UE1).
- The IUT (UE1) consists of all sub-systems of the MCx user service including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE1 is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

A.1.3.2 Test procedure sequence

Table A.1.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		U - S	Message		
1	Check: UE1 establish UDP based session introducing QUIC content	-->	Connection is accepted by MCx_S1 (note)	1	P
2	Check: UE1 establish TCP based session introducing QUIC content	-->	Connection is accepted by MCx_S1 (note)	2	P
NOTE:	If more than one uniflow will be used, then also the destination needs to be checked. A destination can have multiple connection numbers and one connection number can have multiple destinations regarding to the MPF settings.				

A.1.3.3 Specific message contents

Table A.1.3.3-1: MP-QUIC connection from UE1 (step 1, Table A.1.3.2-1; IETF RFC 9000 [43])

Derivation Path IETF RFC 9000 [31]; condition MP-QUIC				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
User Datagram Protocol	UDP	Bearing protocol type		
QUIC IETF				
QUIC Connection Information	Connection Number	Uniflow index	IETF RFC 9000 [31]	
Destination Connection ID				
Source Connection ID				
Token				
Payload		Payload		
CRYPTO		Cryptography part		

Table A.1.3.3-2: MP-QUIC connection from UE1
(step 2, Table A.1.3.2-1; IETF RFC 9897 [32])

Derivation Path: IETF RFC 9897 [32]; condition MP-TCP				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Transmission Control Protocol	TCP	Bearing protocol type		
Options				
Multipath Transmission Control Protocol	Multipath Capable			
Multipath TCP				
Multipath TCP flags				
Senders Key				
Subflow Token	And expected IDSN			
MPTCP analysis			IETF RFC 9897 [32]	
Master flow	Master TCP stream			
Stream index				
TCP subflow stream id	Sub-stream identification			

A.2 Support of IPv6

A.2.1 Test Purpose (TP)

(1)

```
with { Support of IPv6 }
ensure that {
    when { FRMCS Service server is attached to the network }
    then {FRMCS Service server is receiving an IP address with IPv6 and the routing function can
handle the entry }
            }
```

A.2.2 Conformance requirements

[ETSI TS 103 765-1 [28], clause 5.3.6]

"IP Address assignment is required:

a) for public service layer IP address for each Multipath Dataflow Handler; and

b) for each transport domain Multipath Datapath for each Multipath Dataflow Handler.

The 5G Core provides IPv6 prefix during the initial default data connection establishment to each UE. The procedure is IPv6 Prefix Delegation. The IPv6 prefix is assigned via DHCPv6.

As illustrated in Figure 5.3.5-1 (Multipath Client), the transport-related IP addresses on the OB side (IP OB 1.1, IP OB 2.1, etc.) are appended with this prefix. The IPv6 prefix is delegated from the 5G Core directly, thus the IP addresses, that are part of the prefix, are reachable and routable from the 5G Core to the correct UE.

The Multipath Client uses the given IPv6 prefix from the 5GC/UE to define the full transport related IP address on the Multipath Client's side (IP OB 1.1, IP OB 2.1, etc.) by IPv6 stateless autoconfiguration:

- *The transport-related IP addresses on the Multipath Client's side (IP OB 1.1, IP OB 2.1, etc.) are defined locally on the Multipath Client. These IP addresses shall be propagated to the Multipath Gateway so that the Multipath Gateway can populate the mapping table for each Multipath Dataflow Handler to map the correct trackside transport Multipath Datapath IP address to the On-Board transport IP Multipath Datapath address.*

- *The MPF-CTRL of Multipath Client shall assign the IPv6 addresses to the TAP Service Clients interfaces. They may be public addresses (e.g. Global Unicast Addresses, GUA) or Link-Local addresses, based on the specific address plan in use."*

A.2.3 Test description

A.2.3.1 Pre-test conditions

System Configuration:

- MCx Service system consists of all sub-systems of the Common Services Core.
- For the underlying "transport bearer" over which the UE1 (MCx Service system) and the IUT will communicate Parameters are set to the default parameters specific for the basic E-UTRA single simulated cells cell network scenarios, as defined in ETSI TS 138 508 [6] (3GPP TS 38.508) clause 6.3. The simulated Cell 1 shall belong to PLMN1 (the PLMN specified for MCx operation in the MCx configuration document).
- 5GS Common Service Core is supporting IPv6 according to [30].
- MCx Service system is configured to support the following features:
  - Support of IPv6.

IUT:

- IUT (MCx_S1).
- The IUT (MCx_S1) consists of all sub-systems of the MCx Service system including the ability for floor control, pre-arranged group communication functionality and ad-hoc group communication functionality. Ensure the correct configuration.
- UE1 is the acting user.

Preamble:

- The IUT UE is connected to PLMN of the primary 5GS.
- The MCx Service system 1 (MCx_S1) is connected to the IUT.

A.2.3.2 Test procedure sequence

Table A.2.3.2-1: Main behaviour

St	Procedure	Message Sequence		TP	Verdict
		S - U	Message		
	EXCEPTION: Procedure 'MCPTT Server - MCPTT Client / Configuration / Authentication / User Authorization / UE Configuration / User Profile' as described in ETSI TS 136 579-3 [7] (3GPP TS 36.579-3) clause 5.1.3	--		-	-
1	Check: request IP settings on MCx_S1	--	MCx_S1 has allocated an IPv6 address	1	P
2	Check: MCx_S1 send ping request to UE1 using IPv6	-->	MCx_S1 is using ICMPv6 according to IETF RFC 4443 [29]	1	P
3	UE1 reply to ICMPv6 request (note)	<--	UE is using IPv6 format	-	--
NOTE: The reply on the ICMPv6 request indicates, that the routing between UE1 and MCx_S1 is correctly. This means, that following communication regarding FRMCS or MCx communication is also routed correctly in Ipv6.					

A.2.3.3 Specific message contents

Table A.2.3.3-1: IPv6 from UE1
(step 1, Table A.2.3.2-1; IETF RFC 8200 [30])

Derivation Path: IETF RFC 8200 [30]; condition Internet Protocol part				
Information Element	**Value/remark**	**Comment**	**Reference**	**Condition**
Internet Protocol Version 6				
Traffic Class				
Source Address				
Destination Address				
Source SLAAC MAC				
Destination SLAAC MAC				
Stream Index				

History

Version	Date	Status
V1.1.1	September 2026	Publication